

The Relativistic Investigation Bureau

The Murder That Hadn't Happened Yet

Sven Gerald Rupertson — S. G. R.

Dealing with Special and General Relativity — S. G. R.

BOOK ONE

The author is Sven Gerald Rupertson, allegedly.

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Prologue

In Which the Universe Refuses to Behave

The murder had not happened yet. This was, in itself, not especially alarming. Most murders, after all, have the elementary courtesy to occur before the paperwork arrives. The universe had declined to observe this courtesy. It rarely observed any, and never with an apology. In this respect it resembled most government departments, and several of Derek's relatives. The resemblance was not considered flattering by any of the parties involved, though only the relatives had complained. The universe, as usual, had not been available for comment. Its press office had been closed for approximately thirteen point eight billion years, and the answering machine was full. Most murders have the decency to wait until after they happen before becoming administrative problems. The universe had, characteristically, declined to observe this convention. The universe, as usual, had not been consulted. The difficulty was that the Relativistic Investigation Bureau had already received the paperwork.

The time of death had been entered. The location had been entered. The victim had been identified.

The victim was Derek Gent. Derek was alive. He was also the sort of man who noticed discrepancies in official paperwork.

This particular discrepancy was rather difficult to ignore. The paperwork predicted his death three days before it was due to occur. Derek therefore did what any sensible investigator would do.

He checked the date, then the time, then the old clock.

Then the clock again, because the first check had produced a result he found professionally inconvenient. 11:17. The mysterious clock had stopped.

It had stopped at precisely the time printed on the murder notice. This would have been an excellent reason to panic if Derek had not already spent several years working in physics. Physicists are trained to distrust coincidences, measurements, clocks, diagrams, intuition and, when necessary, each other. This training has proved useful more often than is generally admitted, and has ruined more dinner parties than it has saved.

They are particularly suspicious of anything described as obvious. This suspicion has served the profession rather well. It has also made dinner parties more difficult than strictly necessary. Newton discovered that falling objects obeyed laws.

Einstein discovered that space and time did not behave the way Newton had imagined. General Relativity then made the still more inconvenient suggestion that gravity is deeply connected to the geometry of spacetime itself. None of this would have been especially troublesome if the universe had confined its eccentricities to respectable laboratories.

It did not. Clocks could disagree about elapsed time without either old clock being broken. Distances could depend on the observer.

Two observers could disagree about whether separated events were simultaneous. The speed of light could remain the same for all inertial observers. Gravity could alter the rate at which clocks accumulated proper time.

Curvature could reveal itself through the behaviour of neighbouring free-fall paths. And somewhere along the way, an ordinary cucumber became relevant. Nobody had intended this. In the Bureau's experience, the most durable problems almost never began with anyone intending them. Intention, it turned out, was optional.

The universe had simply been uncooperative. It had a long and well-documented history of this. That was how the Relativistic Investigation Bureau came into existence. It was not founded because anyone had dreamed of becoming a civil servant.

Nobody dreams of becoming a civil servant. This is one of the few points on which the entire human species is in complete and suspicious agreement. The universe, for once, appears to share the sentiment — though it has never confirmed this in writing. Bureaus involve stationery, regulations, meetings, filing systems and the possibility that a perfectly respectable physical anomaly may be rejected because Form 17B had been completed in black ink instead of blue. The Relativistic Investigation Bureau existed because Derek Gent had begun asking questions.

Unfortunately, Derek was very good at asking them. There was Trevor Boltzman, who understood difficult mathematics but occasionally required a small verbal shove toward the obvious. There was Penny Gent, who could make tensor calculus sound almost reasonable and who regarded most forms of nonsense as a personal challenge.

There was Julius Barbarian, who maintained that time did not exist and had developed an argument for this that was considerably more coherent than anyone wanted it to be. There was Hermann Minkowski, who regarded spacetime as an entirely adequate explanation of itself and had no patience with people who demanded that it behave more intuitively. And there was Alfred Weinstein, who had somehow acquired the inconvenient habit of being the real inventor of Special and General Relativity in a universe where history insisted on crediting Einstein.

Weinstein regarded this as plagiarism on a cosmological scale. He had attempted to correct the historical record through ordinary academic channels. The academic channels had requested Form 17B.

Weinstein had never fully recovered. Then there was Sherlock. Sherlock was a dolphin. This was considered, within the Bureau, only moderately unusual — a fact that said rather more about the Bureau than about the dolphin. He was paid in fish, at the official rate of 1.7 fish per hour. He was hoping for 1.8. The relevant form had been pending since before the Pre-Transition drawer was

labelled. Most of the Bureau treated him as a remote consultant who occasionally sent useful geometry and then went back to thinking about fish.

Sherlock was also, according to Penny, the least improbable member of the Bureau once one had properly understood General Relativity. Nobody asked him to explain this. The Bureau had learned that explanations often made matters worse.

So when Derek found a murder notice announcing his own future death, he did not immediately conclude that he was being hunted by a time traveller, a quantum ghost or an unusually vindictive railway timetable. He began with the evidence. A stopped clock.

A future date. A murder that had not happened. And, eventually, a distant, mysterious train.

There was only one sensible response. Investigate. Because if the universe has made a mistake, someone has to fill in the form. The universe itself has never shown the slightest inclination to do so. It appears to regard paperwork as a purely human weakness, and possibly a form of performance art.

Preferably before 11:17. The universe, in Derek's experience, rarely granted extensions, and never with an apology.

Chapter 1

The Murder That Hadn't Happened Yet

At precisely 9:17 on a Tuesday morning, Derek Gent received an email informing him that he was dead. He was not particularly surprised. Derek had always suspected that death would arrive by email. In this, as in so many other matters, he had been only slightly ahead of the general population — a position he found both comforting and faintly depressing, which was, on balance, better than the alternatives. The alternatives were rarely an improvement, a fact Derek had verified more than once, usually at personal cost and never with a sense of novelty.

Not because he was especially pessimistic, but because nearly everything important seemed to arrive by email these days. Electricity bills. Insurance renewals. Invitations to webinars. Notifications that someone had changed a password Derek had never used. Death was simply the logical next step. What surprised him was the sender.

The message had come from the Derek Gent Relativistic Investigation Bureau. Derek stared at the screen. The Bureau was not supposed to send him emails. Derek had written the rule himself, which made its violation particularly irritating.

It was his Bureau. More precisely, it was a small private investigative organization occupying two rooms on the second floor of an old office building in London, plus a corridor which the landlord insisted was a communal area and Derek insisted belonged to his Bureau whenever he needed somewhere to put a coat. The Bureau dealt primarily with unusual cases.

This was partly because Derek liked unusual cases and partly because ordinary cases paid terribly. The sign on the frosted glass door read:

DEREK GENT RELATIVISTIC INVESTIGATION BUREAU

Underneath, in smaller letters:

CAUSALITY, CHRONOLOGY & OTHER INCONVENIENCES

And underneath that, added by Penny several years earlier:

NO MURDERS BEFORE 10 A. M. WITHOUT AN APPOINTMENT

Derek had objected. Penny had pointed out that they had never actually received a murder before ten.

Derek had said that this was beside the point.

Penny had stated that was precisely the point.

Derek had lost. He was now looking at an email from the Bureau. The subject line was:

CASE UPDATE: DEREK GENT

He opened it.

Dear Mr Gent,

We regret to inform you that you have been murdered.

The incident will occur tomorrow at 11:03 a. m. The perpetrator has been identified. It is you.

Kind regards,

Derek Gent

Relativistic Investigation Bureau.

Derek read it again.

Then a third time. He glanced at the sender's address. It was correct.

Then he stared at the Bureau's computer. It was his computer.

Everything appeared to be where it ought to be, which was, in Derek's experience, rarely a reliable indicator that everything was. The universe had a talent for looking organised just before it wasn't. The filing cabinet was locked. The kettle was where the kettle normally was, which was not necessarily where it ought to be.

The window was shut, the door was shut, and nothing appeared to have been disturbed.

Derek gazed at the email again, with the particular stillness of a man who has just been handed a problem that refuses to sit neatly in any of the usual categories. The Bureau had categories for almost everything. This was not among them. A new category would no doubt be invented once the damage was complete. The Bureau was excellent at classifying disasters after the fact. Prevention had always been considered a secondary specialism, and occasionally an optional one. The official motto, never written down, was: "We'll know what it was once it's over." No one had ever successfully ordered the motto in reverse.

Derek had once tried to summarise the Bureau's actual remit on a single sheet of paper. The sheet had required three revisions, two appendices, and a footnote that began "In cases where the crime has not yet occurred..." He had eventually settled on a working definition: the Relativistic Investigation Bureau noticed things that ordinary agencies were structurally incapable of noticing, and then attempted to file them before they noticed back.

As for the purpose of everything, the Bureau's unofficial position was that it was probably 42. No one had successfully derived the corresponding question, and the forms required to request it had been missing since before the Pre-Transition drawer was labelled.

There were, of course, children. The Bureau's official position was that they were a long-term staffing solution and should be discouraged from developing an interest in geometry until after their GCSEs. Social security, like spacetime, required a continuous supply of new observers.

London supplied them without comment. The city had always been better at producing new observers than at agreeing what they were observing.

"That's inconvenient," he commented, as if the matter were primarily a filing error.

Across the empty room, Trevor Boltzman looked up. Trevor was twenty-eight, highly intelligent, technically qualified, and possessed of the unfortunate tendency to think about one problem at a time — a trait that made him excellent at mathematics and intermittently disastrous at life. He had a sister, Tabitha, who was generally considered the more dangerous of the two, a fact Trevor mentioned only when cornered, and sometimes not even then. Derek had met her once at a Bureau Christmas function that had refused to end on time. She had asked one question about simultaneous events and then left before anyone could answer it properly.

It made him considerably less useful when confronted with a sentence such as: "I have been murdered."

Trevor removed his glasses — a gesture that, in the Bureau, usually preceded either insight or complete bafflement, with no reliable way of telling which in advance. Derek had stopped trying to predict it years ago. "What?"

"I've been murdered."

"Recently?"

"Tomorrow."

Trevor stared at him.

"Tomorrow?"

"Yes."

Trevor considered this carefully.

"You're telling me you've been murdered tomorrow?"

"Yes."

Trevor nodded.

"Ah."

Derek waited. Trevor put his glasses back on. Derek continued waiting.

"Why did you say 'ah'?"

"I wasn't sure what else to say. 'Ah' seemed to cover the available options."

"How about 'that's impossible'? It has a certain traditional appeal."

"I thought you'd already know that. You usually do."

Derek peered at him.

"Trevor, I have been working in this field for twenty-three years."

"Yes."

"I know the difference between something being impossible and something being highly inconvenient."

"Which is this?"

Derek held up the screen.

"I'm not sure yet."

Trevor stood.

"Tea?"

"Yes."

"Milk?"

"No."

"Sugar?"

"One."

Trevor nodded and walked toward the kitchen. Derek called after him.

"Two."

Trevor stopped.

"Two?"

"Yes."

Trevor continued. Derek watched him disappear. A moment later there was a knock at the door.

Before Derek could answer, the door opened. Penny Gent, Derek's cousin, stepped in without waiting for an invitation. She had never been much given to waiting for invitations, a trait that had served her well and other people less so. She also had the disconcerting ability to look as if she had somewhere more interesting to be, even when she was standing in a room full of stopped clocks. Derek had long since stopped minding. Resistance had proved inefficient. He had tried it once and found the paperwork excessive even by Bureau standards, which was saying something.

She was also one of those people who always appeared to have been designed specifically to make other people lose their train of thought. She was a striking brunette, though Derek had never used the word in her presence because he valued his life and had no particular desire to become the subject of a police report — or worse, a footnote. She wore a dark coat, carried a handbag, and had the calm expression of someone who had already dealt with several irritating people before breakfast and was prepared, if necessary, to deal with several more.

She eyed Derek.

"Morning."

"Morning."

She put her handbag down.

"Why do you look like that?"

"Like what?"

"Like you've just discovered the Inland Revenue has found you."

"I've been murdered."

Penny paused.

"Already?"

"Apparently."

"Who did it?"

"I did."

She nodded.

"Fair enough."

Derek watched her.

"You're taking this remarkably well."

"It's Tuesday."

"That doesn't explain anything."

"It explains most things."

She walked over to the computer.

"Show me."

Derek turned the monitor toward her. Penny read the email. Her expression changed.

Only slightly. Penny was clever. Very clever.

Unlike Trevor, who approached every new piece of information as if it might require a spreadsheet, Penny seemed to absorb things almost before anyone had spoken. She read the mysterious message again.

"Tomorrow at eleven-oh-three."

"Yes."

"Not approximately eleven."

"No."

"Eleven-oh-three."

"Yes."

"That's very specific."

"I thought so."

"And the murderer is you."

"Apparently."

Penny regarded Derek.

"Are you planning to murder yourself?"

"Not consciously."

"Good."

She pulled out a chair.

"I'd hate to think you'd become predictable."

He smiled lightly, somewhat against his own will. Trevor came back carrying three mugs.

"Milk?"

"None," Derek noted.

"How about 'that's impossible'?"

Trevor handed Penny her mug.

"Thanks."

Then he handed Derek the other one. Derek took a sip before looking sharply at Trevor.

"Where's yours?"

"I didn't make one."

"Why not?"

"I thought there were only two of us."

Penny met his gaze. Trevor regarded Penny. Then at Derek.

"Oh."

Derek sighed softly.

"Trevor."

"Yes?"

"That's Penny."

"I know."

"My cousin."

"I know."

"She's been here for approximately ninety seconds."

"Yes."

"And yet somehow you've already forgotten that she exists," countered Derek.

"I haven't forgotten she exists."

"You made two cups of tea."

"I thought you didn't want one."

"I have one."

Trevor stared silently at Derek's mug.

"Right."

Penny smiled, clearly amused. "Don't worry. He's usually this bright."

"I'm sitting right here."

"We know."

Derek turned back to the computer. Something was still bothering him, and it was not the murder itself. Murders, at least, had the virtue of being conclusive. This felt more like an administrative error with fatal consequences — the sort of thing the Bureau specialised in, usually without the fatal part. The fatal part was considered an over-performance. The Bureau preferred its disasters to remain administrative — they were easier to file, and considerably harder to explain at the inquest. The Bureau disliked inquests almost as much as it disliked surprises. Both had a tendency to generate unanswered questions, and unanswered questions had a tendency to generate more forms. The Bureau considered this a feature. Features, in the Bureau's vocabulary, were problems that had learned to fill in their own forms. Some of them had started applying for promotion. One had reportedly made it onto a shortlist before anyone noticed it did not exist. The panel had scored it highly for clarity of objectives.

The murder was, in some respects, straightforward. The date was the problem. He clicked the attachment.

A photograph appeared. For a moment Derek stopped breathing. The room felt briefly smaller, as if the air itself had taken an interest. He recognised the feeling: it was the same one that arrived just before a theory collapsed, or a relationship, or both. Somewhere beneath the professional irritation, something colder and more personal had stirred — the quiet recognition that the face in the photograph was not merely a case, but a deadline. Derek had never liked deadlines that looked back at him. Most deadlines had the decency to remain abstract. The photograph itself felt like a decohered outcome: one possibility selected, the others no longer available for inspection. It had the quiet finality of a decision the universe had made without consulting the minutes. Derek filed the sensation under 'to be examined later' and continued working. He was good at filing things under 'later'. It was one of his more reliable skills, and possibly one of his more dangerous ones. Trevor noticed.

"What?"

Derek turned the monitor toward them. The photograph showed their office. Not a similar office.

Not a room that resembled it. The actual room. The same battered desk.

The same green filing cabinet. The same window overlooking the street. The same framed photograph of Derek's grandfather standing beside a Morris Minor in 1964.

And on the floor, face-down beside the familiar desk, was Derek. There was blood on the back of his shirt. Trevor stared intently.

Penny leaned closer.

"That's you."

"Yes."

"You look dead."

"I should hope so."

"Why?"

"Because it would be a very poor photograph if I didn't."

Trevor zoomed in. "There's something on the desk."

Derek peered at the screen. A small, silent brass railway clock stood beside the computer, looking entirely ordinary and entirely wrong, which was, in Derek's experience, the most dangerous combination. He had the sudden, unwelcome sense that the clock had been waiting for him.

"Fair enough."

Penny examined him. "Yes, you do."

"No."

"You bought it last Thursday."

"I did?"

"You described it as historically significant."

"What is it?"

"An old railway clock."

Derek looked again. The old clock showed 11:03. He inspected the clock on his own desk. 9:21.

He considered his computer: 9:21.

His phone and the wall clock both read 9:21.

He looked back at the photograph. 11:03.

Derek leaned back. "Interesting."

Penny's brow tightened. "What?"

"That photograph wasn't taken tomorrow."

Trevor frowned. "Then when was it taken?"

Derek pointed at the picture. "The clock reads eleven-oh-three."

"So?"

"Tomorrow's murder is at eleven-oh-three."

"Yes."

"And the photograph was apparently taken at the moment of my death."

"Yes."

Trevor waited. Derek glanced at him. Trevor turned back.

"So?"

Derek sighed.

"Trevor, you really must learn to recognise a mystery when it is standing on the table waving its arms."

Trevor stared at the photograph. "Oh."

Penny smiled. "There he is."

Trevor leaned closer. "Hang on."

"What?"

"If that's a photograph of tomorrow —"

"It's not."

"Right."

"If it's a photograph taken tomorrow, then why does the clock show eleven-oh-three?"

"Because that's when Derek dies."

"But the clock here reads nine-twenty-one."

"Yes."

"So, the photograph was taken two hours later."

"Yes."

Trevor wore the pleased expression of a man who had just solved something. "That's what I meant."

"No," Penny replied. "It isn't."

Trevor's expression collapsed with the quiet dignity of a man who has just watched a carefully constructed theory meet reality and lose. It was a look Derek had come to recognise, and almost respect. "Oh."

Derek got up and walked to the window. Outside, gloomy London was doing what London did best: being grey, getting wet, and charging for the privilege.

Also, being expensive and mildly annoying.

A bus stopped, and another stopped behind it; neither seemed to know why. Derek watched the traffic. London was, as usual, being mildly inconvenient.

Penny raised her eyes. "Your impending murder?"

"Other than that."

"Your Bureau?"

"Also other than that."

Trevor thought. "Your cousin?"

Penny gazed at him. Trevor immediately reconsidered his life choices.

But Derek smiled softly. "No."

He pointed toward the street. "Everyone assumes time is the same everywhere."

Trevor frowned. "Isn't it?"

"No, obviously."

"Why not?"

"Because Alfred Weinstein ruined that particular assumption."

Trevor asked hesitantly, "Special Relativity?"

"Yes."

"The Weinstein?"

"There aren't many of them."

Penny sat down. "Would you care to explain?"

Derek picked up a marker and went to the whiteboard.

"This is basic Special Relativity — the part that still manages to surprise people who thought they already understood time. They almost never have, you see. Watch. Time turns out to be less a river than a set of carefully negotiated local agreements between observers who have never actually met... and would probably dislike each other if they did. The agreements are enforced by the speed of light, which accepts no appeals and keeps no office hours. It is the only absolute the Bureau has never successfully filed a complaint against."

Derek paused.

"That last part isn't in the textbooks."

Penny raised an eyebrow.

"You just added it."

"I know," Derek said. "It felt true."

He looked at the board again.

"That's usually how the dangerous ones start. The safe ones come with references."

Trevor wrote the sentence down anyway, under Provisional.

He drew a railway track. Then a mysterious train. "Suppose you're standing on a platform and a train passes you at a very high speed."

"How high?"

"High enough that relativity becomes interesting."

"That's not a unit."

"It's sufficient for the moment."

He drew two lightning bolts. "Lightning strikes the front and rear of the train simultaneously."

Penny nodded silently.

Trevor watched.

"An observer standing on the platform, exactly halfway between the two strikes, sees the flashes arrive simultaneously."

"Right."

"Now imagine someone sitting exactly halfway between the front and rear of the train."

"Okay..."

"But the train is moving."

"Yes. Trains tend to do that."

"So, the passenger is moving toward the flash at the front and away from the flash at the back."

Trevor nodded. "The front flash reaches him first."

"Exactly."

Trevor frowned sharply. "But the lightning still struck at the same time."

Derek smiled. "According to whom?" he asked Trevor.

Penny leaned forward. "Ah."

"Exactly," answered Derek, pointing at her.

"Different observers can disagree about whether two separated events happened at the same time. Neither is wrong. The universe simply refuses to appoint a referee. It's never shown any interest in the job. Applications are not being accepted. They never have been. The position remains permanently vacant. Candidates are discouraged from applying. The discouragement is considered permanent. The universe simply declines to supply a single, universal 'now' — a fact that continues to inconvenience anyone who prefers their reality tidy. Most people prefer it tidy. The universe has never taken this preference into account, and shows no sign of starting. It has, however, been very consistent about this — consistency being one of its few reliable vices. Unreliability would have been more interesting, but less characteristic. The universe preferred to be boring in ways that were fatal."

"Without either observer being wrong."

Trevor studied the photograph. "You're saying there's no universal 'now'."

Derek put the marker down. "Not in Special Relativity. And that matters here more than most people are prepared to notice."

Penny looked again at the vivid photograph. "Because the murder is an event."

"Yes."

"And the timestamp is an observation of that event."

"Yes."

"So, we need to know the reference frame."

Derek nodded. "Now you're thinking like a detective."

Trevor raised a finger. "Wouldn't the photograph itself give us the frame?"

"No."

"Why not?"

"Because a photograph records light."

"So?"

"Light takes time to reach the camera. That's obvious — until you realise how many conclusions quietly depend on pretending it doesn't. The camera, like every other observer, is trapped inside its own light-cone and has never been consulted about the matter. Cameras are surprisingly philosophical once you notice this. Most of them just keep it to themselves."

Trevor stopped. Derek could see the thought arriving. "Ah."

"Exactly."

Penny smiled.

"So, even the photograph isn't necessarily showing us what happened when we think it happened. The camera, like every other observer, is trapped inside its own light-cone."

"Correct." Derek sat down. "And that's where things become interesting."

He enlarged the photograph. There was something in the lower-right corner. A piece of paper.

He zoomed in. It contained three lines.

11:03 — DEATH

11:17 — DEPARTURE

12:04 — ARRIVAL

Below them:

DO NOT TRUST THE CLOCKS.

Penny read it.

"That's ominous."

"Very."

"Could 'departure' mean the train?"

"Perhaps."

"Arrival?"

"Perhaps."

"Death?"

"Almost certainly me."

Trevor shifted uneasily.

Penny turned to him.

"That's a surprisingly weak argument."

"It has served me well."

Derek zoomed in again. The railway silent clock in the photograph wasn't showing exactly 11:03. It showed 11:03:17.

Derek stared intently. He looked at his own clock. 9:27:17.

He checked his phone. 9:27:17. The wall clock.

9:27:17. Penny noticed.

"What?"

Derek pointed.

"Seventeen seconds." He said it quietly, as if the number itself might change if spoken too loudly. In the Bureau, numbers that behaved themselves were considered mildly suspicious.

Trevor looked sharply.

"Same."

Derek nodded.

"That's interesting."

Penny leaned closer.

"Why?"

"Because if this were simply a badly set clock, there's no reason for the seconds to line up."

Trevor thought about that.

"Could someone have set it deliberately?"

"Yes."

Penny fixed him with a look. Trevor immediately reconsidered his life choices.

"Exactly."

Derek opened the Bureau's internal records. The system showed no record of the ancient brass clock. No purchase.

No inventory entry. No photograph. No mention of it.

Derek searched the building's security footage. Nothing. He checked the office camera.

At 9:16, the old desk was empty. At 9:17, the old desk was empty. At 9:18, the desk was empty.

At 9:19, the old desk was empty. At 9:20, the ancient brass clock appeared. Trevor stared.

"How?"

Derek replayed it. The clock simply suddenly appeared. There was no hand.

There was no person and no shadow. One frame showed nothing.

The next showed the clock. Penny watched Derek.

"That's not normal." "No."

"No."

"Could the camera have glitched?"

"Yes."

"Did it?"

"No."

"How do you know?"

"Because the clock casts a shadow."

Penny studied it again, more carefully this time. It did. The shadow moved normally as the light from the window changed.

Derek replayed the footage. The silent clock existed. Then it didn't.

Then it existed again. Trevor's expression deteriorated by degrees.

"Could someone have edited the footage?"

"Yes."

"Could they have edited the photograph?"

"Yes."

"So, how do we know anything is real?"

Derek met his eyes. "Now you're asking the right question."

Trevor didn't look reassured. Derek slowly opened another file.

"Let's start with the clocks."

He took the ancient brass clock from the photograph and compared its mechanism with the Bureau's clocks. They were different. He compared the hands.

Different. He compared the second hand. Identical.

Seventeen seconds. Derek frowned. Penny noticed.

"Another problem?"

"Yes."

"How many are we up to?"

"Seven."

Trevor raised his eyebrows.

"Seven?"

"At least."

Derek turned the clear photograph upside down. There was a small number written on the back. 1047.

He reviewed the case database. Case 1047 did not exist. He searched deleted cases.

Nothing. He searched archived cases. Nothing.

He searched future-dated cases. Nothing. Then the computer made a noise.

A soft chime. A new case appeared.

CASE 1047.

CLIENT: Derek Gent

STATUS: ACTIVE

Derek stared intently. Penny leaned over.

"What does it say?"

Derek scrolled. The case contained only three pieces of information. Client: Derek Gent.

Subject: Murder. Date: Wednesday. Time: 11:03:17.

Derek clicked "Assigned Investigator."

The field was blank.

He clicked "Perpetrator."

Blank.

He clicked "Victim."

Derek Gent.

He clicked "Case Notes."

There was one note. Written in Derek's own handwriting.

DO NOT LET TREVOR NEAR THE STRANGE CLOCK.

Trevor read it.

"I feel slightly insulted."

Derek turned toward him. "Don't."

"I wasn't going to."

"You were."

"I was."

Penny smiled. "This is becoming quite good."

Derek stared at the screen.

"Something is using the Bureau."

"What?"

"Someone—or something—is creating cases inside our system."

"From where?"

"I don't know."

"Can we trace it?"

Derek checked.

"No."

"Why?"

"Because the originating address is..."

He stopped.

"What?"

Derek turned the monitor toward them. The address read:

LOCALHOST.

Trevor frowned.

"That's the computer."

"Yes."

"So, someone is doing it from here."

"Apparently."

Penny surveyed the room. There were three people in it. Derek.

Trevor. Penny. Nobody else.

She folded her arms.

"Well."

Trevor met her gaze.

"What?"

"You've got an interesting problem."

"Which one?"

"One of us is lying."

Derek looked at the three chairs. Then the silent clock. Then the computer.

"Perhaps."

Penny smiled.

"Or one of us hasn't done it yet."

Derek regarded her. The office suddenly felt colder. Not the ordinary London cold — something more precise, as if a draft had found its way in from a place that did not appear on any map. Trevor swallowed.

"You mean..."

Derek didn't answer. The computer chimed again. All three studied the screen.

A new message suddenly appeared.

CASE UPDATE.

PERPETRATOR IDENTIFIED.

Derek opened it. The name appeared.

TREVOR BOLTZMAN.

Trevor stared silently.

"What?"

Derek gave him a measured look. Trevor and Penny exchanged a glance.

"That wasn't me."

"I know," Derek added.

"How do you know?"

"Because you're still here."

Trevor blinked.

"That's your evidence?"

"For the moment."

Penny walked to the window.

"Then who is framing him?"

Derek glanced at the clock. 9:31:17. He stared at the photograph.

11:03:17. He returned to the case file. Then he looked at Trevor.

"Maybe nobody."

Trevor frowned.

"What does that mean?"

Derek walked to the whiteboard. He wrote:

EVENT

Then:

OBSERVER

Then:

FRAME

Then:

TIME

He circled the first three.

"Before we worry about who did it, we need to know what actually happened."

Trevor nodded. "Right."

Penny regarded him carefully. "Do you understand?"

Trevor thought for a while. "I think so."

"Good."

Derek picked up his coat. The decision, once made, felt both inevitable and slightly unreal.

"Then we're going to need to leave."

"Where?"

Derek considered the photograph.

"The clock."

Penny frowned. "The clock?"

"So, whoever put it here expected us to find it."

"What about it?"

"It doesn't belong here."

"So?"

Trevor stood. "All right, then, where are we going?"

Derek held up the photograph. "To the place where it came from."

Penny grabbed her coat. "And where is that?"

Derek turned the photograph over. On the back, beneath the number 1047, another line had suddenly appeared.

LONDON TERMINUS

Platform 7

11:17 - Penny read it.

"That's tomorrow."

Derek nodded. "Yes."

Trevor surveyed the photograph. "And that's after you're dead."

"So it appears."

"So, how are we supposed to investigate it?"

Derek smiled. "We'll have to be there."

Trevor looked concerned. "Tomorrow?"

"Yes."

"At eleven seventeen?"

"Yes."

"After you're murdered?"

"Presumably."

Trevor thought for a moment. "How do we know we'll get there?"

Derek opened the door. "Because someone has already told us that we will."

Penny followed him. Trevor remained where he was.

"Derek?"

"Yes?"

"What if the message is wrong?"

Derek turned back.

"Then we'll have wasted a perfectly good morning."

"And if it's right?"

Derek smiled.

"Then I suppose I'll have to solve my own murder. The words sounded almost reasonable until one considered them."

They stepped into the corridor. Mrs Marsh, the building superintendent, was already there, polishing a non-existent mark on the wall with the air of someone who had been waiting specifically to be inconvenient. She was widely regarded as the most immovable object in the building, and possibly in several adjacent dimensions. She did not look up. "You're tracking geometry in again," she said.

Further down the corridor, Mr Pendleton from Facilities stood with a clipboard and the expression of a man who had already decided the answer was no. He specialised in a different kind of obstruction: the kind that required signatures, countersignatures, and the quiet insistence that whatever you needed was in a different section, a different building, or a different continuum. The heavy door closed behind them with the finality of a decision that had already been made somewhere else. Inside the empty office, the old brass clock appeared on Derek's desk as if it had been waiting for the room to be unoccupied.

Its second hand moved. One. Two.

Three. It reached seventeen. Stopped.

Then continued. But now the clock was showing 11:03. And outside the empty office, somewhere far away, a distant train whistle sounded.

There was no railway empty station anywhere near the building. The whistle sounded again. Then a third time.

The Bureau's computer switched itself on. A final message appeared.

CASE 1047

STATUS: ACTIVE

VICTIM: DEREK GENT

PERPETRATOR: TREVOR BOLTZMAN

WITNESS: PENNY

PHYSICIST: ALFRED WEINSTEIN

REFERENCE FRAME: UNKNOWN

And beneath it, in smaller letters:

DO NOT LET WEINSTEIN GET TO THE EMPTY STATION FIRST.

The message disappeared. The screen went black. Then the computer displayed the date.

Wednesday. 11:03. 17 seconds.

The quiet office was empty. For the moment.

Chapter 2

The Lorentz Transform

By Wednesday morning, Derek Gent had decided that there were two kinds of impossible. The first kind was the ordinary sort. A man could not be in two places at once.

A clear photograph could not normally exist before it was taken. A clock could not run backwards without either being broken or being subjected to a level of engineering that Derek considered unnecessarily ambitious. The second kind was considerably worse.

The second kind was something that was perfectly consistent with the laws of physics but completely inconsistent with common sense. Derek had spent most of his professional life dealing with the second kind. He found it much more expensive.

At 8:36 on Wednesday morning, he was standing in front of the whiteboard in the Bureau while Trevor sat at the old desk and Penny leaned against the window. The brass railway clock sat between them. It had arrived sometime during the night.

Nobody had seen it arrive. Nobody knew where it had come from. And it was still showing 11:03:17.

Derek had written three words on the board.

EVENT

FRAME

TIME

He underlined the first.

"The first thing we need to establish," he explained, "is what actually happened."

Trevor studied the photograph.

"You mean the murder?"

"I mean the event."

"Isn't that the same thing?"

"No."

Trevor considered this.

"Not necessarily. You may be putting the cart before the flaming horse."

"Precisely."

Penny regarded the silent clock.

"We have a photograph of you lying dead."

"Yes."

"We have a timestamp."

"Apparently."

"We have a second timestamp in the Bureau."

"Yes."

"And the two don't agree."

"Not yet."

Trevor frowned.

"Not yet?"

Derek turned.

"That's an important distinction."

"Why?"

"Because we don't know whether they are supposed to agree."

Trevor examined the clock.

"They're both clocks."

"Yes."

"So, they measure time."

"Yes."

"And time is time."

Penny smiled. Derek inspected Trevor. Trevor immediately raised both hands.

"I know."

"You've just contradicted Special Relativity."

"I know."

"Good."

Trevor nodded.

"You were not." Trevor sighed. "I was merely testing the waters." Derek found his eyes. "You were throwing the fish tank into the river."

"You were not."

"No."

Derek returned to the board.

"Let's start from the beginning."

He drew two points.

"Two events."

He labelled one A and the other B.

"These happen somewhere in spacetime. We can describe each event by four coordinates."

He wrote: x, y, z, t Trevor nodded.

"Position and time."

"Exactly."

"And different observers can assign different coordinates."

"Yes."

"Because they're moving relative to one another."

"Yes."

Penny considered the photograph.

"So, the question isn't simply when Derek died."

"No."

"It's when Derek died according to which observer."

"I think so." Penny smiled. "Good. You've hissed all my mystery lectures, and now one has finally landed."

Trevor glanced at Derek.

"But the death itself either happened or it didn't."

"Of course."

"That part doesn't depend on the observer."

"Correct."

Trevor nodded.

"Right."

Derek smiled.

"You've just rescued metaphysics from physics."

"I do my best."

Penny picked up the photograph.

"And the photograph?"

"That's more complicated."

"Because light has to travel from the event to the camera."

"Yes."

"So, the photograph doesn't show the event itself."

"It shows light arriving from the event."

"Which means the image contains information about an earlier event."

"Exactly."

Trevor fell into thought.

"So, when we say the photograph was taken at 11:03, we're already making an assumption about the relationship between the clock and the event."

Derek stared at him.

"Yes."

Trevor smiled.

"That one was easy."

"Don't get used to it."

Penny laughed. Derek walked to the old desk and picked up the ancient brass clock. He turned it over.

There was a small metal plate on the back. No manufacturer's name. No serial number.

Just three letters.

LGV.

Trevor gazed at it.

"Any idea what that means?"

Derek shook his head.

"Not yet."

Penny took the silent clock.

"May I?"

Derek handed it to her. She examined it carefully.

"This isn't a normal railway clock."

"I know."

"The mechanism is unusually precise."

"How precise?"

"More precise than it has any right to be."

Trevor leaned in.

"How do you know?"

Penny smiled.

"Because I opened it."

Derek consulted the clock.

"When?"

"Last night."

"You opened it?"

"Yes."

"Without telling me?"

"You were asleep."

"I was working."

"You were asleep at your desk."

"That was a temporary research posture."

Penny ignored him.

"There are no batteries."

Derek nodded.

"Mechanical?"

"Not exactly."

She handed the silent clock back.

"It has a power source."

"What sort?"

"I don't know."

Trevor took it. He examined the back.

"That's odd."

"What?"

"There isn't enough room."

"For what?"

"For a power source."

Derek took the clock from him. Trevor was right. The casing was too thin.

There was nowhere to put anything substantial. Derek turned it over again. The second hand moved.

Tick. Tick.

Seventeen seconds. Tick.

Trevor watched.

"Why seventeen?"

Derek turned his attention to him.

"I don't know."

Penny watched the photograph.

"It's on the photograph too."

"Yes."

"And the Bureau clock."

"Yes."

"And yesterday's footage."

Derek nodded.

"Yes."

Trevor thought.

"So, seventeen is part of the message."

"Probably."

"Could it be a date?"

Derek shook his head.

"No."

"Reference to a page?"

"Possibly."

"An equation?"

"Possibly."

Penny declared, "Or a velocity."

Derek surveyed her.

"Explain."

"Seventeen kilometres per second. Seventeen percent of something. Seventeen metres per second. There's no shortage of seventeen in physics."

"True."

Trevor studied the clock again.

"Could be 17 seconds of proper time."

Derek stopped. Penny regarded him.

Trevor realized he had announced something interesting.

"That would make the difference meaningful."

"Yes."

Derek took the photograph.

"If seventeen seconds is a proper-time interval..."

"Then we'd need to know which worldline we're talking about," Penny asserted.

Trevor nodded.

"And the gravitational potential."

"Exactly."

Derek examined both of them.

"Which means we're already getting ahead of ourselves."

Penny smiled.

"That's what we do."

Derek put the photograph down.

"Not yet."

He turned to the whiteboard.

"We need the simplest possible explanation first."

Trevor nodded.

"Special Relativity."

"Yes."

Derek wrote: $x' = \gamma(x - vt)$ $t' = \gamma(t - vx/c^2)$

Trevor inspected the equations.

"The Lorentz transformation."

"Yes."

Penny nodded.

"Weinstein."

"Yes."

Trevor met Derek's gaze.

"Do you ever find it strange saying that?"

"Why?"

"That you're using Weinstein's equations to investigate a case involving Weinstein's own theory."

Derek thought about it.

"Not particularly."

"Why not?"

"Because if you want to investigate a crime involving gravity, you use General Relativity."

"Fair."

"If you want to investigate a crime involving relative motion and time, you use Special Relativity."

"And if it involves both?"

"Then you start charging by the hour."

Penny laughed. Derek pointed to the first equation.

"This is the spatial transformation."

Then the second.

"This is the temporal transformation."

Trevor nodded.

"And the important bit is that t' depends on x ."

"Yes."

"Meaning that time in one frame depends partly on where the event occurs in the other frame."

"Correct."

Trevor studied the photograph.

"So, two events that are simultaneous in one frame need not be simultaneous in another."

"Quite."

Penny pointed out, "Which is relativity of simultaneity."

"Yes."

Derek glanced at the photograph again.

"And that is the first thing I want to test."

He opened the Bureau's sophisticated analysis software.

"Suppose the photograph's timestamp is genuine."

Trevor leaned forward.

"All right."

"Suppose the photograph corresponds to an actual event tomorrow at 11:03:17."

"Okay."

"And suppose the camera and the Bureau are moving relative to one another."

Penny nodded.

"Then the time coordinate assigned to that event could be different in the two frames."

"Yes."

Trevor clarified, "But unless they're moving at a substantial fraction of the speed of light, the difference should be tiny."

Derek smiled.

"Exactly."

"How tiny?"

"Very."

"How tiny?"

"Small enough that London Underground timetables are generally a greater source of temporal uncertainty."

Penny smiled.

"So, ordinary motion can't explain a difference of nearly two hours."

"No."

Derek entered the relevant numbers. The computer calculated. He studied the result.

Then he entered them again. Trevor watched.

"What did you get?"

"Nothing."

"Nothing?"

"Exactly."

Derek deleted the calculation.

Penny came over.

"What did you expect?"

"A result."

"Meaning?"

"A physically meaningful relative velocity."

"And you didn't get one?"

"I got one."

"Then what's wrong?"

Derek pointed at the number.

"It's greater than c ."

Trevor stared.

"How much greater?"

"Enough."

Trevor leaned closer.

"That's not a velocity."

"Correct."

Penny stared at Derek.

"So, Special Relativity is telling us the premise is wrong."

"Or the frame is wrong."

"Or the event."

"Or the clocks."

"Or the photograph."

"Yes."

Trevor slowly gazed at the equation.

"Could the photograph be showing a different event?"

Derek turned.

"Explain."

"The photograph may be real, but we assume it shows Derek's death."

"Yes."

"What if it shows Derek at a different event that happens to look like his death?"

Derek considered this. Penny smiled.

"That would be rather unfortunate for Derek."

"Indeed."

Trevor continued.

"Maybe the blood isn't blood."

Derek peered at him.

"Maybe."

"Maybe the body isn't Derek."

"Possible."

"Maybe he's unconscious."

"Possible."

"Maybe the photograph has been staged."

"Possible."

Trevor brightened.

"So, we don't actually know that the photograph is evidence of murder."

Derek nodded.

"Good."

Trevor sat back.

"Then why did you say you'd been murdered?"

Derek eyed him.

"Because the Bureau told me."

Trevor stared intently.

"Right."

Penny mentioned, "We're going to need to question the Bureau."

Derek nodded.

"I was afraid of that." Trevor, for his part, had gone quiet in the specific way that meant he was already three steps ahead and reluctant to share the destination. Some problems, once noticed, had a way of following you home. A few, if noticed early enough, never quite managed to happen at all. He was beginning to recognise the symptoms. The mug on the windowsill was empty again. No one had drunk from it. Others simply rearranged the furniture and waited for you to notice they had always been there. He had never been greedy about problems. Problems, in his experience, multiplied without encouragement. The Bureau was usually good at inventing new problems once the old ones became inconvenient. Derek suspected the universe operated on similar principles. The evidence was, by now, fairly strong. He was beginning to think the universe and the Bureau had more in common than was strictly healthy — both specialised in rules that only became visible after you had already broken them. The symmetry was not comforting. Mrs Marsh, Derek suspected, operated on similar principles, only with better stationery. Mr Pendleton operated on the complementary principle: nothing was his responsibility until it went wrong, at which point it had always been yours.

Derek had long since stopped trying to file London by category. The categories kept moving.

Outside the Bureau, a red Mini Cooper was parked at an optimistic angle. It contained more people than its interior volume should have permitted. Derek counted seven heads and one hand that did not appear to belong to any of them. The seating arrangement looked as if it were violating at least two conservation laws and a local parking by-law. Mrs Marsh walked past without breaking stride. "If that is a singularity," she said, "it is not registered to this address."

Since the paperwork had changed, certain older forms still referred to arrangements that no longer applied. These were kept in a drawer labelled Pre-Transition / Do Not Open. Derek had opened it

once. The forms inside had the air of documents that knew they were out of date and intended to remain that way. The universe, he had noticed, had become slightly harder to interpret since the annexes stopped matching. He was becoming quite good at scheduling problems for other days. It was one of the few forms of time travel the Bureau officially permitted. The unofficial ones were more interesting and strictly forbidden.

The telephone rang. All three watched it.

Derek answered.

"Derek Gent."

A pause. Then a man's voice.

"Good morning."

Derek knew the voice.

"Hermann."

"Yes."

"Where are you?"

"Munich."

Derek surveyed Penny.

"Of course."

Trevor mouthed, "Who?"

Derek waved him quiet.

"What do you want?"

"I understand you have been using the Lorentz transformation."

Derek glanced at the whiteboard.

"Yes."

"And?"

"It doesn't work."

There was silence.

Then Minkowski responded:

"It works perfectly."

"Not on this problem."

"Then you're using it incorrectly."

Derek frowned.

"Are you telling me I don't understand Special Relativity?"

"I'm telling you that you are trying to make a flat-spacetime transformation solve a problem that may not be flat."

Derek met Penny's gaze. She immediately understood. Trevor did too.

"Gravity," Derek opined.

"Perhaps."

"We don't have evidence of gravity."

"You have clocks."

"We always have clocks."

"Not clocks behaving like yours."

Derek regarded the brass railway clock.

"Where are you?"

"Munich."

"Why?"

"Because Alfred is here."

Derek paused.

"Alfred Weinstein?"

"Yes."

"Doing what?"

"Being bored."

"Where?"

"In a patent attorney's office in Schwabing."

Derek examined Penny. Penny raised an eyebrow.

"That's very Weinstein."

Minkowski continued.

"He has been thinking about General Relativity."

"At the moment?"

"Yes."

Derek inspected the silent clock.

"How convenient."

"It is not convenient."

"Why?"

"Because Alfred has just discovered that the clocks in his office are disagreeing."

Derek's expression changed.

"By how much?"

Minkowski gave him the figure. Derek went very still. Penny noticed.

"Same?"

Derek nodded.

"Same."

Trevor stood.

"Same what?"

Derek covered the receiver.

"Time difference."

Trevor frowned.

"Between Munich and London?"

"No."

"Then what?"

Derek considered the clock.

"The same anomaly."

Penny took the phone.

"Minkowski?"

"Yes."

"How accurate are Weinstein's clocks?"

"Extremely."

"Could they be synchronized?"

"They were."

"Could the difference be gravitational?"

"Not at this magnitude."

Penny glanced at Derek.

"What about acceleration?"

Minkowski answered.

"That is precisely the question."

Derek took the phone back.

"Are you saying the clocks in Munich and London are showing the same relativistic anomaly?"

"I'm saying that if you transform the measurements into a common frame, they are related."

Derek thought.

"How?"

"I don't know yet."

"Then why call?"

"Because Alfred wants to speak to you."

Derek wearily sighed.

"Put him on."

There was a click. Then a different voice. Older.

Calmer. Slightly irritated.

"Derek Gent?"

"Yes."

"This is Alfred Weinstein."

"I know."

"Do we know each other?"

"Not personally."

"Hmm."

A pause.

"You're the detective."

"Yes."

"The relativistic detective."

"Yes."

"I've been told you have a clock."

Derek stared at it.

"I do."

"Where did you get it?"

"I don't know."

"That's unhelpful."

"I agree."

"Does it have a brass case?"

"Yes."

"Old railway style?"

"Yes."

"Does it stop at seventeen seconds?"

"Yes."

There was a pause.

"Then you have a problem."

Derek smiled.

"We had already established that."

"No. You have a very specific problem."

"What is it?"

Weinstein took a breath.

"I've seen that clock before."

Derek's expression changed.

"Where?"

"In 2026."

Derek slowly gazed at Penny. Penny met his gaze. Trevor stared.

Derek maintained:

"We're in 2026."

"I know."

"Then what do you mean?"

"I mean I saw it earlier."

"When?"

Weinstein paused.

"Tomorrow."

Nobody spoke. Derek slowly sat down. Penny took the phone.

"Alfred."

"Yes?"

"Are you saying you have already seen the clock?"

"Yes."

"Where?"

"In London."

Penny met Derek's gaze, one eyebrow already halfway toward the ceiling.

"Here?"

"I believe so."

Derek took the phone back.

"Alfred, are you sure?"

"I am very old, Mr Gent, but I am not senile."

"I didn't mean —"

"I know what you meant."

A pause.

Then Weinstein affirmed:

"The interesting thing is that I remember seeing it."

"That's reassuring."

"No."

"Why not?"

"Because the Bureau told me."

Derek closed his eyes and, for a moment, let the phone hum quietly against his ear.

Trevor whispered:

"That doesn't help."

Derek covered the receiver.

"No."

He uncovered it.

"Alfred, what exactly happened?"

"I was standing in the Bureau."

"When?"

"Tomorrow."

"What were you doing?"

"Looking at the clock."

"Anything else?"

"Yes."

"What?"

"You were dead."

Derek watched Penny. Penny took a slow breath. Trevor looked as though he would quite like to leave the country.

Derek confirmed:

"Did you see who killed me?"

"No."

"Why not?"

"Because you were standing up."

Derek blinked.

"What?"

Weinstein continued.

"You were dead."

"Yes."

"But you were standing."

Derek surveyed Penny. Penny stared back.

Trevor insisted:

"That's not possible."

Derek covered the phone.

"Thank you, Trevor."

"Sorry."

Derek uncovered it.

"Alfred, are you sure?"

"Quite."

"Could it have been a relativistic effect?"

"Possibly."

"Could the photograph have been from another frame?"

"Perhaps."

"Could the event be displaced in time?"

"Possibly."

"Could I have been dead in one frame and alive in another?"

There was a long silence.

Then Weinstein emphasized:

"No."

Derek waited.

"Why not?"

"Because that's nonsense."

Derek nodded.

"Good."

"However..."

Derek closed his eyes.

"Being bored." Derek paused. "That sounds like Alfred." Minkowski agreed. "He's a well-boiled icicle when he gets interested."

"You could be dead according to one description of an event and alive according to another if we're actually talking about different events."

"That's what I thought."

"Which means the photograph may be telling you the truth."

"Even if I'm not dead."

"Yes."

Derek considered that.

"That's rather annoying."

"I agree."

Minkowski's voice came faintly from the background.

"Ask him about the train."

Derek heard him.

"What train?"

Weinstein queried:

"Ah."

"What?"

"There is a train."

"What train?"

"The one at the station."

"What station?"

"London Terminus."

Derek froze.

"Platform seven?"

Weinstein was silent.

"Alfred?"

"Yes."

"Platform seven?"

"Yes."

"At 11:17?"

"Yes."

Derek studied Penny. Penny's expression had gone completely serious. Trevor was standing beside the whiteboard.

Derek said:

"We have a photograph."

"I know."

"It reads 11:17 departure."

"I know."

"How do you know?"

"Because I was there."

"When?"

"Tomorrow."

Derek sat back.

"All right."

Weinstein remained silent. Derek asked:

"Who else was there?"

Weinstein hesitated.

"You."

"Alive?"

"Yes."

"Dead?"

"Also yes."

Derek stared at the phone. Trevor stared at Derek. Penny regarded the ceiling.

Nobody spoke.

Finally Derek stated:

"Alfred."

"Yes?"

"I think we have a problem."

Weinstein sighed.

"I was hoping you'd say that."

The line went dead. Derek lowered the phone. Trevor spoke first.

"How can you be alive and dead at the same time?"

Derek examined him.

"I don't know."

Penny commented:

"That's not strictly what Weinstein noted."

Trevor inspected her.

"What did he say?"

"He Derek observed was alive and dead."

"That's the same thing."

"No."

"How?"

Penny walked to the whiteboard. She erased the word TIME. Then wrote:

EVENTS.

"He's seeing two descriptions."

Trevor nodded.

"Two frames?"

"Possibly."

Derek returned to the equations.

"Not if the transformation requires a velocity greater than light."

Trevor thought.

"Then perhaps the events are not connected the way we assume."

"Exactly."

Penny turned.

"So, we're back to the beginning."

"No."

Derek picked up the photograph.

"We're further along."

"Then you have a problem."

"We know the train exists."

"We knew that from the photograph."

"We now know Weinstein saw it."

"And?"

"He saw it tomorrow."

Trevor frowned.

"That doesn't narrow it down."

Derek smiled.

"It does."

"How?"

"Because Alfred Weinstein doesn't make mistakes with relativity."

Trevor nodded.

"True."

"He might misunderstand an unfamiliar physical phenomenon."

"Yes."

"But he won't misunderstand Special Relativity."

"Agreed."

Derek pointed at the silent clock.

"So if Weinstein reads the clock is showing us two different events rather than one event seen from two frames..."

Penny finished:

"Then we should believe him."

Derek nodded.

"Yes."

Trevor glanced at the whiteboard.

"And then?"

"Then we find the second event."

"How?"

Derek stared at the ancient brass clock.

"By following the clock."

Penny smiled.

"That sounds simple."

"It won't be."

"Why?"

"Because the clock isn't telling us where to go."

Derek turned it over. The letters LGV were still stamped into the back. Trevor looked over his shoulder.

"What does LGV mean?"

Derek stared at the initials.

"I don't know."

Penny replied:

"Could be someone's initials."

"Could be."

"A model number."

"Possibly."

"A railway designation."

"Maybe."

Trevor gazed at the silent clock face.

"Could it mean Local Gravitational Vector?"

Derek peered at him. Penny eyed him. Trevor shrugged.

"What?"

Derek smiled.

"That's actually rather good."

A pleased smile appeared on Trevor's face.

"Thank you."

Derek took out his phone. He searched the Bureau's records. Nothing.

He searched railway archives. Nothing. He searched scientific databases.

Nothing. Then the Bureau's computer chimed. A new case file had appeared.

Derek opened it.

CASE 1047

He stared intently. The file was different now.

CLIENT: DEREK GENT

VICTIM: DEREK GENT
PERPETRATOR: TREVOR BOLTZMAN
WITNESS: PENNY
PHYSICIST: ALFRED WEINSTEIN
LOCATION: LONDON TERMINUS
PLATFORM: 7
DEPARTURE: 11:17
REFERENCE FRAME: INERTIAL

Derek stared at the last line. Trevor leaned over.

"Inertial."

Penny nodded.

"That's important."

"Why?"

"Because the Bureau is telling us which frame to use."

Derek nodded.

"And it has chosen an inertial frame."

Trevor watched the photograph.

"Which means Special Relativity."

"Just so."

Penny smiled.

"So, perhaps General Relativity isn't involved after all."

Derek surveyed her.

"Perhaps."

The computer chimed again. A final line appeared.

GRAVITY: REQUIRED.

Derek stared. Penny stared. Trevor stared.

Then Trevor answered:

"That's rather rude."

Derek laughed. For the first time since receiving the murder notice, he felt genuinely cheerful. Someone—or something—was giving them instructions.

The instructions were cryptic. The physics was inconsistent. The future appeared to have leaked into the present.

And Alfred Weinstein was apparently going to witness Derek's death tomorrow while Derek remained alive today. In other words, it was exactly the sort of case for which the Bureau existed. Derek put on his coat.

"Platform seven."

Penny picked up her handbag.

"At eleven seventeen."

Trevor grabbed the photograph.

"Tomorrow."

Derek nodded.

"Tomorrow."

Trevor studied the ancient brass clock.

"What about this?"

Derek picked it up.

"We're taking it."

Penny regarded him.

"Why?"

"Because it clearly wants to go."

Trevor frowned.

"Clocks don't want things."

Derek opened the old door.

"Normally, no."

He examined the clock.

"This one appears to be an exception."

They stepped into the narrow corridor. Derek locked the familiar office. Then stopped.

He looked back through the glass. For just a moment, he thought he saw somebody sitting at his desk. A man.

Old. Grey-haired. Wearing a dark suit.

Derek blinked. The chair was empty. He unlocked the door again.

Trevor inspected him.

"What?"

"Nothing."

"The one at the station." Derek frowned. "What station?" Weinstein answered, "London Terminus." Derek froze. "Platform seven?" Weinstein confirmed. "At 11:17?" "Yes." Derek exhaled. "Right. The ship has left the barn."

The brass clock was in his hand. He considered the chair again. Nothing.

Penny mysteriously appeared in the doorway.

"Coming?"

"Yes."

Derek switched off the light. As the familiar door closed, the computer screen came on by itself. One sentence appeared.

THE LORENTZ TRANSFORMATION IS CORRECT.

Then another.

YOUR INTERPRETATION IS NOT.

And beneath it:

NEXT: FIND THE STRANGE TRAIN.

The screen went dark. In the narrow corridor, Derek stopped. Penny glanced at him.

"What now?"

Derek smiled.

"Now we catch a train."

Trevor frowned.

"We're not going anywhere."

"Not yet."

Derek studied the ancient mysterious brass clock.

"That's the point."

And somewhere, very far away—or possibly very close—the sound of a distant train began to move through the walls.

Chapter 3

The Weight of Time

By Thursday morning, Alfred Weinstein was sitting in a patent attorney's office in Schwabing. He was bored. This was not unusual.

He had once pointed out, forcefully and at some length, that a certain other physicist had also spent his most productive years examining patents in an office very much like this one, and that history had found this charming rather than suspicious. Nobody at the Bureau had found a tactful way to change the subject fast enough.

The patent attorney was explaining a device intended to improve the efficiency of a mechanical regulator. Weinstein had been listening for approximately six minutes. He had understood the first two.

The third had been optimistic. The fourth had involved a spring. The fifth had involved a spring behaving in a way Weinstein considered personally insulting.

By the sixth minute, his mind had left the room. He looked out of the cold window. Munich was busy.

A tram rattled past. A cyclist narrowly avoided a delivery cart.

Someone shouted at someone else.

A dog had apparently decided that the entire afternoon belonged to it. Weinstein watched the movement for a while. Then he quietly stared at the silent clock.

He had been in the quiet office for forty-three minutes. It felt like three hours. He smiled.

That was an interesting thought. The feeling of time passing was obviously psychological. Physical time was another matter.

He gazed at the clock again. Then at his watch. They disagreed.

Only slightly. Weinstein frowned. He checked them again.

The difference was still there. He looked around the office. One mysterious clock was on the wall.

His watch was on his wrist. Nothing unusual about that. He returned to the patent.

The attorney was still speaking. Weinstein lasted another twelve seconds. Then his mind wandered back to the clocks.

Suppose two identical clocks were placed at different heights. Would they remain synchronized? He knew the answer.

No.

General Relativity predicted that a strange clock deeper in a gravitational field would run more slowly than one higher up. The difference under ordinary circumstances was extremely small. But it was real.

And, more importantly, it was measurable. Weinstein picked up his pencil. On the margin of the patent application he wrote:

GRAVITY CHANGES THE RATE OF TIME.

He reread the sentence.

Then added:

THE DIFFERENCE SHOULD BE MEASURABLE.

He stared at it. The patent attorney stopped speaking.

"Herr Weinstein?"

"Yes?"

"Are you following me?"

"Absolutely."

"What did I just say?"

Weinstein eyed the patent. Then at the attorney. Then at his pencil.

"You were explaining the regulator."

"Yes."

"And the spring."

"Yes."

"The spring is wrong."

The attorney stared at him.

"I beg your pardon?"

Weinstein watched the diagram.

"It won't behave as you've described."

"Why not?"

Weinstein surveyed the drawing for a moment.

"Because you've assumed the temperature remains constant."

The attorney blinked.

"Doesn't it?"

"No."

"Oh."

Weinstein turned the paper around and pointed to a line.

"This coefficient changes with temperature."

The attorney leaned over the desk.

"Good heavens."

Weinstein shrugged.

"Well, there we are."

He returned to his mysterious clock. The patent attorney studied the amended diagram. Then at Weinstein.

"You've been listening after all."

"Occasionally."

The attorney smiled.

"I suppose I should be grateful."

"You should."

Weinstein regarded his watch again. The difference between it and the wall clock had increased. He frowned.

That was no longer ordinary. He stood.

"Do you have another clock?"

The attorney looked puzzled.

"Yes."

"Why?"

"I'd like to compare them."

The attorney pointed towards the secretary's desk.

"There is one there."

Weinstein walked over. He compared it with his watch. Then with the wall clock.

Then he examined the positions of all three. His expression changed. The patent attorney noticed.

"Something wrong?"

"Yes."

"What?"

"The clocks are disagreeing."

"They do that."

"Not like this."

The attorney inspected him. Weinstein took out his notebook. He calculated.

Then calculated again. The difference was far too large. He looked at the ceiling.

Then at the floor. The secretary came in.

"Herr Weinstein?"

"Yes?"

"Is everything all right?"

"No."

"What happened?"

Weinstein held up the notebook.

"The clocks are behaving badly."

The secretary glanced at the clocks.

"They seem fine."

"That's what makes it interesting."

He put the notebook away.

"How high is this building?"

"Four floors."

"Exactly?"

"I believe so."

Weinstein nodded. He looked up.

"Which floor are we on?"

"Second."

He thought. Then stared at his watch again. The difference had increased.

He frowned.

"That's impossible."

The secretary smiled.

"I thought you didn't believe in impossible things."

"I don't."

"Then what is it?"

Weinstein considered the question.

"A very large error."

The secretary nodded.

"Ah."

She returned to her desk. Weinstein picked up his coat. The patent attorney looked alarmed.

"Are you leaving?"

"Yes."

"What about the rest of the patent?"

"It can wait."

"The spring?"

"It will continue being wrong while I'm gone."

He left the quiet office. Hermann Minkowski was standing in the empty corridor. Weinstein stopped.

"You knew."

Minkowski gazed at him.

"Knew what?"

"The clocks."

"What about them?"

"They're disagreeing."

Minkowski nodded.

"Yes."

"By a ridiculous amount."

"Yes."

"And you knew."

"Yes."

Weinstein stared at him.

"Well?"

"Well what?"

"What is happening?"

Minkowski smiled.

"That is what Derek Gent is trying to discover."

Weinstein frowned.

"Who?"

"Derek Gent."

"Never heard of him."

"You will."

Minkowski handed him a vivid photograph. Weinstein studied it. The picture showed an office.

A man was lying face-down on the floor. Weinstein stared.

"Who is that?"

"Derek Gent."

"Dead?"

"Apparently."

Weinstein peered at the strange vivid photograph again.

"When?"

"Tomorrow."

Weinstein turned to Minkowski.

"That's inconvenient."

"I thought you'd say that."

Weinstein studied the photograph. Then he noticed the clock. 11:03:17.

His expression changed.

"Where did you get this?"

"From the Bureau."

"What Bureau?"

"Derek Gent's Relativistic Investigation Bureau."

Weinstein watched the photograph. Then at Minkowski.

"And you expect me to believe this?"

"No."

"Good."

"I expect you to analyze it."

Weinstein nodded.

"That is more reasonable."

"What frame?"

Minkowski smiled.

"Exactly."

Weinstein turned to him.

"Don't be pleased with yourself."

"I wasn't."

"You were."

"Perhaps."

Weinstein returned to the strange vivid photograph.

"The timestamp alone tells us almost nothing."

"Correct."

"We need the event coordinates."

"Correct."

"And the observer's frame."

"Correct."

"And the light-travel time between the event and the camera."

"Correct."

Weinstein nodded.

"Well, that's straightforward."

Minkowski smiled.

"Is it?"

"No."

"Fine."

Weinstein looked again. There was something else. An old brass clock.

The same old clock he'd seen in his office. He pointed at it.

"That."

"Yes."

"I've seen that clock."

"Where?"

"In London."

Minkowski was silent. Weinstein regarded him.

"Yesterday."

Minkowski nodded.

"Yes."

Weinstein stared intently.

"Why didn't you tell me?"

"I wanted you to notice."

Weinstein sighed.

"You have an unfortunate tendency to turn information into puzzles."

"It keeps the mind active."

"It also wastes time."

"Not according to relativity."

Weinstein examined him.

"That was terrible."

"I know."

Weinstein handed the photograph back.

"Let's go to London."

Minkowski raised an eyebrow.

"Now?"

"Yes."

"You're not curious about the patent?"

Weinstein looked toward the office.

"No."

"The spring?"

"No."

"The regulator?"

"No."

Minkowski nodded.

"Good."

They walked toward the stairs.

After a moment, Weinstein added:

"There's something else."

"What?"

"The difference between the clocks."

"Yes?"

"It's much too large to be explained by the gravitational field of the Earth at that height."

Minkowski nodded.

"Exactly."

"So, either the clocks are wrong..."

"Yes."

"... or we're not dealing with the Earth's ordinary gravitational field."

"Yes."

Weinstein stopped.

"Then what are we dealing with?"

Minkowski inspected him.

"That is the interesting question."

Weinstein nodded.

"Then it is decided."

Minkowski stared at him.

"What is?"

"That we don't know."

Minkowski smiled.

"Quite."

They continued down the stairs. In London, Derek Gent was standing in the Bureau, staring at the same photograph. Penny stood beside him.

Trevor was at the computer. The three of them had spent the morning testing the Lorentz transformation against the mysterious timestamps. The result had been unsatisfactory.

The mathematics worked. The interpretation didn't. Derek considered the latest calculation.

"We've got a problem."

Trevor looked over.

"What sort?"

"The relative velocity required to transform the two timestamps is greater than the speed of light."

Penny nodded.

"So, the two timestamps cannot simply be the same event described in two inertial frames."

"Correct."

Trevor glanced at the equation.

"Then one of our assumptions is wrong."

"Yes."

"Which one?"

"We don't know."

Penny stared at the photograph.

"Could the photograph be genuine but the event misidentified?"

"Yes."

"Could the clock be genuine but the frame wrong?"

"Yes."

"Could both be genuine and there be another effect?"

Derek gazed at her.

"Yes."

Trevor thought for a moment.

"Gravity."

Derek nodded.

"Possibly."

Surprise briefly overtook Trevor's face.

"You think General Relativity?"

"I think we shouldn't ignore it."

Penny nodded.

"That would be sensible."

Derek studied the silent clock.

Derek studied the silent clock for a long moment, turning it over once in his hands, as though weight alone might explain what angle couldn't.

"What?"

"The effect is too large."

Trevor walked over.

"By how much?"

Derek showed him. Trevor whistled softly.

"That isn't a small correction."

"No."

"Could a strong gravitational field do it?"

"Certainly."

"Do we have one?"

Derek surveyed the empty office.

"Not unless the landlord has installed a neutron star."

Penny eyed the floor.

"Perhaps he has."

Derek smiled.

"Then I want my rent reduced."

The telephone rang.

Derek answered.

"Derek Gent."

A familiar voice explained:

"Good morning."

Derek smiled.

"Hermann."

"Yes."

"Where are you?"

"Munich."

"Still?"

"Yes."

"Why?"

"Alfred is here."

Derek watched Penny.

"That's convenient."

"No."

"What?"

"You don't understand."

"Explain."

"He has found the same anomaly."

Derek straightened.

"The clocks?"

"Yes."

"Same magnitude?"

"Essentially."

Penny's expression changed. She had gone very still, the way she did whenever the ordinary rules of a Tuesday afternoon stopped applying. Trevor surveyed her.

"What?"

She whispered:

"Same anomaly in Munich."

Trevor understood.

Derek declared:

"Put Alfred on."

There was a click. Then a voice.

"Mr Gent."

"Alfred."

"You have the clock."

"Yes."

"I have seen it."

"I know."

"No. I mean I have seen that exact clock."

"Where?"

"In your Bureau."

Derek glanced at the ancient brass clock.

"When?"

"Tomorrow."

Derek sat down. Penny took the phone.

"Alfred, what did you see?"

"I saw the clock."

"And?"

"I saw you."

"Alive?"

"Yes."

Penny studied Derek.

"And?"

"You were also dead."

Trevor stared intently. Derek took the phone back.

"Alfred, let's be precise."

"I always am."

"You saw me alive and dead."

"Yes."

"At the same location?"

"Apparently."

"At the same time?"

"That depends on which time you mean."

Derek softly smiled.

"Good answer."

"Thank you."

Penny announced:

"That's settled, then."

Derek nodded.

"Evidently."

Trevor regarded them.

"That's it?"

Penny examined him.

"What else is there to settle?"

Trevor considered that.

"Fair enough."

Outside, the rain had not let up. Derek glanced once at the window before returning to the phone.

"Alfred, do you know what causes the effect?"

"No."

"Could it be gravitational?"

"Yes."

"Ordinary Earth gravity?"

"No."

"Could it be acceleration?"

"Possibly."

"Could it be spacetime curvature?"

"Possibly."

"Could it be a combination?"

"Could be."

Derek smiled.

"Very helpful."

"I am not there to solve your case."

"Then why are you coming to London?"

There was a pause.

"Because I am curious."

Derek nodded.

"Well, that settles it."

"What does?"

"You're coming."

"Yes."

"When?"

"Tomorrow."

Derek inspected Penny. Penny smiled.

"Of course."

Weinstein continued:

"I'll meet you at the station."

"Which station?"

"London Terminus."

Derek returned to the photograph.

"Platform seven?"

"Yes."

"At 11:17?"

"Yes."

Derek went silent. Trevor watched him. Penny understood immediately.

"That was in the photograph."

"Yes," Derek asserted.

Alfred pointed out:

"I know."

Derek glanced at the brass clock.

"Alfred."

"Yes?"

"Why are you so certain?"

A pause.

"Because I've already been there."

Derek frowned.

"Tomorrow?"

"Yes."

"And you've already seen me?"

"Yes."

"And the clock?"

"Yes."

Derek thought.

"Then we have a causal loop."

"Perhaps."

Penny studied him.

"Don't get excited."

"I wasn't."

"You were."

Derek smiled.

"I was mildly interested."

"That's your version of excitement."

Trevor nodded.

"It is."

Derek ignored both of them.

"Alfred, one more thing."

"Yes?"

"Does General Relativity explain what we're seeing?"

There was a pause.

Then Weinstein answered:

"It explains the principle."

"But not the effect."

"No."

Derek waited. Weinstein continued:

"The gravitational time-dilation effect is real. We understand it extremely well. What we don't understand is why your clocks are showing a difference so large."

Derek nodded.

"So, we're not discovering new relativity."

"No."

"We're applying known relativity to an unknown situation."

"Exactly."

Derek smiled.

"Right, that's that, then."

"What is?"

"The physics isn't broken."

"No."

"The situation is."

"Yes."

"Good."

The call ended. Derek put down the phone. Trevor stared at him.

"We're going to London Terminus tomorrow."

"Yes."

"At 11:17."

"Yes."

"After your supposed murder."

"Yes."

Penny picked up her coat.

"And Weinstein will be there."

"Yes."

Trevor nodded.

"That resolves it, then."

Derek gazed at him.

"Does it?"

Trevor thought.

"Yes."

"Good."

Derek picked up the ancient brass clock. The second hand moved. Tick.

Tick. It reached seventeen.

Stopped. Then started again. Derek peered at it.

"There's still one thing."

Penny paused at the door.

"What?"

Derek turned the silent clock over. The initials LGV were still stamped into the back.

"I want to know what this means."

Trevor eyed it.

"Perhaps Weinstein knows."

Derek nodded.

"Perhaps."

Penny opened the door.

"Ask him tomorrow."

Derek put the silent clock in his coat pocket.

"Good idea."

They left the Bureau. Behind them, the computer switched itself on. A new case note mysteriously appeared.

CASE 1047

REFERENCE FRAME: INERTIAL

GRAVITATIONAL CORRECTION: REQUIRED

Derek's name suddenly appeared beneath it. Then Trevor's. Then Penny's.

Then Alfred Weinstein's. Finally, one more line appeared.

HERMANN MINKOWSKI: CONSULTING

The screen went dark. In Munich, at almost exactly the same moment, the wall clock in the patent attorney's familiar office stopped. Weinstein noticed.

He watched his watch. Then at the clock. Then at the second hand.

It had stopped at seventeen. He smiled. Minkowski noticed.

"What?"

Weinstein held up his watch.

"Same thing."

Minkowski looked.

"Indeed."

Weinstein put the watch away.

"Good. Progress."

"What is?"

"We're going to London."

Minkowski nodded.

"And the patent?"

Weinstein looked sharply toward the quiet office.

"The spring can wait."

They walked toward the strange station. Behind them, in the patent attorney's office, the silent clock started again. Only this time it was running backwards.

Chapter 4

The Man Who Had Mislaid Time

Julius Barbarian lived in a house which appeared to have been designed by someone who had once been told what a house was and had then decided to improvise. It stood on a quiet street in Munich, behind a low wall and a hedge which had clearly given up several years earlier. There was no sign on the gate.

There was, however, a small brass plaque beside the front door.

JULIUS BARBARIAN PHYSICIST

Underneath, in smaller letters:

APPOINTMENTS ARE A HUMAN CONVENTION

Derek Gent read it twice.

"He's going to be one of those people."

Penny surveyed the plaque.

"Almost certainly."

Trevor still stood behind them, holding the ancient brass clock.

"What sort of people?"

"The sort who put philosophical statements on doorways."

Trevor considered this.

"Could be worse."

"How?"

"They could put them on mugs."

Penny nodded.

"Fair point."

Derek pressed the bell. Nothing happened. He pressed it again.

Still nothing. He pressed it a third time. The door quietly opened.

A man stood there. Julius Barbarian was in his late sixties, perhaps early seventies. He was slim, silver-haired, and dressed in a dark jacket, open-necked shirt and rather expensive-looking trousers that had nevertheless acquired a small stain of something orange. He looked at Derek.

Then Penny. Then Trevor. Then the silent clock.

"You're late."

Derek glanced at his watch.

"We're three minutes early."

"Precisely."

Barbarian opened the door wider.

"Come in."

Derek hesitated.

"How did you know who we were?"

"I knew you were coming."

"How?"

"Your car."

"We came by train."

"Exactly."

Derek stared at him. Barbarian smiled.

"Come in."

They entered. The hallway was surprisingly ordinary. Bookshelves occupied both walls.

There were papers everywhere. Not piles of papers, as in an ordinary untidy quiet office. Rather, papers appeared to have been distributed according to some private mathematical principle which Barbarian alone understood.

There were books on physics, philosophy, mathematics and cosmology. There were also several books about gardening. One shelf contained nothing but clocks.

Derek stopped. There were perhaps twenty of them. Pocket watches.

Marine chronometers. Railway clocks. A grandfather clock.

A small digital clock. A sundial. And, inexplicably, an egg timer.

Trevor noticed it too.

"Do you collect clocks?"

"No."

Barbarian took off his jacket.

"I collect things people use to pretend time exists."

Derek regarded Penny. Penny examined him. Trevor inspected the clocks.

"That's a rather strong statement."

Barbarian softly smiled.

"I've been told."

He walked into the sitting room. They followed. A large wooden table occupied the centre of the room.

There were four chairs. Barbarian sat down. The others did likewise.

For several seconds nobody spoke. Then Barbarian looked directly at Weinstein.

"Alfred."

Weinstein, who had been examining a book on the shelf, turned.

"Yes?"

"You've spent most of your life making time behave."

"I've spent most of my life making physicists stop treating it as obvious."

"Better."

Barbarian leaned back.

"Now you're having trouble with it."

Weinstein sat opposite him.

"We have clocks that appear to disagree by an amount that cannot be explained by ordinary gravitational time dilation."

"Yes."

"And the same anomaly appears in London and Munich."

"Yes."

"And the clocks appear to be connected to an event which hasn't happened yet."

"Yes."

"And you invited us here because you think the problem is that we're assuming time exists." Julius Barbarian smiled. "Perhaps." Derek nodded. "Close enough for government paperwork."

Barbarian smiled.

"No."

Weinstein waited. Barbarian picked up a cup of tea.

"I invited you here because I think you are assuming that time is something more than the relationships between events."

Weinstein nodded slowly.

"That's different."

"Very."

Derek leaned forward.

"How?"

Barbarian considered him.

"Tell me what time is."

Derek softly smiled.

"That's a dangerous question."

"It's a simple one."

"No. It's a question designed to make the answer complicated."

"Excellent."

Derek glanced at the silent clock on the wall.

"Time is what clocks measure."

Barbarian smiled.

"No."

"That's at least a useful operational definition."

"Certainly."

"Then what's wrong with it?"

"Nothing."

Derek paused.

"Well, that settles it."

Barbarian laughed.

"I like you."

Penny smiled.

"He's getting that a lot lately."

Barbarian stared at the old clock.

"Clocks measure regular physical processes. Oscillations. Rotations. Atomic transitions. Mechanical motion. You choose a process you believe is sufficiently regular, call it a clock, and compare other changes with it."

"Yes."

"So, a clock doesn't measure some invisible substance called time."

Derek nodded.

"It measures change against change."

"Exactly."

Interest sharpened Trevor's expression.

"That doesn't mean time doesn't exist."

Barbarian turned toward him.

"No. It means we have to be careful about what we mean by 'exists.'"

Trevor nodded.

"Fair enough."

Barbarian smiled.

"Excellent."

He pointed at the table.

"Suppose I place an apple here."

He put an apple on the table.

"Then I move it there."

He moved it.

"The apple has changed position."

"Yes," Trevor clarified.

"Do we need time to describe that?"

Trevor thought.

"We need some ordering."

"Perhaps."

"We need to distinguish the first state from the second."

"Yes."

"Otherwise we can't say it moved."

Barbarian nodded.

"Good."

He picked up the apple.

"But suppose instead of saying the apple moved through time, we simply describe two complete configurations."

He put the red apple back.

"Configuration A."

Then moved it.

"Configuration B."

Trevor nodded.

"The relationship between them tells us that the apple changed position."

"Exactly."

"And the amount of change tells us something about the dynamics."

"Yes."

"So where does time go?"

Barbarian softly smiled.

"That's the question."

Derek leaned back.

"It doesn't necessarily go anywhere."

"Precisely."

Penny gazed at the clocks.

"So, you're saying that reality consists of configurations, and what we call time is the ordering or relationship between them."

"Something like that."

Weinstein folded his hands.

"That is philosophically attractive."

Barbarian raised an eyebrow.

"But?"

"But General Relativity doesn't give us such an easy picture."

"Quite."

"In General Relativity, time is part of spacetime."

"Yes."

"And the geometry of spacetime is dynamical."

"Yes."

"Proper time along a worldline is measurable."

"Yes."

"An observer carries a clock, and the clock measures proper time along that observer's worldline."

"Yes."

Weinstein smiled.

"So, we cannot simply declare time nonexistent and walk away."

Barbarian smiled back.

"I wasn't planning to."

"Good."

"I have a much more irritating plan."

Derek turned to Penny.

"He's definitely one of those people."

Penny nodded.

"Yes."

Barbarian continued.

"General Relativity tells us something extraordinary. There is no single universal time that everyone in the universe must agree upon."

Weinstein nodded.

"Correct."

"In Special Relativity, simultaneity is relative."

"Yes."

"In General Relativity, the situation becomes even richer because spacetime itself is dynamical."

"Yes."

"So, the universe doesn't come with a giant clock mounted above it."

Derek mentioned:

"That would be inconveniently large."

Barbarian smiled.

"Quite."

Trevor eyed the silent clock shelf.

"Then what are clocks actually telling us?"

"Relationships."

"Between what?"

"Between physical processes."

Trevor nodded.

"Right."

Barbarian's smile sharpened.

"As settled as anything ever is."

Derek laughed.

"We've been here ten minutes and you're already using our phrase."

"I adapt quickly."

Penny pointed toward the shelf.

"Then why all the clocks?"

Barbarian watched them.

"Because they're useful."

"But you responded that they don't measure time."

"I opined that they don't measure a substance called time."

"What's the difference?"

Barbarian smiled.

"Quite a large one."

He picked up an old railway watch.

"This tells you how one physical process relates to another. If you synchronize two clocks under specified conditions, move them apart, accelerate one, place them at different gravitational potentials and compare them again, the difference tells you something about spacetime."

Weinstein nodded.

"Exactly."

"So, clocks are not the problem."

"No."

"They are evidence."

"Yes."

"Excellent."

Barbarian put the watch down.

"The problem is the interpretation." Penny nodded. "Exactly. We're skating on thin gravy."

Derek surveyed the brass clock Trevor was holding.

"Which brings us to this."

Barbarian's expression changed. Only slightly. But everyone noticed.

"Yes."

He stood.

"May I?"

Trevor handed him the clock. Barbarian turned it over. He examined the initials.

LGV.

Then the mechanism. Then the face. He remained silent.

Weinstein watched him.

"You've seen it before."

Barbarian looked up.

"Yes."

"When?"

Barbarian smiled.

"That's the wrong question."

Weinstein frowned.

"Don't start."

Barbarian laughed.

"Very well."

He placed the clock on the table.

"I saw it yesterday."

Derek frowned.

"Where?"

"Here."

Derek cast a quick look around.

"You had this clock yesterday?"

"No."

"Then what do you mean?"

"I saw it."

"When?"

"Yesterday."

Trevor frowned.

"That's perfectly ordinary."

Barbarian studied him.

"Is it?"

Trevor thought.

"Yes."

"Why?"

"Because yesterday precedes today."

Barbarian nodded.

"According to what?"

Trevor hesitated.

"According to... Time."

Barbarian smiled.

"Exactly."

Trevor regarded Derek. Derek examined Penny. Penny inspected Barbarian.

"That's rather clever."

"Thank you."

Derek picked up the silent clock.

"Let's be precise."

"Please."

"You saw this clock yesterday."

"Yes."

"The physical clock."

"Yes."

"Not a photograph."

"No."

"Not an identical clock."

"No."

"This exact clock."

"Yes."

Derek frowned.

"And it is currently here."

"Yes."

"Which means either you saw it before it arrived here..."

Barbarian nodded.

"... or the clock has somehow moved backwards through whatever we mean by time."

"Possibly."

"Or your memory is wrong."

"Also possible."

Derek nodded.

"Good."

Barbarian smiled.

"You see the difficulty."

"Yes."

Weinstein checked the silent clock.

"Where did you see it?"

Barbarian pointed at the silent ancient clock.

"Here."

Derek stared intently.

"Here?"

"On this table."

"But it wasn't here yesterday."

Barbarian nodded.

"Not according to your description."

Weinstein smiled.

"Now we're getting somewhere."

Barbarian sat down again.

"This is precisely why I wanted to speak to you."

"Why?"

"Because you understand the mathematics well enough not to be frightened by the philosophical consequences."

Derek smiled.

"That's a generous assessment."

"Is it wrong?"

"No."

"Right."

Weinstein picked up a notebook.

"Suppose we forget the word 'time' for a moment."

Derek nodded.

"We have events."

"Yes."

"Each event has relationships to other events."

"Yes."

"Those relationships can be described geometrically."

"Yes."

"Some observers call one relationship earlier and another later."

"Yes."

"But those labels are not fundamental."

Barbarian smiled.

"Now you're talking."

Weinstein continued.

"And the Lorentz transformation tells us how different inertial observers assign coordinates to events."

"Yes."

"While General Relativity tells us how the geometry of spacetime behaves in the presence of matter and energy."

"Yes."

"And proper time is what a physical clock measures along a worldline."

"Yes."

"So, perhaps what we're calling time is the structure of these relationships."

Barbarian nodded.

"That is a respectable way of putting it."

Derek leaned forward.

"And if the clocks aren't showing different amounts of some universal substance called time..."

Barbarian waited.

"They're showing different relationships between events."

"Exactly."

Derek glanced at Penny.

"Good. Moving on."

Penny nodded.

"Good."

Relief crossed Trevor's face.

"Does it?"

Everyone stared at him. Trevor pointed at the silent ancient clock.

"We still don't know why the relationships are wrong."

There was silence. Barbarian smiled.

"Excellent."

Derek smiled.

"Yes."

Trevor surveyed the room.

"What?"

"Nothing," Derek maintained.

"You all look pleased."

"We are."

"Why?"

"Because you noticed the actual problem."

Trevor nodded.

"Oh."

Penny smiled.

"Well, there we are."

Barbarian still stood.

"Come."

"Where?"

"The garden."

Derek looked sharply toward the window.

"Why?"

"Because I want to show you something."

They followed him through the back door. The quiet garden was larger than Derek had expected. There was a lawn, a small pond, several trees and a stone path.

At the far end quietly stood a tall wooden post. Barbarian walked toward it. At the top was a strange clock.

Derek frowned.

"Another one."

"Yes."

Barbarian gazed at the silent clock.

"This one is synchronized with the one inside."

Derek checked. They were.

"How far apart?"

"About forty metres."

"Same gravitational potential to a very good approximation."

"Yes."

"Then they should remain almost synchronized."

"They do."

Trevor visibly relaxed.

"Good."

Barbarian smiled.

"Now wait."

They waited. Nothing happened. A bird landed on the fence.

A car passed somewhere beyond the garden. The wind slowly moved the leaves. The clocks continued ticking.

Derek peered at his watch. Five minutes passed. Then ten.

Then fifteen. Still nothing. Trevor looked at Barbarian.

"What are we waiting for?"

"Change."

Derek smiled.

"That's a very philosophical answer."

"It's also accurate."

Another five minutes. Then the silent clock on the post stopped. The indoor clock continued.

Derek walked closer. The silent clock had stopped at exactly 11:17:00. He checked his watch.

11:17:00. The indoor clock read 11:17:00. Everything was synchronized.

Derek watched Barbarian.

"Well, that's odd."

"Yes."

Trevor examined the clock.

"Mechanical failure?"

"Possibly," Barbarian affirmed.

Derek shook his head.

"No."

"Why?"

"The second hand isn't jammed."

He pointed. The hand was moving. But it was moving in tiny jumps.

One second. Then a pause. Then another.

Penny looked closely.

"The clock is still running."

"Yes."

"Then why isn't the display advancing?"

Derek frowned. He surveyed the silent clock inside the house. That one had also stopped.

Exactly at 11:17. Trevor checked his phone. 11:17.

His phone had stopped too. Penny checked hers. 11:17.

Stopped. Weinstein checked his watch. 11:17.

Stopped. Everyone looked at one another. Barbarian smiled.

"Interesting."

Derek stared at him.

"Did you arrange this?"

"No."

"Did you know it would happen?"

"No."

"Then why are you smiling?"

"Because now we have something useful."

The seconds did not move. The world did not stop. A bird flew across the quiet garden.

A leaf fell. The wind continued. Derek studied the clocks.

"Everything is moving."

"Yes."

"Yet all the clocks have stopped."

"Yes."

Trevor frowned.

"So, the clocks have failed."

Derek shook his head.

"No."

"How do you know?"

"Because they're all failing identically."

Penny nodded.

"And at exactly the same reading."

Weinstein looked around.

"What's the significance of 11:17?"

Derek thought.

"The train."

Everyone went quiet. The photograph.

11:03 — DEATH.

11:17 — DEPARTURE.

12:04 — ARRIVAL.

Derek took out the photograph. The clock in the photograph showed 11:03:17. He regarded the actual clocks.

11:17:00. Then he examined Barbarian.

"Can you make them start again?"

Barbarian shook his head.

"I don't control them."

"Who does?"

"I don't know."

Derek inspected Weinstein. Weinstein was staring at the sky.

"What?"

Weinstein pointed.

"Look."

They looked upward. Nothing unusual was visible. Blue sky.

Clouds. A plane. Then Derek noticed something.

The plane wasn't moving. Neither was the cloud. The leaves had stopped.

The bird was suspended in the air. Trevor stared.

"That's impossible."

Derek smiled.

"Yes."

Penny considered him.

"Don't say it."

"Why?"

"Because I know what you're going to say."

Derek grinned.

"Well, that's settled."

Barbarian looked remarkably calm. Weinstein wasn't. He silently stared at the suspended plane.

"The physical world has stopped."

Barbarian shook his head.

"No."

"What do you mean, no?"

"Look at the pond."

They did. The surface of the water was perfectly still. But a small insect was sitting on it.

Not moving. Derek took in the room. Nothing moved.

Not the leaves. Not the clouds. Not the distant plane.

Not the bird. Not the water. Then Penny pointed.

"Look at Trevor."

Everyone glanced at Trevor. Trevor was standing perfectly still. He blinked.

Everyone stared intently. Trevor blinked again.

"What?"

Derek smiled.

"You're moving."

Trevor stared at his hands.

"So, are you."

Penny slowly moved her head.

"So, am I."

Weinstein gazed at Barbarian.

"You can move."

"Yes."

"So, can we."

"Yes."

"But everything else is frozen."

"Apparently."

Derek checked his watch. Still 11:17. He eyed the sky.

Still frozen. He watched Barbarian.

"How?"

Barbarian smiled.

"Perhaps time has stopped."

Derek frowned.

"You've been waiting your entire career to say that, haven't you?"

"Possibly."

Trevor glanced around.

"If time has stopped, why are we moving?"

Barbarian nodded.

Derek thought.

"Because our local physical processes haven't stopped."

"Perhaps."

"Then time hasn't stopped."

Barbarian smiled.

"Perhaps."

Weinstein looked annoyed.

"You're enjoying this."

"Immensely."

"Why?"

"Because you're trying to make the word 'time' do too much work."

Weinstein folded his arms.

"Explain."

Barbarian gestured around the quiet garden.

"Suppose there is no fundamental time variable ticking behind reality."

"All right."

"Suppose instead there are physical configurations."

He pointed to the frozen plane.

"One configuration contains that plane in one position."

He pointed to the garden.

"Another contains it somewhere else."

He pointed at the clocks.

"The clocks are physical systems within those configurations."

He pointed at Trevor.

"And so are we."

Trevor nodded.

"Yes."

"So if we move from one configuration to another while the clocks don't advance, the clocks don't provide the ordering."

Derek understood.

"The relationships do."

"Just so."

Weinstein frowned.

"But then what has happened to the time coordinate in the equations?"

Barbarian smiled.

"Nothing."

"Nothing?"

"The equations remain useful."

"Then you aren't denying time in physics."

"I'm denying that the symbol t must correspond to a fundamental substance flowing through the universe."

Weinstein nodded slowly.

"That's more defensible."

"Much."

Penny surveyed the strangely frozen plane.

"So what we're seeing is a physical configuration in which the clocks aren't advancing."

"Yes."

"But we're still changing."

"Yes."

"Then the clocks aren't fundamental."

"No."

"They're just one type of physical process."

"Exactly."

Derek studied the silent brass clock.

"And this one is somehow associated with a different sequence of configurations."

Barbarian nodded.

"Perhaps."

Derek smiled.

"Fine. Next problem."

Trevor regarded him.

"What?"

"We've learned something."

"What?"

"That the clock isn't measuring some universal time."

"Right."

"It is measuring something physical."

"Yes."

"And that physical process can disagree with other physical processes."

"Yes."

Penny smiled.

"Which is exactly what relativity already told us."

Derek nodded.

"Yes."

Weinstein fell silent in thought.

"And yet this is different."

"Why?"

"Because in ordinary relativity, the clocks disagree according to the geometry of spacetime."

Barbarian nodded.

"And here?"

Weinstein pointed at the frozen garden.

"We have no idea what geometry we're dealing with."

Barbarian smiled.

"Now you've reached the interesting part."

The bird suddenly moved. It flew away. The leaves rustled.

The distant plane continued across the sky. The cloud moved. Water rippled in the pond.

All the clocks began ticking again. 11:17:01. Then 11:17:02.

Then 11:17:03. Everyone stood silently. Trevor checked his phone.

11:17:03. Penny checked hers. 11:17:03.

Derek examined the wall clock through the window. 11:17:03. Weinstein inspected his watch.

11:17:03. Barbarian smiled.

"Well."

Derek nodded.

"Well, that's settled."

Trevor took in the room.

"What is?"

"Time exists."

Barbarian shook his head.

"No."

Derek smiled.

"You couldn't resist."

"Not really."

Penny laughed. The clocks continued ticking. Derek studied the photograph again.

11:03 — DEATH.

11:17 — DEPARTURE.

12:04 — ARRIVAL.

Then he noticed something he hadn't noticed before. There was a fourth line. It had been hidden beneath the fold.

He unfolded the paper. It read:

11:17 — ALL CLOCKS STOP

12:04 — ALL CLOCKS START

Derek silently stared. Penny leaned over. Trevor read it.

Weinstein turned to Barbarian. Barbarian's smile disappeared.

"That's new."

Derek raised his eyes.

"Very."

Trevor confirmed:

"But the clocks just stopped."

"Yes."

"And started again."

"Yes."

"Exactly as described."

"Yes."

Penny glanced at the paper.

"So, this wasn't a malfunction."

Derek shook his head.

"No."

"It was an event."

"Yes."

Weinstein nodded.

"And the clocks recorded it."

Barbarian slowly stared at them.

"Or participated in it."

Derek considered that.

"What's the difference?"

Barbarian smiled faintly.

"That is the question."

The ancient brass clock in Derek's pocket began ticking. Once. Twice.

Then it stopped. Derek took it out. The face now read:

12:04. Trevor stared intently.

"But it's 11:17."

Derek gazed at him.

"Apparently."

Penny smiled.

"Well, that's inconvenient."

Derek peered at the silent clock.

"Very."

Weinstein took it from him. He examined the mechanism. Nothing.

No obvious fault. No hidden mechanism. No battery.

No radio receiver. No trick. He eyed Barbarian.

"Do you still think time doesn't exist?"

Barbarian considered the question.

"I think reality is perfectly capable of existing without asking my permission."

Weinstein smiled.

"That's not an answer."

"No."

"Then?"

Barbarian handed the silent clock back.

"Time may be a very useful description of relationships between events."

He paused.

"But I have never been convinced that the universe itself carries a watch."

Derek watched the silent clock.

"Perhaps it doesn't."

"Perhaps."

"Perhaps we're the ones carrying it."

"Exactly."

Derek put the silent clock back into his pocket.

"Well, that's settled."

Trevor smiled.

"What?"

"We've learned something useful."

"What?"

Derek looked toward the frozen sky.

"The next event is at 12:04."

Penny nodded.

"And we're currently at 11:17."

Trevor checked his phone.

"Forty-seven minutes."

Derek smiled.

"Excellent."

Barbarian surveyed him.

"Excellent?"

"We have forty-seven minutes to find out where the train is going."

Barbarian shook his head.

"No."

Derek frowned.

"What?"

"You have forty-seven minutes to find out where the train has already been."

Derek stared at him. Then smiled.

"Right."

Penny picked up her coat.

"London?"

"London."

Trevor stood.

"Platform seven?"

"Platform seven."

Weinstein studied the clock.

"And the train?"

Derek carefully opened the door.

"We'll find it."

Barbarian called after him.

"Mr Gent."

Derek turned.

"Yes?"

Barbarian pointed at the silent clock.

"Don't trust it."

Derek smiled.

"We weren't planning to."

Barbarian nodded.

"Good."

Derek opened the door. Then stopped.

"One thing."

"What?"

"If time doesn't exist..."

Barbarian smiled.

"Yes?"

"What day is it?"

Barbarian regarded him. For the first time since they'd arrived, he seemed genuinely uncertain.

"Ah."

Penny laughed. Trevor laughed. Even Weinstein smiled.

Barbarian examined the clocks around the empty room. Then at the sky. Then at the brass clock in Derek's hand.

"Well," he insisted, "that's rather embarrassing."

And somewhere in London, a distant train began moving toward empty Platform Seven. It had left yesterday. It would arrive tomorrow.

And, according to the timetable, it had never existed at all.

Chapter 5

The Apple Problem

Alfred Weinstein had never entirely trusted apples. This was not because of any defect in the fruit. Apples were perfectly respectable.

They were roundish, nutritious, reasonably portable and, unlike pears, rarely gave the impression that they were about to collapse under their own moral burden. His distrust concerned their reputation. For centuries, apples had been given far too much credit.

One particular red apple, in particular, had been blamed for an astonishing amount of physics. Weinstein was thinking about it on the mysterious train.

He was on his way back to Munich after the meeting with Julius Barbarian, although “on his way back” was perhaps an unnecessarily confident description of the situation.

The mysterious train had stopped. It had stopped between two stations. Nobody seemed particularly concerned.

The conductor had announced that there was a “minor delay due to circumstances.”

Weinstein had spent enough time on trains to know that “circumstances” meant absolutely anything.

He sat by the window. Across from him, a woman was reading a novel. Beside her, a man was asleep.

Two rows ahead, a child was eating an apple. Weinstein watched the child. The child took a bite.

Then another. Then, apparently deciding that the apple had become philosophically unacceptable, dropped it. It fell to the floor.

Weinstein intently watched it. The apple rolled beneath the seat. He stared at the space where it had disappeared.

Then he smiled. Newton. He thought of Newton’s famous ordinary apple.

The story was almost certainly too neat. Science rarely arrived with such excellent stage management. One could imagine the scene.

Newton sits under a tree. An apple falls. Newton looks at it.

Newton thinks about gravity. Physics is transformed. It was a charming story.

It was also misleading.

The apple had not explained gravity.

It had merely demonstrated that something fell. The really interesting question was why. Weinstein leaned back.

Why did the apple fall? Everyone knew the answer. Gravity.

But that was not an answer. It was a name. He had spent most of his life telling physicists not to confuse naming a phenomenon with explaining it.

Newton had described gravity extraordinarily well. The inverse-square law worked. The equations worked.

The predictions worked. The planets obeyed them. The Moon obeyed them.

The apple obeyed them. Everything seemed to obey them. Then General Relativity had come along and changed the question.

The apple was no longer simply being pulled downward by a force. It was following a geodesic through curved spacetime. Weinstein smiled.

The poor apple had been promoted from fruit to geometry. The distant train moved. Then stopped again.

Weinstein wearily sighed.

He looked out the window. Nothing. Then, very suddenly, he had an idea.

It was not a complete idea. It was not even a particularly respectable idea. It was the sort of idea that arrived without knocking, sat down in the middle of one's mind and refused to explain why it was there.

Weinstein sat upright. The ordinary apple. Not the falling.

The falling together. He took out his notebook. He drew the Earth.

Then the apple. Then an arrow. He stopped.

Crossed it out. Drew the Earth. Then the apple.

Then two arrows. One pointing from the apple toward the Earth. One pointing from the Earth toward the apple.

He stared at them.

"Of course."

The woman across the aisle looked up. Weinstein noticed.

"Sorry."

She returned to her book. He continued. Newton's description had made it natural to think of the Earth as pulling the apple.

But General Relativity made a remark more subtle. The apple and the Earth were both responding to the geometry. The apple fell.

The Earth moved too. Tiny amount. Almost impossibly tiny.

But not zero. Weinstein wrote:

THE SIMPLE APPLE DOES NOT FALL TOWARD THE EARTH.

He stopped. That was not quite right. He crossed it out.

THE EARTH AND THE APPLE FALL TOWARD EACH OTHER.

He inspected it. Better. Then another thought struck him.

If the apple and the Earth were both following the geometry, what exactly determined the geometry? Mass and energy. Fine.

But there was something else. The equivalence principle. A freely falling observer did not feel gravity locally.

That was the heart of the matter. An ordinary apple falling from a tree was not being dragged through space by an invisible rope. It was freely falling.

And the tree was preventing it from doing so. Weinstein stopped. He considered the words.

The tree was preventing the simple apple from following its natural path. He smiled.

“That’s rather good.”

The woman looked up again. This time she decided not to ask. Weinstein turned the page.

He drew a tree. Then the apple. Then the ground.

Then he drew the same apple in a lift. He imagined the lift accelerating upward in empty space. The apple would appear to fall.

Yet there would be no gravitational field. Then he imagined the lift sitting on Earth. The apple would fall again.

Locally, the two situations could be indistinguishable. That was the equivalence principle. Weinstein knew this.

Everyone in the Bureau knew this. So why had the thought bothered him? He glanced at his drawing again.

Because the apple was falling toward something. And in the accelerating lift there was no Earth. The apparent downward direction was created by the acceleration.

Gravity could be imitated by acceleration. But perhaps the deeper question was the reverse. Could acceleration be understood as geometry in the same way gravity was?

Yes.

That was already part of General Relativity. So why did the idea feel new? Weinstein tapped his pencil against the notebook.

Then he saw it. Not the answer. The question.

The question was not: Why does the apple fall? The question was:

Why does the apple have the trajectory it does? He wearily sat back. That was better.

Much better. The trajectory of a freely falling object was determined by spacetime geometry. But the geometry was determined by the distribution of matter and energy.

So, the simple apple was both responding to the geometry and, in principle, contributing to it. The apple affected the spacetime through which it moved. Tiny amount.

But real. The Earth affected the apple. The apple affected the Earth.

The Earth affected spacetime. The apple affected spacetime. Everything was interacting with everything else.

Weinstein stared at the page.

"That's going to annoy someone."

The man sleeping beside the woman opened one eye.

"What?"

"Nothing."

The man closed his eye again. Weinstein wrote:

ASK DEREK.

Then underneath:

INVESTIGATE THE ORDINARY APPLE.

He stopped. That sounded ridiculous. He crossed out ordinary APPLE.

Then wrote:

INVESTIGATE THE FALLING BODY.

He stared at it. Too clinical. He crossed out the whole line.

He wrote:

WHY DOES THE APPLE FALL?

He smiled. That was better. He took out his phone.

He called Derek.

Derek answered on the fourth ring.

"Yes?"

"Alfred."

"Where are you?"

"On a train."

"Which one?"

"The Munich one."

Derek paused.

"That's not terribly specific."

"I know."

"Are you going somewhere?"

"Apparently."

"Do you know where?"

"Munich."

"You're already in Munich."

"I know."

There was silence.

Then Derek emphasized:

"Has Julius Barbarian finally broken you?"

"No."

"Fine."

"I have a question."

"Go on."

"Why does an apple fall?"

Derek was silent. Then:

"Are you serious?"

"Perfectly."

"Gravity."

"No."

Derek wearily sighed.

"Don't start."

"I'm not starting anything."

"You've just asked me why an apple falls."

"Yes."

"And you're Alfred Weinstein."

"Yes."

"The inventor of General Relativity."

"Yes."

"The theory that explains why it falls."

"Precisely."

"Then why are you asking me?"

"Because I think the question is wrong."

Derek sat down.

"All right."

Penny was sitting nearby. She looked over. Derek put the phone on speaker.

"Go on."

Weinstein queried:

"Suppose an apple is freely falling."

"Yes."

"Is the apple accelerating?"

"In the Newtonian sense, yes."

"And in General Relativity?"

"It is following a geodesic."

"Exactly."

"So, locally, the apple is not experiencing a force."

"Correct."

"Then why does it appear to accelerate toward the Earth?"

Derek thought.

"Because we are observing it from a non-free-falling frame."

"Yes."

"And the Earth's surface is preventing us from following our own geodesic."

"Yes."

Penny nodded.

"That settles the basic point."

Weinstein smiled.

"Yes."

Derek said:

"But that's not the question."

"No."

"What is?"

Weinstein looked out of the window for a while, watching a bus fail to arrive on schedule.

"The apple and the Earth are both moving."

Derek nodded.

"Yes."

"The apple moves toward the Earth."

"Yes."

"The Earth moves toward the apple."

"Yes."

"Why?"

Derek paused. Penny gazed at him.

Derek stated:

"Because of their mutual gravitational interaction."

Weinstein remained silent. Derek smiled.

"That's the Newtonian answer."

"Precisely."

"So, you want the General Relativistic answer."

"Yes."

Derek leaned back.

"The geometry."

"Yes."

"The stress-energy determines the geometry."

"Yes."

"And the geometry determines the geodesics."

"Yes."

"The apple follows the geodesic."

"Yes."

Derek paused.

"There we are, then."

Weinstein laughed.

"No."

Derek smiled.

"You've just agreed to everything."

"I know."

"Then what isn't settled?"

Weinstein consulted his notebook.

"The apple."

Penny leaned toward the phone.

"What about it?"

"Why that apple?"

Derek frowned, and reached for the ordinary cucumber without quite noticing he had done it.

"Why that particular apple?"

"Yes."

"Why does it matter which apple?"

"That's what I don't know."

Penny smiled.

"Now that's interesting."

Derek nodded.

"Yes."

Weinstein continued.

"The Bureau has been investigating events that appear to depend on which frame we use. Julius Barbarian has been arguing that time may not be fundamental. We have clocks behaving differently. And now I am thinking about an apple."

Derek commented:

"That sounds like a normal Thursday."

"It's Wednesday."

"Right."

Weinstein ignored that.

"Suppose the apple falls."

"Yes."

"Suppose we reverse the process."

Derek paused.

"Reverse it?"

"Imagine reconstructing the trajectory."

"From the ground upward?"

"Yes."

"We'd have to know its position and velocity."

"Yes."

"And the gravitational field."

"Yes."

"And the geometry."

"Yes."

"And the matter distribution."

"Yes."

Weinstein noted:

"Exactly."

Derek waited. Then he understood.

"We'd need the entire local configuration."

"Yes."

Penny eyed Derek.

"So, the apple's trajectory contains information about the geometry."

"Yes."

"And the geometry contains information about the matter."

"Yes."

"And therefore the falling apple is effectively telling us something about its environment."

"Exactly."

Derek smiled.

"Well, that's settled."

Weinstein laughed.

"No."

"What now?"

"Because I've just had another thought."

"Of course you have."

"The Bureau's clocks."

Derek straightened.

"What about them?"

"They are following worldlines."

"Yes."

"They're measuring proper time."

"Yes."

"They are effectively sampling spacetime along those worldlines."

"Yes."

"And if their readings are anomalous..."

"Then the geometry is anomalous."

"Perhaps."

Derek frowned.

"Or the clocks are."

"Yes."

"But we've already ruled out ordinary clock failure."

"Yes."

"So..."

Weinstein paused.

"Investigate the trajectories."

Derek stared at the phone.

"The trajectories of what?"

"Everything."

"Everything?"

"Start with the clocks."

Derek watched Penny. Penny nodded.

"That actually makes sense."

Derek observed:

"We've been treating the clocks as stationary measuring devices."

"Yes."

"Perhaps we should treat them as physical objects following worldlines."

"Exactly."

Trevor suddenly appeared in the doorway. He had clearly been listening.

"That means the clocks themselves are part of the experiment."

Derek surveyed him.

"Yes."

Trevor nodded.

"Splendid. Moving on."

Derek smiled.

"See? He's learning."

Trevor's expression brightened. Then asked:

"What's a worldline?"

Penny laughed.

Derek replied:

"Come in."

Trevor did.

Derek explained:

"In spacetime, an object's history can be represented as a line. Not a line through ordinary space, but through spacetime. If the object moves, its position changes with time. The complete history forms its worldline."

Trevor nodded.

"And a clock follows one."

"Yes."

"So, the clock doesn't merely tell us time."

"Correct."

"It gives us a physical record along its worldline."

"Quite."

Trevor studied the ancient brass clock.

"Then if the clock behaves strangely, we should ask whether its worldline is strange."

Derek smiled.

"Now you're ahead of me."

Trevor nodded.

"That happens occasionally."

Penny regarded him.

"Don't overdo it."

Trevor smiled.

"I won't."

Derek returned to the phone.

"Alfred, what exactly do you want us to investigate?"

Weinstein lowered his gaze at his notebook.

"The apple."

Derek laughed.

"You've come all this way in physics to send me after an apple."

"No."

"Then what?"

"The apple is the model."

"For what?"

"Everything."

Derek became serious. Weinstein continued.

"If a freely falling body is following a geodesic, then its trajectory is telling us about the geometry. If we can observe enough trajectories, we can reconstruct information about that geometry."

Derek nodded.

"Yes."

"And if the Bureau is experiencing an unusual spacetime effect, we should stop staring at the clocks."

"We should watch what moves around them."

"Exactly."

Penny smiled.

"That's actually rather elegant."

Weinstein answered:

"Thank you."

Derek asked:

"Do you have a particular trajectory in mind?"

"Yes."

"Whose?"

Weinstein looked out the empty train narrow window. A landscape passed. Trees.

Fields. Buildings. Everything apparently moving.

Or perhaps the train was moving. Or perhaps both. He smiled.

"Mine."

Derek frowned.

"What?"

"I want you to investigate my trajectory."

"Your worldline?"

"Yes."

"Why?"

"Because something happened in Munich."

"What?"

"My clock stopped."

"We know."

"No."

Weinstein's calm voice became quieter.

"It stopped before the clocks in London stopped."

Derek froze. Penny examined him. Trevor stopped moving.

Derek added:

"When?"

"Eleven sixteen."

Derek stared at the ancient brass clock.

"But ours stopped at eleven seventeen."

"Yes."

"So, there's a one-minute difference."

"Exactly."

Derek thought.

"That could be ordinary synchronization."

"No."

"How do you know?"

"Because I wasn't where you think I was."

Derek frowned.

"What do you mean?"

"I was on the train."

Derek stared intently.

"What train?"

"The train to Munich."

"You were going to Munich."

"Yes."

"You were already in Munich."

"Yes."

Derek quietly closed his eyes.

"Alfred."

"Yes?"

"You were on a train going to Munich while you were in Munich."

"Yes."

"How?"

"I don't know."

Penny smiled.

"That's rather good."

Derek inspected her.

"Is it?"

"Yes."

"Why?"

"Because now we have a genuine problem."

Derek returned to the phone.

"Alfred, where are you now?"

There was a pause.

"On the train."

"Which train?"

"The train to Munich."

Derek frowned.

"But you're in Munich."

"It would seem so."

Trevor looked at Penny.

"Could he be in two reference frames?"

Penny shook her head.

"Not in that sense."

"Right."

Derek explained:

"Alfred, can you see the station?"

"Yes."

"Which one?"

"Munich."

"Are you moving?"

"Yes."

"Relative to the station?"

"Yes."

Derek rubbed his forehead.

"Then what is happening?"

Weinstein declared:

"I don't know."

Derek nodded.

"Excellent. Case closed."

Weinstein laughed.

"Precisely."

Derek glanced at Penny. She was smiling.

"What?"

"You're enjoying this."

"Not the impossible train."

"The investigation."

"Yes."

Derek turned back to the phone.

"Alfred, what does the train clock say?"

"11:17."

Derek stared at the Bureau clock. 11:17. Penny gazed at hers.

11:17. Trevor peered at his phone. 11:17.

All four devices were synchronized.

Derek announced:

"And your watch?"

"11:17."

"The brass clock?"

"11:17."

"All right."

Derek took a breath.

"Now look outside."

"What do you see?"

"Munich."

"Moving?"

"Yes."

"Backward?"

"No."

"Forward?"

"Yes."

Derek nodded.

"Good."

"Why?"

"Because I was beginning to worry."

"About what?"

"That we might have to investigate Munich."

Weinstein laughed. Then his quiet voice changed.

"Wait."

"What?"

"I see something."

"What?"

"An apple."

Derek eyed Penny.

"What sort of apple?"

"Red."

"That's encouraging."

"It is falling."

"Where?"

"From a tree."

Derek smiled.

"Alfred, are you sure you're not just watching an ordinary apple?"

"I am."

"Then what is it doing?"

Weinstein was silent. Derek waited.

Finally Weinstein asserted:

"It's falling upward."

Nobody spoke. Trevor watched Penny. Penny regarded Derek.

Derek pointed out:

"Well."

Penny smiled.

"That settles it."

Trevor nodded.

"Gravity has become unreasonable."

Weinstein's strange voice came through the phone.

"No."

Derek fixed him with a look.

"What?"

"The apple isn't falling upward."

"Then what is happening?"

Weinstein watched through the train window.

"The train is."

Derek stared intently.

"The train is falling?"

"No."

"Moving?"

"No."

Weinstein paused.

"The train is changing."

Derek frowned.

"Changing how?"

"The geometry."

There was silence.

Then Derek clarified:

"Alfred."

"Yes?"

"Investigate the apple."

Weinstein smiled.

"I was hoping you'd say that."

The line went dead. Derek put the phone down. Trevor regarded him.

"So, we're investigating an apple."

Derek nodded.

"Apparently."

Penny picked up her coat.

"When?"

"Immediately."

Trevor's brow furrowed.

"Where?"

Derek examined the ancient brass clock. Then at the photograph. Then toward the door.

"Munich."

Penny smiled.

"Good."

Trevor still stood.

"How are we getting there?"

Derek put on his coat.

"Train."

Trevor inspected him.

"Which train?"

Derek paused.

"The one that's already arrived."

Trevor considered this.

"That sounds difficult."

Derek carefully opened the familiar door.

"Yes."

Penny followed him.

"That's why we have a Bureau."

They left. The quiet office remained empty. On the old desk lay Weinstein's forgotten notebook.

The last thing he had written was:

WHY DOES THE APPLE FALL?

Underneath, in smaller letters:

AND WHY DID IT CHOOSE TODAY?

The mysterious brass clock ticked once. Then twice. Then, very quietly, it began to run backwards.

The Shape of an Event

Weinstein did not immediately go to Munich. He went to Paris. This was not because Paris was on the way.

It wasn't. It was also not because he had suddenly developed an interest in French cuisine, although he had no objection to French cuisine and considerable objections to German railway sandwiches. He went to Paris because Derek Gent had gone to the Louvre.

This was, in itself, suspicious.

Derek had once described museums as "warehouses for dead people who had better PR than most living people."

He had never shown the slightest interest in art. He had once spent twenty minutes looking at a painting in the National Gallery before admitting that he had been facing it upside down.

He had also claimed that the Mona Lisa was “smiling because she knows something.”

Penny had asked him what.

Derek had replied:

“That’s what I’m trying to find out.”

So when Weinstein learned that Derek had abruptly travelled to Paris and gone to the Louvre, he did not waste much time wondering whether Derek had developed an appreciation for Leonardo. He assumed Derek had found something. And knowing Derek, it would be something weird.

Very weird. Weinstein was standing beside the Seine when the thought about the red apple returned. He watched the river.

Water slowly moved continuously beneath the bridges.

Yet there was no obvious thing called “the flow of time” visible anywhere.

There was only change. Water here. Water there.

A boat here. A boat there. A person crossing a bridge.

A person no longer crossing it. Weinstein frowned. Barbarian’s argument was irritating precisely because it was difficult to dismiss.

Time might not be a thing. It might be a way of describing relationships between things. That was not particularly controversial when expressed carefully.

Physics already did something similar. Position was not a substance. Velocity was not a substance.

Temperature was not a fluid. They were descriptions of physical relationships. Perhaps time was another.

But General Relativity complicated the matter. There was spacetime. Four coordinates.

Three spatial. One temporal. The metric.

Curvature. Worldlines. Proper time.

Weinstein stopped walking. He had been thinking about the apple as though it were following a line. But what if an event wasn’t really a line?

What if an event was something with structure? He considered the river. An ordinary apple falling from a tree was not merely a point moving through time. It had a history, a path, and a geometry.

Here. Then here.

It was a collection of positions and relationships. At every point, there was a location in space. And there was a temporal coordinate.

He took out his notebook. He drew three axes. X.

y. Z. Then, beside them, he wrote:

t. He stared at it. Then crossed out t.

Underneath, he wrote: τ . Proper time.

He stared intently again.

Then he added:

imaginary? He immediately frowned.

"No."

That was too careless. Imaginary time was useful mathematics. But that did not mean the universe literally contained an imaginary fourth direction.

A Wick rotation could turn a Lorentzian metric into a Euclidean-looking one. It was a mathematical transformation. It could make certain calculations beautifully behaved.

It did not mean clocks suddenly ran through the complex numbers. Weinstein smiled.

"Not yet, anyway."

He quietly closed the notebook. Then opened it again. There was something about the four-dimensional description that bothered him.

Not the mathematics. The interpretation. If an event was specified by three spatial coordinates and one temporal coordinate, then a complete event was a point in spacetime.

Fine. But the apple did not experience the whole spacetime diagram. The apple experienced its own worldline.

A sequence of events. Except Barbarian would object to the word sequence. Because sequence implied time.

Weinstein frowned. Then perhaps the worldline was not a thing moving through time. Perhaps it was simply the complete geometric object.

A curve. All at once. He stopped again.

That was uncomfortable.

If the complete worldline existed as a geometrical object, then asking where the apple “was going” was merely a way of describing the relationship between different points on the curve.

The red apple did not need to know about the future. The geometry did. Weinstein raised his eyes.

The Eiffel Tower was visible in the distance. He ignored it. This was more interesting.

Suppose the universe could be represented as a four-dimensional structure.

Then “past”, “present” and “future” might be properties of how an observer experienced different parts of that structure.

Not necessarily ingredients of the structure itself. He wrote:

THE EVENT DOES NOT HAPPEN IN TIME.

He paused.

Then added:

THE EVENT IS IN SPACETIME.

He stared at the sentence. That was better. Much better.

And suddenly he understood why the clocks in London and Munich were so troubling. They might not be disagreeing about time. They might be sampling different geometrical relationships.

The clocks were innocent. The geometry was not. Weinstein put the notebook away.

Then his phone rang. It was Derek.

Weinstein answered.

“Where are you?”

“Paris.”

“I know.”

A pause.

“You know?”

“Yes.”

“How?”

“You’re at the Louvre.”

Another pause.

“How do you know that?”

“Penny.”

"Of course."

"Why are you there?"

"Because I found something."

"What?"

"I'll tell you when you get here."

"I'm already in Paris."

"Good."

"Where are you?"

"Inside."

"The Louvre?"

"Yes."

Weinstein glanced at the enormous building.

"Why?"

"Because I think Leonardo knew something."

Weinstein quietly closed his eyes.

"About what?"

"Time."

Weinstein opened his eyes.

"Leonardo da Vinci?"

"Yes."

"The painter?"

"Among other things."

Weinstein wearily sighed.

"All right."

"Come inside."

"I am."

"Excellent."

The line went dead. Weinstein entered the Louvre. It took him less than ten minutes to find Derek.

He was standing in front of a painting. Not the Mona Lisa. A much less famous work.

Derek was staring at it with the concentration of a man trying to solve a murder. Weinstein approached.

"What are you looking for?"

Derek didn't turn.

"Geometry."

"Of the painting?"

"Of the event."

Weinstein stared at the painting. It was beautiful. It was also completely irrelevant.

Probably.

"What did you find?"

Derek pointed.

"Look at the perspective."

Weinstein looked.

"Yes."

"Notice anything?"

"No."

"Good."

Weinstein gazed at him.

"Good?"

"Yes. I wanted to know whether you'd see it immediately."

"See what?"

"The problem."

Weinstein examined the painting again. Then he understood.

"The vanishing point."

Derek smiled.

"Exactly."

Weinstein looked closer.

"That's not unusual."

"No."

"But the way the scene is constructed..."

"Yes."

"It isn't simply perspective."

"No."

"It's a coordinate system."

Derek nodded.

"Just so."

Weinstein stared at the painting. The objects appeared to converge toward a point. A mathematical point.

Not a physical object. A representation. A transformation.

A way of describing a three-dimensional scene on a two-dimensional surface. He smiled.

"Interesting."

Derek finally turned.

"Now look at the other one."

They suddenly moved to another painting. Then another. Derek had marked nothing.

He didn't need to. He simply pointed. Weinstein looked.

The same conceptual structure suddenly appeared. Not identical. But related.

Different projections. Different perspectives. Different ways of representing the same kind of spatial relationships.

Weinstein suddenly understood why Derek had brought him there.

"A coordinate system depends on the observer."

"Yes."

"And the underlying scene doesn't."

"Usually."

Weinstein gave him a measured look.

"Usually?"

Derek smiled.

"We're dealing with my murder."

"Fair enough."

They moved to a quieter gallery. Derek stopped.

"Now imagine that the paintings aren't two-dimensional."

"Go on."

"Imagine each painting is a slice through a three-dimensional object."

Weinstein nodded.

"Yes."

"Different observers see different slices."

"Yes."

"But the object itself is larger than any one slice."

"Yes."

Derek eyed him.

"And now add time."

Weinstein became still.

"Three dimensions of space."

"Yes."

"And one temporal dimension."

"Yes."

"Then an event is not merely a point moving through time."

"No."

"It's a point in four-dimensional spacetime."

"Yes."

Derek smiled.

"Well, that settles it."

Weinstein watched him.

"What does?"

"We've finally found the shape of the problem."

Weinstein shook his head.

"Not quite."

"Why?"

"Because you're still thinking of time as the fourth dimension in the same way as the three spatial dimensions."

Derek sharply frowned.

"What's wrong with that?"

"Nothing mathematically."

"Then?"

"The metric."

Derek waited. Weinstein continued.

"The temporal coordinate doesn't behave like an ordinary spatial coordinate. The sign is different."

Derek nodded.

"Lorentzian geometry."

"Exactly."

"So, not an ordinary four-dimensional Euclidean space."

"No."

"Spacetime has a different geometry."

"Yes."

Derek returned his attention at the painting.

"Could we represent it visually?"

"Only imperfectly."

"Why?"

"Because we're trying to draw four dimensions on three-dimensional paper."

Derek smiled.

"That sounds like the sort of thing the Bureau was designed for."

Weinstein laughed. Then Derek's expression changed.

"There's something else."

"What?"

"Look at the people."

Weinstein did.

"Yes?"

"They aren't simply objects in space."

"No."

"They are events."

Weinstein surveyed him.

"Careful."

"Why?"

"Because a person is a worldline, not a single event."

Derek smiled.

"Precisely."

"And you're about to tell me you've found something."

"I have."

"What?"

Derek pointed toward the gallery.

"That man."

A security guard stood near the doorway. Weinstein looked.

"Yes?"

"He walked past us three times."

"So?"

"He walked past us three times without ever appearing in the same place twice."

Weinstein frowned.

"That's normal."

"No."

"Why?"

"Because the Louvre corridor is a straight line."

Weinstein looked. It was. Derek continued.

"I watched him."

"So?"

"He passed us."

"Yes."

"Then he passed us again."

"Yes."

"Then again."

"Yes."

"And each time he was carrying a different newspaper."

Weinstein stared.

"That's odd."

"Very."

"Could be different guards."

"No."

"How do you know?"

"Same face."

"Same clothes?"

"Exactly."

Weinstein studied the guard again. The man turned. For a fraction of a second, his eyes met Weinstein's.

Then he walked away.

Derek mentioned quietly:

"That's our man."

Weinstein frowned.

"Who?"

"I don't know."

"That's rather inconvenient."

"Yes."

Derek smiled.

"But I know where he goes."

"Where?"

Derek pointed toward a staircase.

"Down."

They followed. The staircase led beneath the museum. The further they went, the older the building became.

Eventually they reached a corridor with no tourists. No paintings. No signs.

Only stone walls. At the end still stood a door. Derek stopped.

"That's odd."

"What?"

"There wasn't supposed to be a door here."

Weinstein regarded it.

"Are you sure?"

"I've been here before."

"Why?"

"Long story."

"Fair enough."

Derek opened the heavy door. Inside was a small room. And in the middle of it stood a table.

On the table was a strange clock. Brass. Railway style.

11:03:17. Weinstein stared intently.

"The same clock."

"Yes."

"How did it get here?"

Derek shook his head.

"I don't know."

Weinstein approached. There was something beside the silent clock. A sheet of paper.

He picked it up. Three coordinates were written on it. $X = 3$

$y = 4$ $z = 12$ Then:

$\tau = ?$ Weinstein stared. Derek came closer.

"Well?"

Weinstein softly smiled.

"Now we have a problem."

Derek nodded.

"Good."

"Why?"

"Because problems are what we investigate."

Weinstein examined the silent clock. Then the coordinates.

Then the empty space where the fourth coordinate should have been.

He responded:

"The three spatial coordinates are specified."

"Yes."

"But proper time isn't."

"No."

"Someone has given us a three-dimensional event."

Derek frowned.

"An event needs four coordinates."

"Exactly."

"So, whoever left this knows that."

"Yes."

Derek inspected the silent clock.

"And they're asking us to find the fourth."

Weinstein nodded.

"Yes."

Penny's calm voice suddenly came from the doorway.

"Then I suppose we ought to."

They both turned. Penny stood there. Trevor was behind her, looking rather pleased with himself.

Derek stared.

"How did you get here?"

Penny smiled.

"Train."

Derek looked at Trevor.

"And you?"

"Also train."

Derek nodded.

"Right. Next."

Penny glanced at the clock.

"No."

"What?"

"That's not the weird part."

She pointed at the paper.

"The weird part is the date."

Weinstein dropped his eyes. He hadn't noticed it. At the bottom of the page was a date.

Tomorrow. Derek stared at Penny. Then at Trevor.

Then at Weinstein. Weinstein slowly smiled.

"Now," he opined, "we're getting somewhere."

And above them, in the Louvre, a clock struck twelve. Below them, the brass railway clock struck eleven seventeen. For the first time, Weinstein wondered whether the problem was not that the clocks disagreed about time. It was that reality had decided to play the pot calling the kettle fabulous.

Perhaps they disagreed about which event they belonged to.

Chapter 6

The Fourth Coordinate

For several seconds nobody offered anything. The room beneath the Louvre was small, stone-walled and entirely too quiet. There was a table.

There was a clock. There were four people. And there was a piece of paper asking them for a value of τ .

Derek allowed himself a brief, private observation: this was becoming a pattern. A city, a clock, a number that refused to add up. He kept the observation to himself, mostly because Trevor would have agreed, and agreement on that particular point was unlikely to help anyone. What was new, this time, was the rate. In Munich the discrepancy had grown by roughly a seventeenth of a second per hour. Here, beneath the Louvre, it was growing by nearly a full second. Whatever was causing this, it was not slowing down.

Trevor was the first to speak.

"That's proper time."

Weinstein nodded.

"Yes."

Derek gazed at the paper.

"Which means?"

Trevor pointed at the three numbers.

"Three spatial coordinates have been supplied. The fourth coordinate is missing."

"Correct."

"And they're asking us to calculate it."

"Apparently."

Trevor considered the problem.

"Do we know the metric?"

"No."

"The worldline?"

"No."

"The gravitational field?"

"No."

"The initial conditions?"

"No."

Trevor peered at the paper again.

"Then we can't calculate it."

He paused.

"Unless that's the point."

Derek looked up.

"What do you mean?"

"If whoever wrote this knew we couldn't solve it without the missing information, they weren't actually asking us to calculate τ . They were showing us that τ is missing. That's not a question. That's a diagnosis."

Derek studied him for a moment.

"That's rather good, Trevor."

Trevor looked pleased, and then faintly suspicious of looking pleased.

"Thank you."

Penny, however, had not moved. She was looking at the silent clock.

"There's something else."

Derek followed her gaze.

"What?"

"Look at the time."

"11:17."

"Precisely."

"Yes."

"That's when the clocks stopped."

Derek nodded.

"And restarted."

"At 12:04."

"Yes."

Penny pointed to the paper.

"Then perhaps τ isn't asking us for a number."

Trevor turned to her.

"What else could it ask for?"

"An interval."

Trevor frowned.

"Between what?"

Penny pointed at the mysterious clock.

"Two events."

Weinstein smiled.

"Exactly."

Derek slowly watched him.

"So, the missing coordinate isn't necessarily a timestamp."

"No."

"It's proper time between events."

"Perhaps."

Trevor studied the paper again.

"Then we'd still need the worldline."

"Yes."

"We don't have one."

"We have several."

Trevor raised his eyes.

"Whose?"

Weinstein pointed at the silent clock.

"The clock."

Trevor nodded.

"Right."

Derek picked up the clock.

"Then let's start with the obvious."

He turned it over.

"How does this thing know when to stop?"

Nobody answered.

He examined the mechanism. There was nothing unusual. No obvious electronics.

No radio receiver. No battery. No hidden switch.

The brass casing was old and scratched. The movement was mechanical. Penny took it from him.

"May I?"

"Of course."

She opened the casing. Trevor leaned over. Weinstein watched.

The mechanism was intricate. Beautiful, even. Tiny gears.

Springs. Jewelled bearings. A balance wheel.

An escapement. Penny examined it for several seconds.

"There's nothing here."

Derek smiled.

"That's becoming a recurring feature."

"I mean literally."

She held the movement toward the light.

"There's no mechanism capable of producing the behaviour we've seen."

Trevor nodded.

"Then the clock isn't doing it."

Penny surveyed him.

"What?"

"The clock isn't deciding when to stop."

"Go on."

"It's responding to something."

Derek nodded.

"Exactly."

Weinstein took the clock.

"The clock is a physical system."

"Yes."

"It has a worldline."

"Yes."

"It measures proper time along that worldline."

"Yes."

"So if it stops while everything else continues..."

"Then something about the physical conditions along its worldline has changed."

Trevor turned his attention to him.

"Or the clock itself has changed."

"Yes."

"Could it have been accelerated?"

Weinstein nodded.

"In principle."

"Enough to create a significant time dilation?"

"Not without enormous acceleration."

"How enormous?"

"Enough that you'd notice."

Derek smiled.

"That seems useful."

"Yes."

Penny regarded the clock.

"But the clock didn't move."

"Not in any ordinary sense."

"So, acceleration isn't the explanation."

"Not ordinary acceleration."

Trevor thought.

"Gravity?"

Weinstein nodded.

"Possibly."

Derek pointed toward the ceiling.

"We're beneath the Louvre."

"Yes."

"Could there be something beneath us?"

Penny examined him.

"Such as?"

"A very large mass."

Trevor smiled.

"How large?"

"Large enough to stop clocks."

"That would be rather noticeable."

"Probably."

Weinstein shook his head.

"Not necessarily."

Everyone quietly inspected him. He continued.

"A gravitational field doesn't have to announce itself by dropping buildings on people."

"Fair."

"And gravitational time dilation depends on the metric, not merely on some crude idea of gravitational force."

"Meaning?"

"Two clocks at different gravitational potentials can accumulate different proper times."

Trevor nodded.

"But not by forty-seven minutes in an ordinary room."

"Correct."

Derek considered the clock.

"Then we're back to the same problem."

"Yes."

He put the old clock down.

"Unless..."

Nobody spoke. Derek glanced at the paper. Three spatial coordinates.

A missing proper time. He walked to the wall.

"What if the room itself is the coordinate system?" Derek stared at the doorway. "Then we've been barking up the wrong sausage."

Trevor frowned.

"Explain."

Derek pointed to the floor.

"We're somewhere in three-dimensional space."

"Yes."

"Suppose the numbers describe a point."

"Yes."

"Then the missing τ describes which event occurs at that point."

Trevor nodded.

"Yes."

"So, perhaps the paper isn't telling us where to look."

Penny finished:

"It's telling us when."

Derek smiled.

"Exactly."

Weinstein shook his head.

"Not quite."

Derek turned.

"What?"

"The word 'when' is precisely what we should avoid."

"Why?"

"Because we're trying to determine whether the events are ordered by a fundamental time or merely related geometrically."

Derek nodded.

"Fair."

Penny smiled.

"So what should we call it?"

Weinstein studied the equation.

"Which event."

There was silence. Trevor nodded slowly.

"That's better."

Weinstein continued.

"Suppose the point $x = 3$, $y = 4$, $z = 12$ corresponds to a particular location."

"Yes."

"Then τ identifies a particular event along a worldline."

"Yes."

"But if the geometry is unusual, there may be more than one relevant event associated with that spatial location."

Derek stared at him.

"More than one?"

"Yes."

"At the same place?"

"Perhaps."

"How?"

"Different times."

Derek smiled.

"That's a rather circular answer."

"Only if you assume time is fundamental."

Derek laughed.

"Fair enough."

Trevor walked over to the wall. He took the piece of paper and copied the numbers. Then he drew three axes.

x. Y. Z.

He marked the point.

"Here."

Derek nodded.

"Now what?"

Trevor added a fourth axis.

τ .

"Here."

He drew a curve.

"The clock's worldline."

Weinstein nodded.

"Yes."

Trevor continued.

"If the clock stops at one point and restarts at another..."

"Then its worldline has a strange section."

"Quite."

Penny leaned over.

"Or we've misunderstood what stopping means."

Trevor stared at her.

"What do you mean?"

"If the hands stop but the clock's physical state continues changing..."

Trevor's eyes widened.

"Then the clock's reading has stopped, but the clock hasn't."

"Yes."

Derek studied the movement.

"That's interesting."

Trevor nodded.

"The clock could be experiencing proper time normally while its display remains fixed."

Weinstein smiled.

"That is much more interesting."

Derek gazed at Penny.

"Can we test it?"

Penny picked up the silent clock.

"Probably."

"How?"

"Compare the mechanism's state before and after."

Trevor nodded.

"Balance wheel frequency."

"Yes."

"Escapement."

"Yes."

"Gravitational potential."

"Yes."

Derek looked impressed.

"How long?"

Penny shrugged.

"Ten minutes."

Trevor smiled.

"We'll need a laboratory."

Penny surveyed the room.

"We have one."

Derek surveyed the room.

"This is the Louvre."

"Yes."

"Not CERN."

"No."

"Then where exactly is the laboratory?"

Penny pointed at the old clock.

"Here."

Derek smiled.

"Of course."

They spent the next half hour examining the silent clock. There was nothing supernatural about it. That was, in some ways, the most unsettling thing.

The balance wheel oscillated normally. The spring behaved normally. The escapement behaved normally.

The gears moved. Everything mechanical suddenly appeared perfectly ordinary. And yet the silent clock had stopped.

Weinstein watched the movement.

"There's no anomaly."

Trevor nodded.

"None."

Derek looked at the face.

"Then why did it stop?"

Penny turned the clock over.

"It didn't."

Derek frowned.

"The hands stopped."

"Yes."

"But the movement didn't."

"Correct."

Trevor looked excited.

"That means the hands aren't coupled to the movement in the ordinary way."

Derek watched him.

"How?"

Trevor examined the mechanism.

"There must be an intermediate mechanism."

"Which we can't find."

"Correct."

Weinstein nodded.

"Then perhaps it isn't mechanical."

Derek surveyed the clock.

"It certainly looks mechanical."

"Appearance isn't evidence."

Penny smiled.

"That's something Derek usually maintained."

Derek looked offended.

"I say it more elegantly."

"Naturally."

Weinstein turned the silent clock over again. The initials were still there.

LGV.

He stared at them. Then he took out his notebook.

"LGV."

Derek nodded.

"We've seen it before."

"Yes."

"What does it mean?"

"We don't know."

Weinstein wrote the letters. Then, beneath them, he wrote:

LOCAL?

GRAVITATIONAL?

VECTOR?

Penny studied the page.

"That's what Trevor suggested."

Weinstein regarded Trevor.

"Did you?"

Trevor nodded.

"Yesterday."

"Interesting."

Trevor smiled.

"Thank you."

Weinstein looked again.

"Local Gravitational Vector."

Derek shook his head.

"No."

"Why?"

"Because a vector requires a direction."

"Yes."

"And we've got no indication that the initials represent one."

"Fair enough."

Penny affirmed:

"Could be initials."

"Whose?"

Nobody knew. Weinstein examined the clock. Then at the paper.

Then at the three coordinates. He suddenly stood.

"Let's go upstairs."

Derek inspected him.

"Why?"

"Because I want a map."

"Of the Louvre?"

"Yes."

They returned to the main museum. A staff member directed them toward a map of the building. Weinstein examined it.

Then the underground room. Then the coordinates. He frowned.

"These aren't Louvre coordinates."

Derek leaned closer.

"How do you know?"

"Because the scale is wrong."

Trevor studied the map for a long moment, tracing one finger slowly along the river.

"Then what are they?"

Weinstein took a pencil. He marked three points.

"Not ordinary Cartesian coordinates."

"Then?"

"Relative coordinates."

Derek glanced at him.

"Relative to what?"

Weinstein stared at the map. Then smiled.

"That's the question."

Penny pointed at the silent clock.

"Could the clock be the origin?"

Weinstein nodded.

"Possibly."

Trevor stared at the numbers.

"Three, four, twelve."

He paused.

"That's a Pythagorean triple."

Derek gazed at him.

"What?"

Trevor pointed.

"Three squared plus four squared equals five squared."

Then he added:

"Five squared plus twelve squared equals thirteen squared."

Weinstein peered at the numbers. Derek smiled.

"So?"

Trevor shrugged.

"Nothing necessarily."

Penny nodded.

"Don't force patterns."

Trevor nodded.

"I wasn't."

Derek met his gaze.

"You absolutely were."

"Perhaps slightly."

Weinstein watched the coordinates again, tapping the pencil twice against the desk before setting it down.

"Three, four, twelve."

He wrote: 13. Then:

5. Then: 12.

He stared intently.

"Thirteen."

Derek waited.

"What about it?"

"The spatial distance from the origin is thirteen."

"Assuming Euclidean space."

"Yes."

"And?"

Weinstein surveyed the silent clock.

"The missing coordinate may be related to the invariant interval."

Derek's expression changed.

"Go on."

Weinstein took the pencil.

"In Special Relativity, the spacetime interval is invariant."

He wrote: $ds^2 = -c^2dt^2 + dx^2 + dy^2 + dz^2$ Trevor nodded.

"Depending on sign convention."

"Yes."

Weinstein continued.

"If we know the spatial separation..."

He pointed at the coordinates.

"... and the interval..."

He looked at the missing τ .

"... we can infer the temporal relationship."

Derek nodded.

"But we don't know the interval."

"No."

"Then?"

Weinstein smiled.

"Perhaps we do."

He pointed toward the Louvre.

"The clock."

Derek frowned.

"The clock gives us proper time."

"Yes."

"So, τ is the proper-time coordinate."

"Yes."

"And the spatial distance is thirteen."

"Yes."

"So..."

Weinstein stopped.

"Nothing."

Derek laughed.

"That's a very dramatic pause for nothing."

"We're missing one more piece."

"What?"

"The nature of the interval."

Penny nodded.

"Timelike, spacelike or null."

Weinstein regarded her.

"Exactly."

Trevor smiled.

"If the interval is timelike, the events can be causally connected."

"Yes."

"If spacelike, they can't."

"Correct."

"If null, they're connected by light."

"Yes."

Derek examined the silent clock.

"Then we need to determine which one this is."

Weinstein nodded.

"And that may tell us what the clock is doing."

Derek smiled.

"Right, that's that, then."

Trevor inspected him.

"What?"

"We have a method."

Penny nodded.

"And a destination."

Derek frowned.

"Destination?"

She pointed toward the map.

"Those coordinates."

Weinstein looked. They corresponded to a location in the museum. A gallery on the upper floor.

Derek softly smiled.

"Right."

They climbed the stairs. The gallery was crowded. Visitors moved between paintings.

Tour guides spoke quietly. Cameras suddenly appeared everywhere. Nothing looked unusual.

Derek checked the coordinates.

"We're here."

Weinstein looked around.

"Where?"

"Somewhere in this room."

Trevor checked the numbers again.

"Three metres east."

"From what?"

Trevor looked embarrassed.

"Right."

Derek wearily sighed.

"Origin."

"Yes."

"Which is?"

They considered the clock. Then at one another. Penny pointed.

"The Louvre entrance."

They walked. The three coordinates took them toward a large painting. Derek stopped.

"This is it."

Weinstein studied the painting. It showed a landscape. A road.

Trees. Mountains. A distant building.

Nothing unusual. Then he noticed the perspective. The vanishing point.

The same feature Derek had noticed earlier. Weinstein stepped closer.

"The coordinate system."

Derek nodded.

"Yes."

Trevor glanced at the painting.

"The spatial geometry is encoded in the perspective."

"Exactly."

Penny pointed at the horizon.

"And there."

A tiny ancient clock was visible in the distant building. Weinstein stared intently. It showed 12:04.

Derek took the ancient brass clock from his pocket. Its hands had suddenly moved. 12:04.

Trevor checked his phone. 12:04. Penny checked hers.

12:04. The gallery strange clock showed 12:04. For several seconds, nobody spoke.

Then Derek confirmed:

"Well, that's settled."

Weinstein stared at him.

"What is?"

"We've found the event."

Trevor smiled.

"The fourth coordinate."

Weinstein shook his head.

"No."

They gazed at him. He pointed at the painting.

"We've found the projection of the event."

Derek stared at the painting.

"And the event itself?"

Weinstein returned his attention to the silent clock.

"That may be somewhere else."

The painting suddenly changed. Not dramatically. Not with a flash.

Nothing so vulgar. The tiny clock in the distant building simply moved. 12:04.

12:05. Then the entire scene shifted. The road was different.

The trees were different. The building was gone. The people in the gallery continued talking.

Nobody else seemed to notice. Derek stared.

"Did you see that?"

Penny nodded.

"Yes."

Trevor nodded.

"Yes."

Weinstein smiled.

"Good."

Derek turned to him.

"Why are you pleased?"

"Because now we know."

"Know what?"

Weinstein pointed at the painting.

"The event isn't moving through time."

Derek waited.

"It's changing the geometry."

The brass clock in his hand began ticking backwards. And somewhere beneath Paris, another clock began ticking forwards. The two clocks were now measuring different proper times.

Neither was wrong. And, for the first time, Weinstein understood why the question on the paper had not been:

WHEN?

It had been: τ =? The answer was not a date.

It was a path. A worldline. And somebody had just drawn them one.

Weinstein did not leave the gallery. Everyone else seemed to have moved on. Visitors wandered past the painting.

A guide explained something about perspective to a group of tourists.

Someone took a vivid photograph. A child complained that he was hungry. The universe continued behaving as though nothing particularly remarkable had happened.

Weinstein remained in front of the painting. He was thinking. Derek knew better than to interrupt him.

Penny knew better still. Trevor, after several years of experience, had learned that there were two kinds of silence from Weinstein. There was the ordinary silence.

And there was the dangerous silence. This was the second kind. Weinstein watched the painting.

Then at the ancient brass clock. Then at the three coordinates. Three spatial dimensions.

One temporal coordinate. A geometry. A worldline.

An invariant interval. And curvature. He suddenly took out his notebook.

Derek watched.

"What have you got?"

Weinstein didn't answer. He drew three axes. Then curved them.

Derek leaned closer.

"What is that?"

"Space."

"That's not very flat."

"No."

"Should it be?"

Weinstein smiled.

"That's the question."

He drew another curve. Then another. He began to imagine spacetime itself.

Not a rubber sheet. Not a picture of a planet sitting in a depression. Something more fundamental.

A four-dimensional geometry. The curvature was not necessarily in some larger space. The geometry itself was curved.

An observer moving through it would experience that geometry as reality. Weinstein wrote:

CURVATURE.

Then underneath:

MORE CURVATURE.

Then:

HOW MUCH IS TOO MUCH?

He stopped. Derek slowly read it.

"That's rather philosophical."

"No."

Weinstein lifted his gaze.

"It's physics."

"Fair enough."

Weinstein continued.

"If spacetime can be curved..."

"Yes."

"And matter and energy determine that curvature..."

"Yes."

"Then there must be some limit."

Trevor frowned.

"Why?"

Weinstein surveyed him. Trevor waited. Weinstein considered the question.

"Perhaps there isn't."

Trevor nodded.

"Right."

Weinstein continued.

"But suppose there is."

Derek smiled.

"That's rather different."

"Yes."

Weinstein drew a curve that became increasingly steep.

"Imagine curvature increasing."

"Yes."

"More and more."

"Yes."

"At some point our ordinary description becomes inadequate."

Penny nodded.

"Because the equations stop giving sensible physical answers."

"Perhaps."

"Or because an invariant diverges."

"Yes."

"Or because a geodesic can't be continued."

"Yes."

Weinstein studied the drawing.

"And that..."

He paused.

"... is where we call it a singularity."

Trevor regarded him.

"You're defining a singularity as 'too much curvature'?"

Weinstein shook his head.

"No."

He crossed out the phrase.

"A singularity isn't simply a region where spacetime is very curved."

He wrote:

BREAKDOWN OF THE CLASSICAL DESCRIPTION.

"That's better."

Derek nodded.

"Much."

Weinstein looked again at his drawing.

"But the intuitive idea remains."

"Which is?"

"If curvature increases without bound, the classical theory can cease to provide a meaningful description."

"Yes."

"And that's precisely the sort of thing we're looking for."

Derek's expression changed.

"Why?"

Weinstein pointed at the ancient brass clock.

"Because we have a clock behaving as though it has crossed from one geometry into another."

Penny frowned.

"Or stopped measuring the same proper time."

"Exactly."

Trevor examined the painting.

"So, you're thinking there's an extreme gravitational region somewhere."

"Perhaps."

"Near Paris?"

"No."

"Then where?"

Weinstein smiled.

"That's what Derek is going to investigate."

Derek wearily sighed.

"Of course I am."

Weinstein inspected him.

"You went to the Louvre."

"Yes."

"Why?"

"Because I followed a clue."

"Right."

"You knew that."

"I suspected."

"And now?"

"Now I suspect you found something else."

Derek smiled.

"I did."

"What?"

Derek pointed at the painting.

"Look at the landscape."

Weinstein did.

"The geometry changes."

"Yes."

"Not the painting?"

"No."

"The projection."

"Just so."

Derek took out a photograph. It showed the same painting. But the perspective was subtly different.

The vanishing point had shifted. Weinstein studied both images.

"The projection has changed."

"Yes."

"Without the underlying objects changing."

"Yes."

Weinstein frowned.

"That's impossible."

Derek smiled.

Penny immediately insisted:

"Don't."

Weinstein laughed.

"Fine."

He looked again.

"If the projection changes while the objects remain..."

He stopped.

"The coordinate system has changed."

"Yes."

"And if the coordinate system changes..."

"The metric might."

Weinstein nodded.

"Exactly."

Trevor looked between them.

"So, we're not looking at a strange painting."

"No."

"We're looking at a change in the geometry used to describe it."

"Yes."

Trevor thought.

"Then perhaps the painting isn't showing us a place."

Weinstein turned.

"What is it showing us?"

Trevor pointed.

"An event."

There was silence. Derek smiled.

"Good. Moving on."

Penny nodded.

"Good."

Weinstein turned to Trevor.

"You may keep him."

Derek laughed. Trevor smiled.

"I'll take that as a compliment."

"You should."

Weinstein suddenly turned back to the painting. The idea of a curved event returned to him. An event wasn't merely a point.

It was a point in spacetime. But if spacetime itself was curved, then the relationship between neighbouring events was not Euclidean. The geometry mattered.

The interval mattered. The worldlines mattered. And if the curvature became extreme enough, the familiar description could break down.

Weinstein wrote one word.

SINGULARITY.

Derek read it.

"That's a rather dramatic word."

"It's precise."

"Is it?"

"In the appropriate mathematical sense."

Derek smiled.

"Well, that's reassuring."

Weinstein continued writing.

"Black holes."

"Yes."

"Cosmological singularities."

"Yes."

"Geodesic incompleteness."

"Yes."

Derek glanced at him.

"You're thinking about the clock."

"Yes."

"Could a clock pass near a singularity?"

Weinstein quietly stared at the brass clock.

"Not in any ordinary sense."

"Why?"

"Because a singularity isn't simply a very strong gravitational field that you can safely approach."

"Fair."

"The classical spacetime description itself becomes pathological."

"Meaning?"

"You can't simply ask what the clock does at the singularity. And you especially can't ask what it does if the singularity is only one end of a wormhole."

Derek nodded.

"Because the theory has stopped telling us."

"Exactly."

Penny gazed at the clock.

"Then perhaps the clock isn't near one."

Weinstein nodded.

"Perhaps."

"Perhaps it's telling us that the geometry between two events has become extreme."

"Yes."

Trevor studied the three coordinates.

"Then the missing τ could be the proper-time difference along a worldline approaching that region."

Weinstein turned sharply.

"Yes."

Trevor continued.

"And if the proper time remains finite while something else becomes infinite..."

"Then we could have a singular behaviour."

"Yes."

Weinstein smiled.

"Very good."

Trevor brightened. Derek peered at Penny.

"He's becoming dangerous."

"He was always dangerous."

Trevor ignored them.

"So what do we do?"

Weinstein quietly closed his notebook.

"We find out where the curvature becomes extreme."

Derek nodded.

"How?"

Weinstein eyed the painting.

"The same way we found the event."

"The coordinates?"

"Yes."

"And the clock?"

"Yes."

"And the Louvre?"

"Yes."

Derek cast a quick look around.

"Then we've got a problem."

"What?"

"We don't have enough coordinates."

Weinstein smiled.

"No."

He pointed at the painting.

"We have four."

Derek looked. Three numbers had been written on the paper.

He had assumed the fourth coordinate was missing.

It wasn't. A tiny number was visible beneath the ancient brass clock in the painting. It had been almost completely hidden by the frame.

Derek leaned closer.

"What's that?"

Weinstein looked. The number was: Trevor stared intently.

"Thirteen."

Penny nodded.

"That's the distance."

Weinstein shook his head.

"No."

"Why not?"

"Because we're assuming Euclidean space."

Trevor nodded.

"Right."

Weinstein took out his pencil.

"If this is spacetime, the invariant interval isn't simply the Euclidean distance."

He wrote: $ds^2 = dx^2 + dy^2 + dz^2 - c^2 dt^2$ Then stopped.

"Unless we're using the opposite sign convention."

Derek smiled.

"Does it matter?"

"Not for the physical content."

"Well, that's settled."

Weinstein laughed. He looked again at the painting. Three.

Four. Twelve. Thirteen.

And beneath them, almost invisible: ∞ Derek stopped smiling.

Penny saw it. Trevor saw it. Weinstein stared intently.

"That's not a coordinate."

"No," Derek emphasized.

"What is it?"

Weinstein watched the symbol.

"Infinity."

Trevor frowned.

"That would suggest a divergent quantity."

"Yes."

"Curvature?"

"Possibly."

"Proper time?"

"Possibly."

"Interval?"

"Maybe."

Derek surveyed him.

"Which?"

Weinstein didn't answer. He was staring at the painting. Then he smiled very slowly.

"Now we have our singularity."

Derek gave Penny a measured look. Penny regarded Trevor. Trevor examined Weinstein.

And somewhere, far beneath the Louvre, an ancient brass clock began ticking. Not forward. Not backward.

But sideways. At least, that was what the silent clock appeared to think it was doing.

Derek did the arithmetic without quite meaning to. If the discrepancy kept growing at this rate, by the end of the week the clocks would not merely disagree about the time. They would start disagreeing about the day.

Chapter 7

The Cold Rain, the Underground and the Problem with Poetry

Derek Gent had been given a difficult assignment. He was to investigate the London Underground on rainy days.

This was not, in itself, unusual. The London Underground was already difficult on most days. On rainy days it became something else entirely.

People arrived wet, irritated and vaguely convinced that the entire transport system had been personally designed to inconvenience them. Trains became late. Platforms became slippery. Umbrellas became weapons. Shoes acquired the consistency of soup. Derek had been instructed to find out whether anything in the Underground, specifically on rainy days, could prevent Special Relativity and General Relativity from both being true at the same time.

He had read the instruction three times. Then he had called Weinstein.

"Alfred."

"Yes?"

"Are you aware that you've sent me to investigate the London Underground?"

"Yes."

"On rainy days."

"Yes."

"To determine whether it contains something that makes Special Relativity and General Relativity mutually incompatible."

"Yes."

Derek paused.

"Fine. Next problem."

"Good."

"One question."

"Yes?"

"Why?"

Weinstein had been silent for several seconds.

Then he had queried:

"Because time doesn't exist."

Derek had waited.

"That's your explanation?"

"Not quite."

"What is the rest?"

Weinstein had wearily softly sighed.

"I'm beginning to suspect that time is a bookkeeping device."

"Like accounting?"

"Exactly."

"That sounds dangerous."

"It is."

"And the Underground?"

"Also dangerous."

Derek had considered this.

"Anything else?"

"Rain."

"Right."

And that was how, at 4:18 on a wet Thursday afternoon, Derek Gent found himself standing on a London Underground platform with a notebook in one pocket, a train ticket in the other, and an increasingly damp left shoe. He looked up at the distant station sign. He inspected the rainwater dripping from the entrance.

He considered the arriving empty train. Then he glanced at his notebook. At the top of the page he had written:

INVESTIGATION 1047-B

Underneath:

QUESTION:

Can London Underground infrastructure, under rainy conditions, prevent SR and GR from simultaneously describing reality? Under that he had written:

SECONDARY QUESTION:

Why? And underneath that:

THIRD QUESTION:

What the hell does poetry have to do with it? Derek closed the notebook. A woman standing beside him glanced at the page.

She stared at him. Then at the page. Then back at him.

“Work?”

Derek nodded.

“Unfortunately.”

She smiled sympathetically.

“Important?”

“Apparently.”

She stepped onto the empty train. Derek followed. The train doors quietly closed.

The carriage moved. Derek sat down. Across from him, a man was reading a newspaper.

The headline concerned the weather. Derek gazed at it. Then at the cold rain running down the window.

Then at his notebook. He quietly opened it again. He wrote:

Start with SR. Special Relativity was not difficult in principle. Two observers in relative inertial motion could assign different coordinates to the same event.

Time intervals and spatial intervals could differ between frames. The speed of light remained invariant.

The Lorentz transformation connected the descriptions.

Nothing about that contradicted ordinary experience. It merely contradicted the assumption that everyone was entitled to the same old clock. General Relativity went further.

Gravity was not merely a force acting across space. Spacetime itself had geometry. Matter and energy influenced that geometry.

Freely falling objects followed geodesics. Clocks followed worldlines and measured proper time along them. Again, nothing inherently contradictory.

Indeed, General Relativity contained Special Relativity locally. In sufficiently small regions of spacetime, where curvature could be neglected, freely falling observers recovered the familiar rules of SR. Derek stopped writing.

That was the obvious answer. SR and GR weren't enemies. GR contained SR as its local limit.

So why had Weinstein sent him into the Underground? Derek looked at the wet floor. A drop of water rolled along the rubber mat.

He watched it. It moved. Stopped.

Moved again. He smiled. Perhaps the question wasn't whether SR and GR could both be true.

Perhaps the question was whether the way they described time could be reconciled with the strange behaviour they were observing. The train entered a narrow tunnel. The lights flickered.

Derek eyed his watch. Normal. He checked the empty station strange clock.

Normal. He checked the silent clock on the train. Normal.

Three clocks. Three agreements. He wrote:

Everything disappointingly normal. The train stopped.

A recorded voice announced the empty station.

Derek lifted his gaze. He had been expecting something. He wasn't sure what.

Perhaps the carriage would suddenly become six-dimensional. Perhaps the passengers would all age at different rates. Perhaps a small black hole would appear near the emergency familiar door.

Nothing happened. A man sneezed.

Derek softly sighed.

He got off. Rainwater dripped from his umbrella as he climbed the stairs. At street level he stopped.

The cold rain was heavier now. He watched the sky. Then at the pavement.

Then at the Underground entrance. The cold rain was falling. The trains were moving.

The clocks were ticking. Everything seemed perfectly ordinary. Which, Derek had learned, was often when things became interesting.

He went back underground. This time he chose an empty station with a long escalator. He quietly stood still while it carried him downward.

He quietly watched the steps. They moved beneath him. He watched the handrail.

It moved at a different speed relative to the walls. He surveyed the people. They were moving relative to him.

The entire station was a small laboratory of relative motion. He smiled.

"Special Relativity."

A man beside him looked over.

Derek pretended he had said something else.

At the bottom, he walked toward the platform. Then he stopped. There was a puddle.

A large one. It had formed where rainwater had been carried down by passengers. Derek studied it.

Then at the moving escalator. Then at the rails. Then at the tunnel.

An idea occurred to him. The Underground was full of reference frames. Passengers.

Trains. Platforms. Escalators.

Ventilation systems. Water droplets. Everything moved relative to everything else.

And yet the laws of physics were supposed to work for all of them. SR did not say that only one frame was correct. GR did not say that either.

So what could possibly go wrong? Derek took out his notebook. He wrote:

SR: local inertial frames. Then: GR: curved spacetime.

Then: Question: what happens where the Underground is not locally inertial? He stopped.

That was better. A distant train accelerating away from a station was not an inertial frame. A train braking into one wasn't either.

A passenger standing on the platform was supported by the Earth. The platform was therefore not a freely falling frame. The Earth itself was rotating.

The entire planet was orbiting the Sun. The Sun was moving through the galaxy. The galaxy was moving relative to other galaxies.

There was no shortage of motion. Derek smiled. The universe had an extraordinary sense of humour.

He put the notebook away. Then he heard a quiet voice behind him.

"You're thinking too hard."

He turned. A woman was standing there. She was perhaps forty, dressed in a dark coat, holding an umbrella.

Derek slowly regarded her.

"Excuse me?"

"You've been staring at that puddle for five minutes."

"Have I?"

"Yes."

"Did you time me?"

"No."

"Then how do you know?"

She smiled.

"Because I was staring at you."

Derek considered this.

"That's fair."

She pointed at his notebook.

"Physics?"

"Yes."

"What sort?"

"Relativity."

She nodded.

"Special or General?"

Derek stared at her.

"Both."

"Good."

She smiled.

"Then you have a problem."

Derek laughed.

"So it appears."

She walked away. Derek watched her disappear into the crowd. He stood still.

Then he opened his notebook. He wrote: Find woman.

Then crossed it out. He wrote: Find out why she knew.

Then crossed that out too. Finally: Probably irrelevant.

He put the notebook away. He was learning. Not everything was a clue.

Sometimes a woman on the Underground simply knew what relativity was. It was London. Anything was possible.

He did not write her down as irrelevant a second time.

The empty train arrived. Derek boarded. He sat down.

And then, for no particular reason, he began thinking about poetry. The thought had been there since Weinstein's phone call. Time didn't exist.

It was merely a way of preventing everything from happening at once. The phrase had been absurd. But the more Derek thought about it, the less absurd it became.

Consider a poem. The words had an order. A line followed another line.

A word followed another word. Change the order and the poem changed. Not merely its meaning.

Its rhythm. Its structure. Its sound.

Derek imagined reading a poem in which every word was spoken simultaneously. He immediately understood the problem. It would sound like a fart.

A very literary fart, perhaps. But a fart nonetheless. He smiled.

He wrote:

TIME PREVENTS POETRY FROM SOUNDING LIKE A FART.

He stared at it.

Then added:

Possibly its greatest achievement. A man sitting beside him looked over. Derek closed the notebook.

The train entered another deep tunnel. The lights flickered again. And this time something happened.

Derek's watch stopped. He examined it. 11:17.

He inspected the mysterious clock. 11:17. The carriage display.

11:17. Everyone else's phones. Several passengers checked them.

11:17. Derek sat up. The mysterious train continued moving.

A woman coughed. Someone turned a page. The mysterious train rocked.

Everything was happening. But the clocks had stopped. Derek smiled.

"Right."

He looked around. Nobody else seemed particularly alarmed. A man checked his phone again.

Still 11:17. Derek looked through the window. The tunnel walls moved past.

He stood. He walked toward the end of the carriage. The train continued.

Then he noticed something else. The cold rain. He could see it through a small window near the door.

Rainwater was running down the outside of the mysterious train. But the drops suddenly appeared to be moving differently. Not backwards.

Not forwards. Simply differently. Derek stared.

The mysterious distant train emerged from the deep tunnel. Daylight flooded the carriage. The clocks jumped.

11:18. Everyone continued reading. Nobody noticed.

Derek sat down. He opened his notebook. He wrote:

11:17 → 11:18 Then: ancient Clock stopped for approximately?

He looked at the second hand. It had not advanced. But the train had travelled several kilometres.

Derek thought. The clock had measured proper time. The mysterious train had followed its worldline.

The passengers had followed theirs. The Earth had followed its own. All of them had continued through physical processes.

Yet the displayed time had stopped. That was not necessarily a failure of time. It was a failure—or transformation—of the relationship between a physical clock and the events around it.

Derek smiled.

“Weinstein.”

He called. No answer. He called again.

Still nothing. Then a message appeared.

WEINSTEIN: WHAT?

Derek typed: Clock stopped at 11:17. The reply came immediately.

WEINSTEIN: WHERE?

LONDON UNDERGROUND.

A pause.

WEINSTEIN: RAIN?

Derek looked out the window. Cold rain was still falling.

YES.

The reply:

WEINSTEIN: GOOD.

Derek stared. He typed:

GOOD?

The response:

YES. NOW FIND OUT WHAT THE WATER IS DOING.

A small official notice was taped to a pillar: RAIN IS NOT TO BE CONSIDERED EVIDENCE WITHOUT FORM 17B-W. Beneath it, in different handwriting: Especially if it falls sideways.

Derek studied the notice for a moment. Someone in the Bureau had clearly been here before him, or after him, or in a reference frame that made the distinction unhelpful. He photographed the notice, filed it under Provisional Geometry, and only then looked back at the rain.

Derek glanced at the cold rain. Then at the window. Then at the track.

Then at the puddles on the lonely platform as the train approached the next station. He understood. Water was not merely water.

It was matter moving through spacetime. Each droplet followed a trajectory. Each trajectory depended on its initial conditions and the geometry.

If there was something strange about the geometry, the water would reveal it. The Underground had thousands of natural test particles. Rain provided millions.

Derek got off the train. He stood beneath the gloomy station canopy. Rain poured beyond it.

He quietly watched. Droplets fell. Most followed ordinary paths.

But some struck the pavement. Some hit umbrellas. Some bounced.

Some ran along walls. Some disappeared into drains. Every one of them had a history.

A worldline. Derek suddenly understood what Weinstein had been getting at. The apple.

The clock. The train. The cold rain.

They were all the same problem. Different physical systems. Different worldlines.

Different measurements. One geometry. He opened his notebook.

He wrote: If SR and GR are both true, then every local inertial observer must recover SR. Then:

GR determines how those local frames fit together globally. Then: Therefore, if the Underground appears to produce a contradiction, the contradiction must be in our description, not in the theories.

He stopped. That was it. That was what Weinstein wanted him to find.

Unless... Derek stared at the cold rain. Unless there was a region where the assumptions needed to connect the local frames failed.

A singularity. A boundary. Something pathological.

Something where the geometry itself ceased to behave normally. He thought of the Louvre. The painting.

The changing perspective. The missing coordinate. The infinity.

He thought of the clock. 11:17. Then 12:04.

Then the train. Then the cold rain. He smiled.

Perhaps the Underground wasn't causing the contradiction. Perhaps it was revealing it. He called Weinstein.

This time Weinstein answered.

"Yes?"

"I think I understand."

"Fine."

"SR and GR don't conflict."

"No."

"SR works locally."

"Yes."

"GR tells us how those local inertial frames fit into curved spacetime."

"Yes."

"And if we find something that appears to make them incompatible..."

"We've misunderstood something."

"Precisely."

There was a pause. Derek smiled.

"Well, that's settled."

"Yes."

"There's just one problem."

"What?"

"I think the Underground has a singularity. Or something that wants to be a wormhole and hasn't decided yet."

Weinstein looked pained.

"Please do not invent new topology without filing a form."

Derek nodded.

"I'll put it under Pending Geometry."

Weinstein did not look reassured. Pending Geometry was already three drawers deep.

Mrs Marsh appeared at the end of the platform as if she had always been part of the architecture. "If that is a hole," she said, "it is not one of mine. Mine are all accounted for and properly signed." She pointed at a small official sticker on a nearby pillar. "No unauthorised singularities." Derek found himself almost respecting the sticker. It had the same finality as a decohered photograph.

On the windowsill sat a white mug. It was the same mug that had been in the office that morning, except that it was now empty when it should have been half full, and the tea stain inside had moved three centimetres to the left. No one commented. In the Bureau, objects that quietly disagreed with themselves were considered ordinary until they filed a complaint.

Mr Pendleton appeared as if summoned by the word "unauthorised." He made a note on his clipboard. "If it requires a form, it is not my form. If it does not require a form, it is still not my form." He looked faintly satisfied, as though he had just closed a stable door on a horse that had never existed.

Weinstein was silent. Then:

"Where?"

Derek lowered his gaze to the platform. At the far end, beyond the lights, a distant strange train was approaching. He gazed at the rain.

Then at the clock. 11:17. Then he peered at the dark tunnel.

"That's what I'm trying to find out."

Weinstein remained silent. Derek waited.

Finally Weinstein stated:

"Don't go into the tunnel."

Derek smiled.

"Why?"

"Because I know you."

"That's not a scientific reason."

"No."

"Then?"

"Because the last time you claimed 'I'm trying to find out', you ended up in the basement of the Louvre."

Derek considered this.

"Fair."

He hung up. The train arrived. The doors opened.

Derek stepped inside. He eyed the clock. 11:17.

The doors quietly closed. The mysterious train suddenly moved into the tunnel. And the cold rain continued falling outside.

Derek watched the droplets through the window. One of them ran upward. He stared.

Then another. Derek smiled.

“Well.”

He took out his notebook. And wrote:

THE COLD RAIN IS FALLING UP.

He thought about it.

Then added:

Or the train is falling down. He paused. Then, beneath that:

Or neither. The mysterious train entered darkness. And somewhere ahead, something in the geometry of London appeared to be waiting for him.

Chapter 8

The Open-Door Principle

There was a rule at the Bureau. It was not written down.

It did not appear in the employee handbook.

There was, in fact, no employee handbook, largely because Derek considered employee handbooks “a form of administrative poetry.”

Nevertheless, everyone knew the rule. If an office door was open, the occupant had vanished into virtuality. Obviously. Penny considered this. “That’s the sort of rule that sounds sensible until it starts wearing a hat.”

There was no point knocking. There was no point calling. There was no point standing in the doorway shouting someone’s name.

An open old office door meant that the person in question had ceased, for all practical purposes, to occupy ordinary spacetime. They might be in another room. They might be in London.

They might be in Munich. They might be in a completely different reference frame. They might even, on occasion, be dead.

The Bureau had learned not to worry. If the door was quietly closed, however, the situation was entirely different. Then the person was in there.

Obviously. They were either working hard... or shagging, equally hard.

There was no third possibility worth discussing. Derek had first formulated the principle after Penny had disappeared for three hours. He had found her office door open.

He had checked the kitchen. Nothing. He had checked the laboratory.

Nothing. He had checked the basement. Nothing.

He had checked the train timetable. Nothing. Eventually he had returned to her office.

The door was still open. Trevor had suddenly appeared beside him.

“Where’s Penny?”

“Virtuality.”

Trevor nodded.

“Right.”

They went back to work. Three hours later Penny had emerged from Derek’s office carrying a notebook. Derek had watched her.

"Where were you?"

"Working."

"With the door closed?"

"Yes."

Derek nodded.

"There we are, then."

Penny had softly smiled. Trevor had surveyed the two of them. Then at the quietly closed old door.

Then at Derek. Then at Penny. He had wisely remained silent.

The rule had subsequently become part of Bureau culture. And on Friday morning, Derek's office door was open. Penny's was quietly closed.

Trevor's was open. Weinstein's was quietly closed. Nobody found this unusual.

Trevor walked down the narrow corridor carrying three cups of tea. He stopped outside Derek's office. The door was open.

He looked inside. Empty. He nodded.

"Virtuality."

He continued. Penny's heavy door was closed. Trevor knocked.

"Come in."

He opened it. Penny was sitting at her desk. She was reading a report.

Trevor put down her tea.

"Working hard?"

"Yes."

"Good."

He left. Penny waited until the door closed. Then smiled.

The Bureau was a peculiar place. Derek returned twenty minutes later. He walked through the front door carrying a wet umbrella.

Penny glanced up.

"Where have you been?"

"Underground."

"Did you find anything?"

"Yes."

"What?"

"Rain."

Penny waited.

"Anything else?"

"Yes."

"What?"

"It was falling upward."

Penny nodded.

"There. Sorted."

Derek took off his coat.

"Exactly."

He put the umbrella in the stand.

"Except I don't know whether the rain was falling upward."

"Or?"

"The train was falling downward."

Penny nodded.

"Or?"

"Neither."

"Good."

Derek fixed her with a look.

"You're getting rather good at this."

"I've had excellent training."

He smiled. Then he noticed her familiar door. It was open.

Derek frowned.

"Where's Trevor?"

"Virtuality."

"Of course."

He looked toward Trevor's office. His familiar door was open too. Derek nodded.

"Both at once?"

"Apparently."

"That's unusual."

"Not really."

Derek considered this.

"Fair enough."

He walked to the whiteboard. There were equations everywhere.

Lorentz transformations.

Metric tensors. A sketch of a geodesic. A badly drawn simple apple.

An empty train. A cloud. And, in large letters:

THE RAIN IS NOT OBLIGED TO FALL DOWN.

Derek stared at it.

"Did you write this?"

"No."

"Trevor?"

"Probably."

Derek regarded the open office door.

"Virtuality."

Penny nodded. Derek picked up a marker. Underneath the sentence he wrote:

NOR IS THE TRAIN OBLIGED TO KNOW WHERE IT IS GOING. He put down the marker. Then the telephone rang.

Penny answered.

"Relativistic Investigation Bureau."

A pause. She examined Derek.

"It's for you."

"Who?"

"Hermann."

Derek took the phone.

"Minkowski."

"Derek."

"Yes?"

"We've found something."

Derek smiled.

"Where?"

"London."

"Good."

"What do you mean, good?"

"That's where we are."

"Yes."

"Then you've saved us a journey."

There was silence.

"We're in Munich."

Derek paused.

"Ah."

Penny inspected him.

"Virtuality?"

Derek nodded.

"Apparently."

Minkowski continued.

"The problem concerns your clocks."

Derek became serious.

"What about them?"

"We've compared the London readings with the Munich readings."

"And?"

"They don't disagree."

Derek frowned.

"That's good."

"No."

"Why not?"

"Because they should."

Derek sat down slowly, the chair creaking in a way that felt, this week, unnecessarily editorial.

"Explain."

"The clocks are following different worldlines."

"Yes."

"Different gravitational potentials."

"Yes."

"Different motions."

"Yes."

"So, their proper times should differ."

"Yes."

"But they don't."

Derek frowned.

"Then they're synchronized."

"Exactly."

"That's not a problem."

"It is if they have never been synchronized."

Derek stopped. Penny considered him.

"Go on."

Minkowski commented:

"Someone has synchronized the clocks."

"When?"

"We don't know."

"How?"

"We don't know."

"Why?"

"We don't know."

Derek smiled.

"Well, that's settled."

"What's settled?"

"We know almost nothing."

Minkowski sighed.

"That's not what I meant."

Derek glanced at Penny. She smiled the particular smile she reserved for moments when she already knew how the conversation would end.

"Fair enough."

Minkowski continued.

"There is something else."

"Of course."

"We found a reference to the Louvre."

Derek's expression changed.

"Where?"

"In a file that should not exist."

"What's in it?"

"Your name."

"Anything else?"

"Alfred's."

"Penny?"

"Yes."

"Trevor?"

"Yes."

Derek paused.

"And Julius Barbarian?"

There was silence.

"Yes."

Derek stared at Penny.

"That's interesting."

"Very."

Minkowski noted:

"The file is dated tomorrow."

Derek leaned back.

"Excellent. Case closed."

Minkowski sounded irritated.

"Would you stop saying that?"

"Sorry."

"No, you're not."

"Not particularly."

There was another pause.

Then Minkowski observed:

"You should examine your own office."

Derek looked toward the narrow corridor, where the light seemed, if anything, slightly reluctant to reach the far end.

"My office?"

"Yes."

"Why?"

"Because the file indicates something is there."

"What?"

"We don't know."

Derek stood. He walked toward his familiar office. The door was open.

He stopped. Penny followed him.

"What?"

"Nothing."

He looked inside. The office was empty. The old battered desk was empty.

The chair was empty. The computer was off. Derek smiled.

"Virtuality."

Penny nodded.

"Obviously." Penny fixed him with a look. "You're not the brightest bulb in the toolbox today."

Derek stepped inside. On the desk was a single sheet of paper. He picked it up.

It contained one sentence.

IF THE FAMILIAR DOOR IS OPEN, DON'T GO IN.

Derek stared. Penny stared. Trevor suddenly appeared behind them.

"What's that?"

Derek took in the room.

"Where did you come from?"

Trevor pointed at his own office.

"The door was closed."

Derek nodded.

"Ah."

Trevor gazed at the paper.

"What does it say?"

Derek handed it to him. Trevor read it. Then looked at the open door.

"That's confusing."

Penny nodded.

"Yes."

Trevor thought.

"Perhaps we're not supposed to be here."

Derek smiled.

"Excellent."

A pleased smile appeared on Trevor's face.

"I've solved it."

"Not quite."

"Oh."

Derek eyed the paper again. Then at the door. Then at the clock.

11:17. The ancient clock stopped. All three watched.

The second hand froze. The computer switched on. A message quietly appeared:

DOOR STATUS: OPEN

Then:

OCCUPANT STATUS: VIRTUAL

Derek frowned.

"Who's the occupant?"

The computer answered.

DEREK GENT

Derek stared intently. Trevor stared. Penny stared.

Derek dropped his eyes at himself.

"I'm here."

Penny nodded.

"Yes."

Trevor considered this.

"Apparently you're both here."

Derek watched him.

"Both?"

Trevor pointed at the screen.

"Virtual Derek."

Then at Derek.

"Actual Derek."

Derek smiled.

"Well, that's inconvenient."

The screen changed.

DOOR CLOSING IN 10 SECONDS

Derek surveyed the door.

"Why?"

The countdown began.

Penny replied:

"Perhaps you should leave."

Derek nodded.

"Good idea."

They stepped backward. The heavy door slammed shut. The clocks restarted.

11:18. The computer went dark. For several seconds nobody moved.

Then Derek studied Penny.

"What does a closed door mean?"

Penny smiled.

"Someone's in there."

Derek nodded.

"Obviously."

Trevor regarded the door.

"Who?"

Derek opened it. The quiet office was empty. He stared.

Then quietly closed it again.

"Me, apparently."

Penny laughed. Derek examined the closed door.

"Right."

He knocked. From inside came Derek's own quiet voice.

"Come in."

Trevor's mouth fell open. Penny looked delighted. Derek carefully opened the door.

His office was still empty. Except for himself. He stood in the doorway.

His own voice came from somewhere inside.

"You're late."

Derek stared into the empty room.

Then answered:

"Well, that's settled."

The distant voice answered:

"Not remotely."

Chapter 9

The Relativistic Ride

The first important observation was that Derek Gent had apparently not quite murdered himself yet. This was reassuring.

It was also becoming slightly embarrassing. The Bureau had announced his death with considerable confidence. There had been a photograph, a timestamp, a location, and enough circumstantial evidence to make most people reasonably uncomfortable about their own mortality. Yet Derek remained alive.

He found this increasingly difficult to reconcile with the paperwork. He was standing in his office, staring at the quietly closed door. From inside came his own voice.

"You're late."

Derek stared at the empty room. Penny stood behind him. Trevor stood beside her.

Nobody spoke.

Finally Derek added:

"I don't think I've murdered myself yet."

Penny nodded.

"Apparently not."

Trevor inspected the photograph.

"That's encouraging."

Derek glanced at him.

"Is it?"

"Well, if you had already done it, we'd have a rather awkward problem."

"We already have an awkward problem."

"True."

Derek returned to the photograph. The dead man was unmistakably him. Same face.

Same clothes. Same regrettable tendency to look as though he had just been interrupted while thinking about something complicated. Yet he was still standing.

Alive. Annoyed. And, at least for the moment, in possession of all the usual body parts.

Derek put the photograph down.

"Perhaps the photograph is from the future."

Penny nodded.

"Obviously."

"Or another event."

"Yes."

"Or another worldline."

"Possibly."

Trevor's brow furrowed.

"Or another Derek."

Derek glanced at him.

"There's only one of me."

Trevor nodded.

"That's what makes it difficult."

Derek wearily sighed.

"Thank you."

"You're welcome."

The office door remained quietly closed. Which meant Derek was in there. Obviously. Trevor looked at Penny. "Or we're being mildly inconvenient again."

Penny quietly stared at the door.

"Although technically..."

Derek raised a hand.

"No."

"What?"

"Let's not complicate the door."

"Fair."

He looked out of the window. Rain was falling. It had been raining for most of the day.

This was not particularly surprising. It was London. Rain in London was not weather so much as a local administrative policy.

Derek watched the drops run down the glass.

"Rain."

Trevor gazed at him.

"Yes."

"We keep coming back to rain."

"Yes."

"Why?"

"Because it keeps falling."

Derek smiled.

"Not always."

Trevor peered at him. Derek pointed at the window. Trevor followed his finger.

The drops were falling normally. Downward. He frowned.

"That's disappointing."

"Give it time."

Penny picked up the ancient silent brass clock. It was reading 11:18. She turned it over.

"Still no explanation."

Derek nodded.

"Then perhaps we're asking the wrong question."

"Which one?"

"Why the clock stops."

Trevor gave him a measured look.

"What should we ask?"

"Why does it start again?"

Penny smiled.

"That's better."

Derek nodded.

"And why does it always happen at 11:17?"

Trevor explained:

"Because something happens at 11:17."

"Yes."

"And what is that?"

Derek watched the photograph.

"The train."

Penny nodded.

"Platform seven."

"Exactly."

Trevor looked toward the window.

"Then why are we still here?"

Derek smiled.

"We're not."

Penny understood immediately.

"We're taking the Underground."

Trevor surveyed her.

"In the rain?"

"Yes."

"Why?"

Derek reached for his coat without quite deciding to. Some choices simply finish making themselves.

"Because Weinstein wants a relativistic ride."

Trevor nodded.

"That sounds expensive."

"It's the Underground."

"Ah."

They left the Bureau. Derek locked the front door. Then stopped.

He looked at the office doors. Penny's was closed. Trevor's was open.

Derek's was quietly closed. He nodded.

"Good."

Penny regarded him.

"What?"

"Everyone's where they ought to be."

Trevor glanced over his shoulder.

"My office is empty."

Derek shrugged.

"Virtuality."

"Obviously."

They walked into the cold rain. The Underground was busy. It was always busy.

But on a rainy afternoon it was something else. Wet coats. Wet shoes.

Wet umbrellas. Wet people. The smell of rainwater, electricity and human resignation.

Derek descended the stairs. He examined the crowds.

"Excellent."

Trevor inspected him.

"What?"

"Test particles."

Penny smiled.

"Thousands of them."

"Quite."

Derek opened his notebook.

"Every passenger is following a worldline."

Trevor nodded.

"Every train as well."

"Yes."

"The station."

"Yes."

"The Earth."

"Yes."

"The rain."

"Yes."

Penny declared:

"And you."

Derek considered her.

"Yes."

"Don't forget yourself."

"I wasn't planning to."

They reached the platform. The train arrived. Derek glanced at his watch.

11:12. He stared at the station clock. 11:12.

He gazed at the train display. 11:12.

"Five minutes."

Trevor nodded.

"Until 11:17."

Derek glanced down the tunnel.

"Yes."

Penny smiled.

"What are we expecting?"

Derek considered this.

"Nothing."

"Good."

"That's usually when something happens."

They boarded. The doors quietly closed. The train moved.

Derek sat beside the window. Trevor sat opposite. Penny stood holding the rail.

The train accelerated. Derek intently watched the deep tunnel.

"Now."

Trevor found his eyes.

"Now what?"

"Acceleration."

"Ah."

Derek took out his notebook.

"The train is not an inertial frame."

"No."

"Passengers are accelerating with it."

"Yes."

"But locally, each passenger can still regard themselves as approximately inertial over a sufficiently small region."

"Yes."

"Which is where Special Relativity works."

"Yes."

Penny nodded.

"And General Relativity tells us how the local inertial frames fit together."

Derek softly smiled.

"Exactly."

Trevor slowly eyed the tunnel.

"So, there's no contradiction."

"Not ordinarily."

"And this train isn't going to disprove either theory."

"Not ordinarily."

Trevor smiled.

"Case closed. Probably."

The mysterious train entered a station. Brakes applied. Derek felt the deceleration.

He watched the passengers. Everyone leaned slightly forward. The mysterious train stopped.

Then started again. Derek watched his watch. 11:14.

Normal.

"Again."

Trevor surveyed him.

"Again what?"

"Acceleration."

Trevor wearily sighed.

"Yes."

They continued. 11:15. 11:16.

Derek watched the clock. The second hand moved smoothly. 11:16:50.

11:16:55. 11:16:58. 11:16:59.

11:17:00. The second hand stopped. The train continued moving.

Derek smiled.

"Right."

Penny studied her phone. 11:17. Trevor regarded the mysterious train clock.

11:17. The carriage display. 11:17.

A passenger checked his phone. 11:17. Another passenger.

11:17. Derek examined Penny.

"All stopped."

She nodded.

"Yes."

Trevor surveyed the room. Nobody else seemed concerned. The train continued through the tunnel.

Derek stood. He walked toward the window. Outside, the tunnel wall slid past.

He watched it. Then he noticed something. The mysterious train was moving.

The silent clock was not. He smiled.

"That's interesting."

Trevor joined him.

"What's interesting?"

"The train is still changing position."

"Obviously."

"Yet its clock isn't advancing."

"Yes."

"So, the train's spatial coordinates are changing while its displayed temporal coordinate isn't."

Trevor nodded.

"That doesn't mean its proper time has stopped."

"Correct."

Penny came closer.

"The clock could simply be failing."

"Except all of them stopped."

"Yes."

"At exactly the same reading."

"Yes."

"And we've already seen this elsewhere."

"Munich."

"The Louvre."

"Yes."

Trevor frowned.

"So, the anomaly isn't in the clocks."

Derek smiled.

"That's what I'm thinking."

"What is it in?"

"The relationship between the clocks and the events."

Penny nodded.

"Good."

Derek inspected the rain through the window. The tunnel ended. Rain streaked across the glass.

He watched one droplet. It slid downward. Then sideways.

Then upward. Derek blinked.

"Did you see that?"

Penny looked.

"Yes."

Trevor looked.

"Yes."

Derek smiled.

"Right. Next."

Trevor frowned.

"What?"

"The rain is misbehaving again."

The mysterious train entered another tunnel. The droplet disappeared. Derek consulted his watch.

11:17. Still. The strange train continued.

He felt the acceleration. He thought about Special Relativity. Then General Relativity.

Then Barbarian. Time wasn't necessarily a thing. It might be a relationship between events.

The train didn't need a universal time. It had a worldline. The passengers had worldlines.

The cold rain had worldlines. The clocks had worldlines. The Earth had a worldline through spacetime.

Everything had a history.

Perhaps "time" was simply how they described the relationships among those histories.

Derek smiled. Then thought about poetry. He couldn't help it.

If time were removed completely, a poem would be a collection of words with no ordering. Every word would be there. But all at once.

The idea was ridiculous. It would sound like a fart. A highly educated fart.

Possibly a French fart. But a fart nonetheless. Derek wrote in his notebook:

TIME MAY BE THE THING THAT STOPS POETRY FROM BEING A FART. Trevor looked over his shoulder.

"What are you writing?"

"Physics."

Trevor read it.

"That's not physics."

"It will be."

Penny looked. She smiled.

"Actually, he's got a point."

Trevor stared at her.

"How?"

"A poem requires ordering."

"Yes."

"Ordering requires distinguishing one event from another."

"Yes."

"And our experience of that ordering is what we call temporal succession."

Trevor nodded slowly.

"That's fair."

Derek looked pleased.

"Well, that's settled."

Trevor smiled.

"Yes."

The train slowed. Derek glanced at the station sign. They had arrived at the next station.

He checked the silent clock. Still 11:17. The station clock also read 11:17.

Derek stepped onto the platform. He looked down. Water was running along the edge.

He crouched. A stream of rainwater moved toward the drain. Derek watched.

The water reached the drain. Then stopped. The mysterious train doors closed.

The train departed. The water remained. Derek stood.

Penny stared at him.

"What?"

"The water."

"What about it?"

"It stopped."

"Because the train left?"

"No."

"Then why?"

Derek shook his head.

"I don't know."

Trevor studied the water.

"It could be blocked."

"Yes."

"It could be debris."

"Yes."

"It could be the gradient."

"Yes."

Trevor smiled.

"Well, that's settled."

Derek laughed.

"Not quite."

He slowly gazed at the tunnel. Then at the station clock. Then at the water.

"Let's see what happens at 12:04."

Penny's expression changed.

"The restart."

"Exactly."

They waited. The cold rain continued. The empty station remained busy.

Trains arrived. Trains departed. People came and went.

Nobody paid any attention to the three people standing beside a puddle. At 12:03, Derek checked his watch. Still 11:17.

At 12:03:30, nothing happened. At 12:03:50, nothing. At 12:03:59—

The water moved. The clock jumped. 11:17 became 12:04.

The strange station clock did the same. Derek's watch did the same. Penny's phone jumped.

Trevor's phone jumped. The train arriving at the cold platform seemed to hesitate for the briefest fraction of a second. Then everything continued.

Derek stared at the silent ancient clock. 12:04. He peered at the puddle.

The water was flowing normally. He eyed Penny.

"Well."

She smiled.

"That's settled."

Trevor nodded.

"The clocks weren't broken."

"No."

"They were waiting."

"Yes."

"For what?"

Derek lowered his gaze the tunnel.

"For the next event."

A distant train approached. Its headlights suddenly appeared in the darkness. Derek watched it come closer.

Then he noticed the destination display.

The gloomy London Terminus.

cold Platform 7.

Derek's face changed.

"That's our train."

Trevor watched the display.

"We're underground."

"Yes."

"The train is coming from the wrong direction."

"Yes."

Penny studied the silent clock. 12:04. Derek smiled.

"Of course."

The train stopped. The doors opened. Inside, the carriage was empty.

Completely empty. No passengers. No driver.

No conductor. Just seats. And at the far end of the carriage was a single brass clock.

11:17. Derek stared. Trevor stared.

Penny silently stared. Then Derek stepped aboard. Penny grabbed his arm.

"Derek."

He surveyed her.

"Yes?"

"The door is open."

Derek smiled.

"Yes."

"So, whoever's in there..."

"Has vanished into virtuality."

"Obviously."

Derek looked into the empty carriage.

"Then we'd better find out where they've gone."

He stepped inside. The doors slowly closed behind him. The strange train began to move.

And somewhere ahead, beneath London, something was waiting at Platform Seven. It had been waiting since 11:17. It would remain there until 12:04.

And, according to Derek's watch, it was already tomorrow. The Flight Derek fell asleep on the empty train.

This was unfortunate. Not because sleeping on the Underground was particularly dangerous, although it was generally considered unwise to trust one's neck with the suspension system of a London carriage. It was unfortunate because Derek had a dream.

In the dream, he was on an aeroplane. He was already late. This was especially irritating because Derek had been absolutely certain he had left with plenty of time.

He checked his watch. 11:17. He checked it again.

11:17. He looked at the departure board. His flight had already departed.

Derek stared.

"But I was here."

The airport appeared to have no sympathy. He ran. He reached the gate.

quietly closed. The attendant regarded him.

"Sorry."

"How long ago?"

"About forty minutes."

"Impossible."

She shrugged.

"It happens."

Derek examined his watch. 11:17. He inspected the airport clock.

12:04. He frowned.

"Ah."

The attendant smiled.

"Yes."

"Different reference frame?"

She considered him.

"Different airline."

Derek stared.

"What?"

"You booked Ryanair."

He glanced at the ticket. Ryanair. He frowned.

"I usually fly British Airways."

"That may be your problem."

Derek stared at her.

"Why?"

She smiled.

"British Airways uses Greenwich Mean Time."

"Ryanair uses?"

"Whatever time gets you to the gate."

Derek considered this.

"That's not how time zones work."

"No."

"Or Special Relativity."

"No."

"Or airlines."

"Apparently."

She handed him another ticket. Derek gazed at it. His seat number was written in large letters.

He stared.

"What's this?"

"Your seat."

"I have a seat."

"Yes."

"Where?"

She pointed toward the aircraft.

"On the plane."

"Obviously."

She smiled.

"And please put your seat in the upright position."

Derek dropped his eyes. He was suddenly sitting in the aircraft. A very attractive flight attendant was leaning over him.

She softly smiled.

"Your seat, sir."

Derek glanced down. His seat was reclining. He looked back at her.

"Yes."

"Upright, please."

Derek adjusted it. She nodded approvingly.

"Much better."

Then she walked away. Derek watched her go. He glanced at his watch.

11:17. He looked at the aircraft strange clock. 12:04.

He stared intently.

"Well, that's not good."

The captain's voice came over the speaker.

"Ladies and gentlemen, we're currently travelling at approximately thirty-seven thousand feet."

Derek looked out of the narrow window. The Earth was below. He thought about proper time.

He thought about gravity. He thought about acceleration. He thought about SR.

He thought about GR. Then he thought about the flight attendant. Perhaps the problem was not relativity.

Perhaps the problem was Ryanair. He quietly closed his eyes. When he opened them, the aircraft was empty.

The flight attendant was gone. The pilot was gone. The passengers were gone.

The plane was still flying. Derek watched the clock. 11:17.

He looked out of the window. The Earth was still below. Then he noticed something.

The clouds were moving backwards. Derek stood. He walked toward the cockpit.

The door was open. Obviously, the pilot had vanished into virtuality. He looked inside.

Nobody. The instrument panel was running. The plane continued flying.

Derek surveyed the controls. Then at the clock. Then at the empty cockpit.

"Right, that's that, then."

The aircraft suddenly descended. Derek grabbed the nearest seat. The flight attendant appeared beside him.

"Sir?"

Derek studied her.

"Yes?"

"Your seat."

"What about it?"

"Upright."

Derek lowered his gaze. His seat was reclining again. He adjusted it.

She smiled.

"Thank you."

Then she disappeared. Derek surveyed the room. The aircraft was empty again.

He regarded the clock. 11:17. Then 12:04.

Then 11:17. Then 12:04. He understood.

The aircraft wasn't travelling through a single universal time. Its clocks were describing different relationships between events. The flight had not been late.

The arrival event had not happened "later."

The entire question of later and earlier had become frame-dependent. Derek smiled.

"That's actually quite beautiful."

Then he examined the airline ticket.

It announced:

ARRIVAL: YESTERDAY

Derek stared.

"Right."

The distant plane landed. The doors opened. Derek stepped out.

He was back in the London Underground. He woke with a start. The mysterious train was still moving.

Penny was sitting opposite him. Trevor was asleep beside the window. Penny inspected Derek.

"You were talking."

Derek rubbed his eyes.

"What did I say?"

She smiled.

"You made a remark about an airline."

Derek checked the clock. 11:17. Still.

He looked out the window. Cold rain streaked across the glass. Then he noticed that his seat was reclined.

He stared at it. Penny glanced at him.

"What?"

Derek slowly pushed the seat upright.

"Nothing."

Penny smiled.

"Lovely. Next?"

Derek stared at the rain.

"Not quite."

He took out his notebook. Under the previous entry he wrote: NOTE: Never book a relativistic flight with Ryanair.

Then, after a moment's thought: British Airways probably isn't much better. He closed the notebook.

The distant train continued into the darkness. And somewhere ahead, the clock was still waiting for 12:04.

Chapter 10

The Ordinary Cucumber Problem

Penny had a theory. It was not a particularly complicated theory.

Nor, for once, did it involve spacetime. She needed an ordinary cucumber. She was standing in an organic fruit market in London, looking at a crate of them while a man in a green apron explained the virtues of soil.

"The important thing," he asserted, "is that they're grown naturally."

Penny picked one up.

"Naturally?"

"Without synthetic pesticides."

"Right."

She examined it.

"Are cucumbers fruits?"

The man hesitated.

"Botanically, yes."

Penny nodded.

"Excellent."

"But culinarily—"

"Don't."

He stopped. Penny put the cucumber in her basket. She had no particular interest in the botanical classification of cucumbers.

She had an interest in what they could be used for. And at the moment, she needed one for an experiment. Nothing sexual.

Nothing particularly mysterious. Just a cucumber. The vendor watched her.

"Anything else?"

"No."

"Tomatoes?"

"No."

"Courgettes?"

"No."

"Organic aubergines?"

"No."

He looked disappointed.

"Just the cucumber."

"Very well."

Penny paid. She left. It was raining.

Of course it was. London had apparently decided that weather was unnecessary and rain was sufficient. She walked back toward the Bureau with the cucumber under her coat.

She crossed the street. A bus went past. Water sprayed from its wheels.

She watched the droplets scatter. She smiled. Everything had a frame.

Everything had a worldline. Everything had relationships. Even cucumbers.

Perhaps especially cucumbers. She reached the Investigation Bureau. Trevor was sitting at the familiar front desk.

He gazed at the cucumber. Then at Penny. Then at the cucumber again.

"What is that?"

"A cucumber."

"I can see that."

"Then why ask?"

Trevor nodded.

"True."

He peered at it again.

"What are you doing with it?"

"Experiment."

"What sort?"

"Relativistic."

Trevor nodded.

"Of course."

Penny walked past him. He followed.

"Is it dangerous?"

"No."

"Explosive?"

"No."

"Quantum?"

"No."

"Gravitational?"

"Possibly."

Trevor stopped.

"With a cucumber?"

Penny turned back.

"You'd be surprised what can be gravitational."

Trevor considered that.

"Fair enough."

They entered the laboratory. Derek was there. His familiar office door was quietly closed.

Obviously. Trevor pointed toward it.

"He's in there."

"Working?"

"Probably."

Penny smiled.

"Or shagging."

Trevor wearily sighed.

"Please don't."

"Why?"

"Because I know what you're going to say."

Penny smiled.

"Then you have excellent predictive powers."

She placed the ordinary cucumber on the laboratory table. Trevor eyed it. Derek's voice came through the quietly closed door.

"What are you two doing?"

Penny called back.

"Science."

"Good."

A pause.

"Carry on."

The door remained closed. Trevor nodded.

"He's working hard."

"Obviously."

Penny picked up the cucumber.

"Right."

Trevor watched.

"What exactly are you testing?"

Penny watched the cucumber. Then at the silent clock. Then at the equipment.

"Whether an ordinary object can help us detect the strange change in the local frame that Derek encountered on the Underground."

Trevor's expression changed.

"Ah."

"Exactly."

She placed the ordinary cucumber on a small platform. The platform was mounted on a sensitive motion sensor. Trevor leaned over.

"You're using it as a test mass."

"Yes."

"Why a cucumber?"

Penny surveyed him.

"Because it's convenient."

"Why not a metal sphere?"

"Because we've already tested those."

"And?"

"Nothing."

"And cucumbers?"

"We haven't tested one."

Trevor nodded.

"That's one less variable, then."

Penny smiled.

"Precisely."

She switched on the apparatus. The cold platform began to move. The cucumber rolled slightly.

Then stopped. Trevor watched the graph. A smooth curve suddenly appeared on the monitor.

Then— A tiny discontinuity. Trevor frowned.

“That’s odd.”

Penny nodded.

“Very.”

The platform returned to its starting position. She reset the experiment. Again.

The same thing happened. Smooth acceleration. Tiny discontinuity.

Smooth acceleration. Trevor stared at the screen.

“Same position.”

“Yes.”

“Same acceleration?”

“Almost exactly.”

“Same discontinuity?”

“Yes.”

He studied the cucumber.

“Perhaps cucumbers are strange.”

Penny smiled.

“That’s what I’m hoping.”

They replaced the cucumber with a wooden block. They ran the experiment. The discontinuity remained.

They replaced the block with a brass sphere. Same result. Then a glass sphere.

Same result. Then a bag of sugar. Same result.

Trevor regarded the collection of objects.

“The cucumber is innocent.”

Penny nodded.

“Completely.”

“Poor thing.”

Penny picked it up.

“You can go back to being a fruit.”

She put it beside the apparatus. Trevor examined the graph.

"It's not the test object."

"No."

"It's not the apparatus."

"Apparently not."

"It's not the acceleration."

"Not ordinary acceleration."

Trevor inspected her.

"What does that mean?"

Penny sat down.

"Imagine you're in a train."

"All right."

"The train begins accelerating."

"Yes."

"You feel yourself pushed backward."

"Yes."

"But if you're inside a sufficiently small room, you could describe the effect as though there were a gravitational field pointing backward."

Trevor nodded.

"The equivalence principle."

"Just so."

"Locally, acceleration and gravity can be indistinguishable."

"Yes."

"But only locally."

"Right."

"Because a real gravitational field can produce tidal effects."

Penny smiled.

"Exactly."

Trevor considered the graph again.

"So, we're looking for something that isn't simply acceleration."

"Yes."

"Something that changes from one location to another."

"Yes."

"Tidal effects."

"Possibly."

Penny stood. She moved the sensor lonely platform a metre to the left. They ran the test.

The discontinuity occurred. She moved it a metre to the right. Again.

She moved it upward. Again. Trevor stared.

"Everywhere."

"Yes."

"Then it's not a local effect."

Penny shook her head.

"Not necessarily."

"Why?"

"Because our laboratory is tiny compared with the scale of the geometry we're investigating."

Trevor nodded.

"So, the curvature could be approximately uniform across the room."

"Exactly."

"Then we'd need a larger baseline."

Penny smiled.

"Now you're thinking."

Trevor glanced at the ceiling.

"The Underground."

"Yes."

"The whole train."

"Yes."

"Perhaps the whole station."

"Yes."

"The whole city?"

"Could be."

Trevor looked impressed.

"That would make the experiment rather inconvenient."

Penny nodded.

"That's why we have Derek."

They looked toward Derek's closed office door. Trevor smiled.

"He's in there."

"Yes."

"Working hard."

"Obviously."

"Or—"

Penny pointed at him.

"Don't."

Trevor shut up. Penny walked to the whiteboard. She drew a distant empty train.

Then a series of small local frames around it.

"Imagine the train moving through spacetime."

Trevor nodded.

"At any particular point along its worldline, we can construct a local inertial frame."

"Yes."

"Inside that sufficiently small region, Special Relativity works."

"Yes."

"But as we move farther along the train—or farther through space—the local frames don't necessarily line up globally."

"Because of curvature."

"Exactly."

She drew a second frame. Then a third. The axes were slightly rotated.

Trevor stared at them.

"So, the contradiction between SR and GR is only apparent."

"Yes."

"SR describes each local frame."

"Yes."

"GR describes the geometry relating those frames."

"Precisely."

Trevor nodded.

"Then what are we actually looking for?"

Penny pointed at the tiny discontinuity in the graph.

"That."

"Why?"

"Because something has happened to the relationship between our local frame and the physical system we're measuring."

Trevor frowned.

"Could it be a coordinate effect?"

"Yes."

"Meaning nothing physical has happened?"

"Possibly."

"Then why does the apparatus detect it?"

"Because coordinate effects can change what different observers assign to an event."

Trevor nodded.

"But measurable quantities must remain invariant."

"Correct."

Penny pointed to the graph.

"So, the next question is whether that discontinuity survives a change of coordinates."

Trevor softly smiled.

"That's clever."

"Thank you."

"How do we test it?"

"We change observers."

Trevor stared at her.

"How?"

Penny pointed at the door.

"Derek."

Trevor nodded.

"Of course."

He went toward the door. Then stopped. It was still closed.

He knocked. Nothing. He knocked again.

Still nothing. Penny looked over.

"Working hard."

"Evidently."

Trevor opened the door. Derek was sitting at his desk. He looked up.

"What?"

"We need you."

"For what?"

"An experiment."

Derek gazed at the ordinary cucumber. He stared intently. Then at Penny.

Then back at the cucumber.

"No."

Penny laughed.

"It's not about the cucumber."

"Good."

"It's about the reference frame."

"Better."

She showed him the graph. Derek studied it.

"When does the discontinuity occur?"

"At the same position every time."

"Relative to the apparatus?"

"Yes."

"Not the room?"

"No."

"The room itself?"

"We haven't tested that."

Derek nodded.

"Then that's what we need."

He stood. Penny smiled.

"You're coming?"

"Yes."

Trevor's eyebrows rose.

"Just like that?"

Derek shrugged.

"It's interesting."

They went outside. The cold rain had become heavier. Derek looked at the street.

Then at the laboratory. Then at the ordinary cucumber.

"Bring it."

Penny held it up.

"Seriously?"

"Absolutely."

"Why?"

Derek smiled.

"If we're going to do relativistic physics in London, we may as well have a proper control."

Trevor inspected the cucumber.

"What's the control?"

"The cucumber."

"Why?"

"Because everyone expects us to lose our minds."

Trevor nodded.

"Fair."

They placed the apparatus on a trolley. Then pushed it into the street. The cold rain struck the sensor.

Water ran down the equipment. Penny covered it. Derek watched the traffic.

Buses. Cars. Bicycles.

Pedestrians. Umbrellas. Everything moving.

Everything with its own worldline. He watched the cold rain.

"Perfect."

Penny surveyed him.

"Why?"

"Because now we've got millions of test particles."

Trevor smiled.

"Rain."

"Yes."

"Again."

"Again."

They started the apparatus. The trolley moved down the pavement. The graph suddenly appeared.

Smooth. Then — The discontinuity.

Derek stopped.

“Where?”

Penny pointed.

“Exactly here.”

Derek cast a quick look around. They were beside a distant station entrance. He checked the map.

“Which line?”

“Central.”

Derek studied the entrance. Then at the silent clock. 11:17.

He went still. Penny noticed.

“What?”

Derek pointed. The station clock had stopped. 11:17.

Trevor regarded his phone. 11:17. The sensor clock.

11:17. The ordinary cucumber rolled slightly on the trolley. Then stopped.

Derek smiled.

“The cucumber has found the reference frame.”

Trevor stared.

“That’s not possible.”

“No.”

“Then what?”

Derek dropped his eyes into the Underground entrance.

“I think we’re looking at the same event.”

Penny nodded.

“11:17.”

“Yes.”

“The clocks stop.”

“Yes.”

"The rain behaves strangely."

"Yes."

"And now our test mass detects a discontinuity."

"Yes."

Trevor examined the cucumber.

"So, the cucumber isn't the cause."

"No."

"It's a witness."

Derek softly smiled.

"Exactly."

Penny inspected the Underground entrance.

"Then perhaps we should go down."

Derek nodded.

Trevor wearily sighed.

"Again?"

Derek turned his attention to him.

"Did you have another appointment?"

"No."

"Good."

They descended. The cold rain continued falling above them. The cucumber came too.

And, somewhere deep beneath London, an empty train was approaching a station at exactly 11:17. Derek quietly glanced at the silent clock. It stopped.

The train arrived. The doors opened. For one second, Derek could see through the carriage.

There was nobody inside—except at the far end, sitting calmly by the window, was Derek Gent. Derek stared at himself.

The other Derek tipped his head back. Their eyes met. The other Derek smiled.

Then he raised an ordinary cucumber. Derek stared at Penny. Penny gazed at Trevor.

Trevor peered at the ordinary cucumber in Penny's hand. Then back at the mysterious train.

"Well," he pointed out.

"That's rather settled."

Derek shook his head.

"No."

He stepped toward the carriage.

"That isn't settled at all."

Chapter 11

The Hologram Problem

Derek was still staring at himself. The other Derek was still sitting on the mysterious train. The train was still sitting in the distant station.

The empty station clock still read 11:17. The cucumber was still in Penny's hand. Trevor was still trying very hard not to say anything stupid.

This was proving difficult. The other Derek raised the ordinary cucumber again. Derek raised an eyebrow.

"Why has he got a cucumber?"

Penny regarded him carefully.

"Why have you got a cucumber?"

"I haven't."

She held hers up.

"No. You have."

Derek glanced down. He was holding a cucumber. He stared at it.

Then at Penny. Then at the other Derek.

"Ah."

Trevor sighed.

"That clears everything up."

The other Derek smiled. Then the train doors closed. The train departed.

Derek ran after it.

"Wait!"

The empty train disappeared into the deep tunnel. Derek stopped. He watched the empty track.

Then at the cucumber in his hand.

"Right."

Penny walked over.

"You've got a cucumber."

"Apparently."

"Where did you get it?"

"I don't know."

"You were holding it."

"I know."

"Then you got it somewhere."

"That's annoyingly logical."

Trevor surveyed the dark tunnel.

"Perhaps we should ask whether the other Derek is real."

Derek met his gaze.

"Excellent question."

Penny nodded.

"How would we know?"

Trevor thought.

"If he can interact with things..."

"He can."

"If he has a worldline..."

"Apparently."

"If he has proper time..."

"Unknown."

"If he can eat a cucumber..."

Derek regarded him.

"Why would that prove anything?"

Trevor shrugged.

"It would at least be satisfying."

Penny laughed. Derek examined the cucumber. Then at the dark tunnel.

Then at the clock. 11:17. He suddenly smiled.

"Maybe we've been asking the wrong question."

Penny nodded.

"Again?"

"Yes."

"What is it this time?"

"Not whether the other Derek is real."

"Then?"

"Whether reality itself is real in the way we think it is."

Trevor frowned.

"That's rather broad."

"Yes."

"How broad?"

Derek inspected the ceiling.

"Potentially four-dimensional."

Penny smiled.

"Or more."

Derek nodded.

"Or fewer."

Trevor looked sharply concerned.

"Fewer?"

"Don't worry."

"That's not reassuring."

Derek began walking.

"Come on."

"Where?"

"Back to the Bureau."

"Why?"

"Because I have an idea."

Penny followed. Trevor followed Penny. The cucumber came too.

The idea was not Derek's.

That was the first thing he admitted when they reached the Bureau.

"The holographic principle."

Trevor wearily sat down.

"That's not your idea."

"I know."

"Who proposed it?"

"Several people contributed to the development."

"Fine."

"Gerard 't Hooft and Leonard Susskind are particularly important."

Trevor nodded.

"And Witten?"

"Edward Witten is important to the later theoretical development connecting holography and string theory."

Penny considered Derek.

"You've been studying."

"I have."

"That's worrying."

Derek ignored her. He went to the whiteboard.

"The basic idea is astonishing."

He drew a cube. Then a flat surface around it.

"Imagine that all the information describing what happens inside a volume can, in principle, be encoded on the boundary."

Trevor nodded.

"The number of degrees of freedom scales with area rather than volume."

"Exactly."

"Black-hole entropy is the clue."

"Yes."

Derek wrote:

$$S \propto A$$

Then:

"Not volume."

Trevor nodded.

"Interesting."

Penny glanced at the equation.

"So, the universe could be described by information living on a boundary."

"Possibly."

"Like a hologram."

"That's the analogy."

Derek turned.

"But we need to be careful."

"Why?"

"Because saying 'the universe is a hologram' is much stronger than saying that certain theories have holographic descriptions."

Trevor nodded.

"So, we shouldn't actually claim we're living inside a giant cinema screen."

"Correct."

Penny looked disappointed.

"That would have made the paperwork easier."

Derek smiled.

"And there are no strings attached."

Trevor stared at him.

"Really?"

Derek smiled.

"No."

Trevor groaned. Penny laughed. Derek continued.

"String theory is one route by which holographic duality becomes extraordinarily powerful. The famous example is the AdS/CFT correspondence."

He wrote: $\text{AdS} \leftrightarrow \text{CFT}$

"A gravitational theory in a higher-dimensional spacetime can be equivalent to a non-gravitational quantum theory living on its lower-dimensional boundary."

Trevor nodded.

"So, gravity appears on one side but not the other."

"Exactly."

"That's extraordinary."

"Yes."

"And if that works..."

"Then perhaps spacetime itself is emergent."

Penny stared at him.

"Emergent from what?"

"Information."

"Quantum degrees of freedom."

"Relationships."

"Entanglement."

"Possibly."

Derek stepped back. The board was now covered. Spacetime.

Curvature. Worldlines. Information.

Entanglement. Holography. He gazed at the ordinary cucumber.

"And now we have a problem."

Trevor softly sighed.

"What?"

"If spacetime is emergent..."

"Yes?"

"Can Special Relativity still be true?"

Trevor thought.

"Why not?"

"Can General Relativity?"

"Why not?"

"Can both?"

"Why not?"

Derek smiled.

"Quite."

Penny nodded.

"The laws we observe don't necessarily have to disappear merely because there's a deeper description."

"Correct."

Trevor consulted the board.

"The temperature of a gas is real even though it's emergent from microscopic motion."

"Exactly."

"Fluid pressure is real."

"Yes."

"Sound waves are real."

"Yes."

"Yet none of those are fundamental microscopic objects."

"Correct."

Derek pointed at the word SPACETIME.

"So, perhaps spacetime could be like that."

Penny nodded.

"Real without being fundamental."

"Exactly."

Trevor considered it in silence.

"Then SR could describe the local behaviour of the emergent spacetime."

"Yes."

"And GR could describe its large-scale geometry."

"Yes."

"While the deeper holographic theory describes the underlying information."

"Exactly."

Derek smiled.

"Well, that's settled."

Penny regarded him carefully.

"Not quite."

Derek paused.

"Why?"

"Because we still have the cucumber."

He looked down. The cucumber was sitting on the table. Derek stared at it.

"Yes."

Penny picked it up.

"If spacetime is emergent..."

"Yes."

"And the cucumber is made of atoms..."

"Yes."

"And the atoms are described quantum mechanically..."

"Yes."

"And the geometry in which those atoms exist might itself be emergent..."

"Yes."

"Then the cucumber is an emergent object in an emergent spacetime."

Derek smiled.

"That sounds unnecessarily complicated."

Penny shrugged.

"It's a cucumber."

Trevor nodded.

"That simplifies things."

Derek laughed.

"Does it?"

"Yes."

"How?"

"We know it exists."

Derek watched him.

"Do we?"

Trevor held up the cucumber.

"Yes."

"How?"

Trevor considered this.

"It's here."

Derek smiled.

"Excellent epistemology."

Penny took the ordinary cucumber from him.

"Perhaps we should test it."

Derek looked suspicious.

"How?"

"Measure it."

"With what?"

"A scale."

"Fine."

She weighed it. Trevor recorded the result. Penny measured its length.

Then circumference. Then temperature. Then mass distribution.

Derek intently watched.

"What are we proving?"

"Nothing."

"Good."

"We're establishing that the cucumber has observable properties."

"Yes."

"And if the holographic description is correct, those properties must ultimately be encoded in the underlying degrees of freedom."

Derek nodded.

"Yes."

"So, the cucumber can be described in more than one way."

"Like spacetime."

"Just so."

Trevor fell into thought.

"Could the cucumber itself have a holographic description?"

Derek smiled.

"Of course."

"Really?"

"In principle."

Trevor surveyed it.

"That's extraordinary."

Derek nodded.

"The information describing the physical system can have different representations."

Penny smiled.

"Just don't tell the cucumber."

"Why?"

"It may become arrogant."

Derek checked his watch. Still 11:17. Somewhere, he suspected, a different representation of the same eleven minutes was already causing trouble.

Chapter 12

Everything Everywhere All at Once

Penny had decided that Derek was becoming unbearable. Not because he was wrong. That would have been easier.

It was because he was usually right. There was something deeply irritating about a man who could spend an afternoon discussing curved spacetime, singularities and the holographic principle and then, when Penny pointed out an obvious practical difficulty, simply say:

“Yes.”

As though the universe had just been waiting for him to catch up. So, Penny went to the cinema. Alone.

This was not particularly unusual. What was unusual was the film she chose.

Everything Everywhere All at Once.

She settled into her seat with popcorn and a drink and watched the opening credits. Then she saw her. Penny sat upright.

“Blimey.”

Jamie Lee Curtis. She had survived Halloween after all. Apparently, having spent several films being pursued by a man who refused to die, she had decided that the next sensible career move was to become an IRS employee and spend her time tormenting bewildered Asians across multiple universes.

Penny smiled.

“Americans.”

She took a handful of popcorn. The film began. Within minutes there were universes.

Then other universes. Then other versions of the same people. Then universes in which perfectly ordinary things had become completely ridiculous.

Penny watched. She became increasingly interested. Not because the film was physically accurate.

It wasn't. Nor was it trying to be. But the underlying idea was strangely familiar.

Different possible realities. Different versions of people. Different histories.

Different outcomes. Different choices. A universe in which one thing happened.

Another in which it didn't. Penny thought about Derek. She thought about the clock. She thought about the distant train that apparently existed in one description of reality and didn't in another. And about Weinstein's question...

If spacetime itself could be emergent, what exactly did it mean for there to be one universe? She leaned back. The film continued.

Michelle Yeoh's character moved through wildly different versions of reality. Penny smiled. "Right."

A woman sitting several seats away glanced at her. Penny lowered her strange voice.

"This is actually rather relevant."

The woman looked away. Penny continued watching. The film's multiverse was not the same thing as the multiverse physicists sometimes discuss.

That distinction mattered. But it was useful as an intuition pump. Suppose reality could be described in many different ways.

Suppose an observer occupied one particular history. Suppose other histories existed in some mathematical sense. Then the question became:

What was fundamental? The particular history? The collection of histories?

The mathematical structure connecting them? Or something deeper still? Penny thought about holography.

If a higher-dimensional gravitational description could be equivalent to a lower-dimensional quantum description, then perhaps what looked sharply like an enormous four-dimensional reality from one perspective could be represented by something radically different from another. Not a movie. Not a projection in the everyday sense.

A mathematically equivalent description. Two apparently different descriptions. One underlying physics.

She smiled.

"Now we're getting somewhere."

On screen, the characters continued jumping between realities. Penny watched Jamie Lee Curtis.

"Still alive."

She shook her head.

"She really does have remarkable career longevity."

Then another universe suddenly mysteriously appeared. Penny frowned. Something about it bothered her.

Not the physics. The logic. If every possible version of a person existed, what distinguished one version from another?

Their history. Their state. Their relationships.

Their information. Penny sat forward. Information.

That was the word. Everything came back to information. A physical system could be described by its state.

A clock had a state. An ordinary cucumber had a state. A person had a state.

A train had a state. A universe had a state. And perhaps the geometry of spacetime itself emerged from relationships between quantum states.

She took out her phone. She typed: Holography: information on boundary → bulk geometry?

She typed a second line, read it back, and deleted it. It had sounded profound for exactly four seconds before revealing itself as nonsense.

Then: Could different descriptions correspond to same physics? Then:

SR and GR survive because they are emergent descriptions? She paused.

Then added:

Ask Alfred. She put the phone away. On screen, Jamie Lee Curtis was once again doing something completely unreasonable.

Penny smiled.

“Americans.”

A few minutes later she laughed. Then laughed again. Then stopped.

Because something on the screen had triggered a thought. The characters weren't merely travelling through space. They were switching between descriptions of reality.

The important thing was not that they suddenly moved from one physical universe to another. The important thing was that the information defining the character changed. Different memories.

Different relationships. Different physical histories. Different versions of the same person.

Penny thought about worldlines. A worldline was a history through spacetime. But if spacetime itself was emergent, perhaps the deeper theory did not contain worldlines at all.

Perhaps worldlines were emergent too. That would mean that Derek wasn't fundamentally a line through spacetime. He was a particular physical state, embedded in a network of relationships.

His apparent history was one way of describing that state and its correlations. Penny smiled.

"That explains the other Derek."

Then she frowned.

"No, it doesn't."

If Derek was simply a state rather than a line, there was nothing in the mathematics that said there could only be one of him. Two identical states, two separate histories, no contradiction anywhere in the equations. That was the trouble with elegant explanations. They had an alarming tendency to explain slightly too much.

The woman beside her looked over again.

Penny whispered:

"Sorry."

The film continued. Penny watched. Suddenly the multiverse stopped being merely funny.

It became useful. Not as physics. As a question.

If there were many mathematically possible descriptions of reality, what made one description equivalent to another? What had to remain unchanged? The answer, she realised, had to involve observables.

Physics couldn't depend on what coordinates someone chose. It couldn't depend on whether someone described the world using one mathematical language or another. The measurable predictions had to agree.

That was already familiar from relativity. Coordinates could change. Physics didn't.

A Lorentz transformation changed the coordinates assigned to events.

It didn't change the invariant interval. A different coordinate system could make the mathematics look different. The physics remained.

Perhaps holographic duality was a much more radical version of the same philosophical lesson. One description could have gravity and curved spacetime. Another could have quantum fields without gravity.

Yet they could encode the same physics. Penny sat back.

"Bloody hell."

The woman beside her looked again. Penny smiled.

"Physics."

The woman nodded uncertainly. On screen, everything became even more absurd. Penny laughed.

Then she suddenly remembered something. The cucumber. She had left it at the office.

“That’s the straw that broke the camel’s toaster.” Penny turned to him. “I don’t think that proverb has ever existed.” Derek nodded. “Neither does the train, apparently.”

A mysterious message from Derek.

WHERE ARE YOU?

Penny typed: Cinema. Three dots appeared.

WHAT ARE YOU WATCHING?

She replied:

Everything Everywhere All at Once.

The reply came almost immediately.

WHY?

Penny smiled. She typed: Because you keep talking about holography and I wanted to see what the Americans think it means.

There was a pause. Then:

AND?

Penny stared at the screen. She typed: Jamie Lee Curtis survived Halloween.

Another pause. Then:

THAT’S YOUR PHYSICS CONCLUSION?

Penny smiled.

No.

She gazed at the screen. Then typed: My physics conclusion is that if different descriptions can encode the same underlying reality, SR and GR don’t need to be fundamental to be true.

Derek replied:

Correct.

Penny smiled. Then another sudden message mysteriously appeared.

ALSO, JAMIE LEE CURTIS DID NOT “SURVIVE HALLOWEEN” IN THE PHYSICS SENSE.

Penny frowned. She typed: You know what I meant.

I don’t. Americans. A pause.

Then: That settles it. Penny laughed.

She put the phone away. Then she noticed something. On the screen, a character was holding a cucumber.

Penny silently stared. Surely not. She leaned forward.

There it was. A cucumber. She laughed.

"You're having a laugh."

The woman beside her studied the screen.

"What?"

"The cucumber."

"What about it?"

Penny stared intently.

"Nothing."

She reached into her handbag. Her own cucumber was still there. She looked from the ordinary cucumber in her bag to the cucumber on the screen.

Then back again. She frowned. The one on the screen was not identical.

Of course it wasn't. That would have been ridiculous. Still...

Penny took a vivid photograph of the screen. She sent it to Derek.

LOOK.

His reply came almost immediately.

I'M LOOKING.

Then:

THAT'S A CUCUMBER.

Penny smiled.

Yes.

So? She typed: Nothing.

She paused. Then: Yet.

She put the phone away. The film continued. Penny watched the characters collide with one absurd reality after another.

Then she thought about the Bureau. About Derek. About Weinstein.

About Barbarian. About the clocks. About the cold rain.

About the train. About the other Derek. Perhaps the Relativistic Investigation Bureau had been approaching the problem backwards.

They had assumed that something strange was happening inside spacetime. What if the strange thing was the relationship between different descriptions of spacetime? What if the clocks weren't moving incorrectly?

What if they belonged to different descriptions? What if the mysterious train wasn't travelling through a strange geometry? What if the train was the strange geometry?

Penny wearily sat very still. Then she smiled. She took out her phone.

She wrote one final sudden message to Derek. Don't investigate the clocks. His reply came:

Why? She typed: Investigate the descriptions.

A pause. Then: Explain.

Penny examined the film. Then at the ordinary cucumber. Then at the rain running down the cinema window.

She wrote: Maybe the universe doesn't have one story. She thought for a moment.

Then added:

Maybe it has several descriptions that all have to agree on what can actually happen.

Derek replied:

Now you're talking. Penny smiled. She put the phone away.

On the screen, everything was happening everywhere all at once. Penny watched for another minute.

Then quietly mentioned:

"Americans."

She ate another piece of popcorn.

"That settles it."

And somewhere in London, a clock stopped at 11:17. Again.

Chapter 13

The Station That Wasn't Quite There

The first thing Derek noticed was that the gloomy station had twelve doors. The second thing he noticed was that it had thirteen. The third thing he noticed was that Trevor was counting them.

"How many?"

"Twelve."

Derek pointed.

"Thirteen."

Trevor counted again.

"Twelve."

Penny looked.

"Thirteen."

Trevor looked again.

"Twelve."

Derek nodded.

"Good."

Trevor sharply frowned.

"Good?"

"At least we're all seeing the same disagreement."

They were standing in a strange corridor beneath London. It looked like an ordinary Underground corridor. This was, naturally, suspicious.

The tiles were ordinary. The fluorescent lights were ordinary. The smell was ordinary.

Even the damp patch on the wall was reassuringly ordinary. But the narrow corridor was wrong. Not dramatically wrong.

That would have been easier. Dramatically wrong things announce themselves. They have the decency to be obvious.

This narrow corridor was merely wrong enough to make a sensible person uncomfortable. Derek walked back to the entrance and eyed the sign. There was no station name.

Only:

PLATFORM 7

Penny watched it.

"We're already on Platform Seven."

Derek nodded.

"Apparently."

Trevor looked behind them. There was no Platform Seven. There was only the narrow corridor.

"Perhaps it's a service corridor."

Derek considered this.

"Then why does it have a platform number?"

Trevor shrugged.

"British Rail?"

"It's the Underground."

"Even worse."

Penny smiled.

"Glad that's cleared up."

Derek ignored them. He took a tape measure from his coat. Trevor surveyed him.

"You carry a tape measure?"

"I investigate things."

"You carry one everywhere?"

"Not everywhere."

"Where don't you carry it?"

Derek thought.

"The shower."

Penny studied him.

"Why not?"

Derek stared.

"Because I have limits."

They measured the strange corridor. Forty-seven metres.

Exactly.

Derek wrote it down. Then he walked from one end to the other. Forty-seven metres.

He walked back. Forty-seven metres. Penny did the same.

Forty-seven metres. Trevor did the same. Forty-seven metres.

Derek nodded.

"Good."

Trevor regarded him.

"What's good about forty-seven metres?"

"Nothing."

"Then why —"

"Because it's the same."

Trevor wearily sighed.

"Right."

Derek examined the doors.

"Now let's count them again."

They did. Twelve. Then thirteen.

Then twelve. Then thirteen. Penny folded her arms.

"Perhaps we're counting different things."

Derek smiled.

"Excellent."

He pointed to the first door.

"Describe it."

"Brown."

"Number?"

"One."

"Next?"

"Two."

"Next?"

"Three."

They continued. Every door was ordinary. Every number was ordinary.

Every distance between them was ordinary. And yet, somehow, the corridor contained either twelve or thirteen doors depending on who was doing the counting. Derek took out a notebook.

"Trevor."

"Yes?"

"Walk to the end."

Trevor did.

"How many?"

"Twelve."

"Penny?"

"It's not rocket surgery." Trevor fixed him with a look. "That is not reassuring."

"Derek?"

He counted.

"Twelve."

Penny smiled.

"So, it's me."

Trevor inspected her.

"You're the problem."

"It would seem so."

Derek shook his head.

"No."

They gave him a measured look.

"The problem is that we're assuming the number of doors is an invariant."

Trevor frowned.

"A what?"

"Something every observer must agree on."

Penny nodded.

"Like the spacetime interval."

"Exactly."

Trevor glanced at the doors.

"So, door-counting isn't invariant."

"Not if the corridor isn't behaving like ordinary Euclidean space."

Trevor looked suspicious.

"Are you saying we've found curved space?"

"Perhaps."

"Underground?"

"Yes."

"In London?"

"Yes."

Trevor considered this.

"That seems unnecessarily ambitious."

Derek smiled.

"It's London."

"Fair."

They slowly moved farther down the narrow corridor. Derek stopped beside one of the doors.

"Open it."

Trevor did. There was a small room inside. Two chairs.

A table. Two clocks. Nothing else.

Derek entered. Penny followed. Trevor remained in the doorway.

"What?"

Derek stared at the clocks. One read: 11:17

The other: 12:04 Trevor stared silently.

"Again?"

Derek nodded.

"Again."

Penny stepped inside.

"Are they broken?"

Derek checked them.

"No."

"Different mechanisms?"

"No."

"Different batteries?"

"No."

He compared them carefully. Identical. Same make.

Same model. Same calibration. Same temperature.

Same reference frequency. Same everything. Except the time.

Trevor gazed at the two clocks.

"They're less than two metres apart."

"Yes."

"And they're disagreeing by forty-seven minutes."

"Yes."

Trevor considered this.

"That's not good."

"No."

"Could one be in a different gravitational field?"

Derek smiled.

"Good question."

He took out an instrument. They measured the local gravitational field. Nothing unusual.

They measured again. Nothing. Derek moved the 12:04 clock three metres down the corridor.

The difference changed. Trevor's eyebrows rose.

"Did you see that?"

"Yes."

Derek moved it back. The difference shrank. Penny peered at him.

"That's gravitational."

"Something gravitational."

"But the field is nowhere near strong enough."

"Correct."

"So?"

Derek eyed the two clocks.

"The important thing may not be the strength of the gravitational field."

Penny nodded.

"The variation."

"Exactly."

Trevor looked confused.

"Explain."

Derek leaned against the table.

"Imagine two people falling toward Earth."

"All right."

"They are both accelerating toward the centre of Earth."

"Yes."

"But the person slightly to the left is falling in a slightly different direction from the person slightly to the right."

"Because they're pointing toward different parts of the Earth."

"Exactly."

"So, their paths slowly change relative to each other."

"Yes."

"That's a tidal effect."

"Precisely."

Penny nodded.

"Geodesic deviation."

Derek smiled.

"Precisely."

Trevor looked from one mysterious clock to the other.

"So, gravity isn't just about how strong it is at one point."

"Correct."

"It's also about how it changes across space."

"Yes."

"And that's curvature."

"Now you're getting it."

Trevor smiled.

"Well, that's one theory down."

The lights flickered. All three looked up. The lights went out.

For one second there was complete darkness. Then they came back on. The clocks were still there.

The chairs were still there. The table was still there. But the narrow corridor outside had changed.

Derek stepped out. There were fourteen doors. Trevor counted.

"Fourteen."

Penny counted.

"Thirteen."

Derek counted.

"Fourteen."

They stared at each other. Trevor pointed at Penny.

"You're doing it again."

Penny smiled.

"Apparently."

Derek watched the corridor.

"Something has changed."

"What?"

"The geometry."

Trevor frowned.

"How can you see geometry?"

"You don't see it directly."

"Then how do you know?"

"Because measurements don't fit together the way they should."

Penny nodded.

"That's the important bit."

She walked to the end of the narrow corridor. Then back.

"Forty-seven metres."

Derek measured.

"Forty-seven."

Trevor measured.

"Forty-seven."

Derek slowly surveyed the doors.

"So, the length hasn't changed."

"No."

"The number of doors has."

"Yes."

"The clocks have changed."

"Yes."

"The gravitational gradient has changed."

"Yes."

Trevor looked increasingly uncomfortable.

"And you're saying all of this can happen because spacetime is curved?"

Derek nodded.

"That's what General Relativity permits."

"But not arbitrarily."

"Correct."

"The geometry has to satisfy Einstein's equations."

"Yes."

"So, this isn't magic."

"No."

Trevor looked at the fourteen doors.

"It looks remarkably like magic."

Derek smiled.

"That's because you haven't seen enough mathematics."

Penny laughed. They continued. At the far end of the corridor was a wall.

No door. No sign. Nothing.

Derek touched it. Solid. Penny knocked.

Solid. Trevor knocked. Solid.

Derek stepped back.

"Move."

Trevor did. Derek took three steps backward. Then ran toward the wall.

Penny shouted:

"Derek!"

He hit the wall. Except he didn't. He went through it.

Penny stared intently. Trevor stared. Derek disappeared.

There was silence. Trevor regarded Penny.

"He's gone."

Penny nodded.

"Yes."

"Should we be worried?"

"No."

"Why not?"

"His office door was closed."

Trevor nodded.

"Ah."

A voice came from the other side of the wall.

"Come through."

Penny walked toward the wall. She touched it. Her hand went through.

She smiled.

"Interesting."

Trevor looked horrified.

"I don't think that's a property walls normally have."

Penny examined him.

"Nothing does until it does."

She walked through.

Trevor sighed.

Then followed. On the other side was another strange corridor. Much longer.

Much darker. And, unlike the first strange corridor, it contained no doors at all. At the far end was a station platform.

A sign read:

COLD PLATFORM 7

Derek stood beside it. Trevor joined him.

"Where are we?"

Derek took in the room.

"Underground."

"We were already underground."

"Yes."

"So?"

Derek pointed upward. They could hear rain. Trevor looked confused.

"We're beneath London."

"Yes."

"And it's raining above us."

"Yes."

"How can you hear it?"

Derek smiled.

"Good question."

Penny inspected the floor. Water was running along the edge of the platform. She crouched.

"It isn't just rain."

Derek joined her.

"What?"

"The water."

She pointed. It was flowing toward the drain. Then it stopped.

Completely. Not slowed. Stopped.

A drop hung at the edge of the drain. Trevor crouched.

"Is that normal?"

"No," Derek responded.

The drop moved backward. Then forward. Then sideways.

Then fell into the drain. Trevor stared.

"Right."

Derek stood.

"The local geometry is doing something strange."

Penny considered him.

"Can we distinguish that from a coordinate effect?"

"Not yet."

"How?"

"Find something invariant."

Trevor glanced around.

"The cucumber."

They all stared at him. He shrugged.

"You've used it before."

Penny softly smiled.

"Good thinking."

She took the cucumber from her bag. Derek stared.

"You brought it?"

"Obviously."

"Why?"

"You never know."

Trevor nodded.

"That resolves it, then."

They placed the ordinary cucumber on the platform. Derek measured its length. Exactly thirty-one centimetres.

Penny measured it. Exactly thirty-one centimetres. Trevor measured it.

Thirty-one centimetres. Derek nodded.

"Stationary."

"Yes."

"Now let's move it."

He placed it on a small wheeled platform. They pushed it along the track. Penny measured again.

Thirty centimetres. Trevor frowned.

"Length contraction."

"Exactly."

"Special Relativity."

"Yes."

Derek pushed it faster. Twenty-eight centimetres. Trevor smiled.

"Still Special Relativity."

"Yes."

Then they stopped. The cucumber returned to thirty-one centimetres. Perfectly ordinary.

Penny smiled.

"Excellent."

Derek nodded.

"Now we know the cucumber works."

Trevor stared at him.

"What does that mean?"

"It obeys relativity."

"Of course it does."

Derek glanced at the empty strange station.

"Now we need to find out whether the geometry itself is changing."

He placed the cucumber at one end of the platform. Then another.

The measured lengths changed slightly depending upon position. Trevor stared at the readings.

"That's not length contraction."

"No."

"Because the cucumber isn't moving."

"Correct."

"So, the spatial geometry is changing."

"Yes."

Penny stared at the track.

"Could the gravitational field be responsible?"

"Possibly."

"The tidal field?"

"More likely."

Trevor took in the room.

"How much curvature would we need?"

Derek smiled.

"More than London normally provides."

"That's reassuring."

"Unfortunately..."

Derek pointed down the track. The rails were bending. Not physically.

They remained straight to the eye. But the measured distance between them was changing. Penny stared intently.

"That's curvature."

Derek nodded.

"Yes."

Trevor swallowed.

"So, the Underground has become a General Relativity experiment."

"Apparently."

"Is that safe?"

Derek considered it.

"Probably."

Trevor waited.

"Probably?"

Derek smiled.

"Would you prefer 'possibly'?"

"No."

"Then probably."

The station lights flickered again. The clocks all stopped. 11:17.

Penny gazed at her phone. 11:17. Trevor checked his.

11:17. Derek checked his watch. 11:17.

The cucumber began to tick. Trevor looked at it.

"That can't be good."

Derek smiled.

"Nothing is good at 11:17."

A train appeared in the tunnel. No headlights. No sound.

It simply suddenly appeared. Penny stood.

"Where did that come from?"

Derek studied the rails.

"That's the wrong question."

"What is the right one?"

"Which worldline did it follow?"

Trevor stared at him.

"The train has a worldline?"

"Everything physical does."

"Even the cucumber?"

"Yes."

Penny watched the ordinary cucumber.

"Even the cucumber."

Trevor nodded slowly.

"Splendid. Moving on."

The train approached. It stopped. The doors opened.

Nobody got off. Nobody got on. Inside, the carriage was completely empty.

Except for a single seat. And on the seat was another cucumber. Derek stared at it.

Penny stared at Derek. Trevor stared at Penny. Nobody offered anything.

Then Derek smiled.

"Well."

Penny folded her arms.

"Don't."

"I haven't offered anything."

"You were about to."

Derek surveyed the second cucumber.

"All right."

He stepped into the carriage. Penny followed.

Trevor wearily sighed.

Then followed them. The doors quietly closed. The mysterious train began to move.

Behind them, the station disappeared. Ahead, the tunnel curved. And somewhere between London and somewhere that London had apparently forgotten to put on the map, the clocks began moving again.

11:18. 11:19. 11:20.

Derek studied his watch. It still read 11:17. He smiled.

"There. Progress."

The mysterious distant train accelerated into the darkness. And for the first time, Derek began to suspect that the strangest thing about the empty station was not that its geometry was curved. It was that something appeared to be curving it on purpose.

Chapter 14

The Distant Train That Arrived Before It Left

The narrow corridor ended at cold Platform Seven. Derek stopped.

"Of course."

Penny regarded the sign.

Platform 7

Trevor examined the timetable.

"There isn't a Platform Seven."

"The early bird gets the worm," Trevor observed. Penny stared at him. "The early bird gets the worm." Trevor nodded. "Yes. Obviously. I was testing the bird."

"Excellent."

Trevor stared at him.

"Why is that excellent?"

"Because if there were a Platform Seven, this would be considerably less interesting."

Penny looked sharply down the platform. Before she could say anything, a woman's voice came from behind them.

"If you're planning to take that train, I'd recommend knowing where it goes first."

All three turned. The woman standing at the end of the lonely platform looked entirely too calm for someone who had just appeared on a railway platform that officially did not exist. She wore a dark coat, carried a small overnight bag, and had the expression of someone who had arrived expecting trouble and was mildly disappointed to find that it had already started.

Trevor stared intently.

"Tabitha?"

She smiled.

"Hello, little brother." She said it as if the relationship were still under review.

Derek studied Trevor.

"Your sister?"

"Unfortunately," Trevor opined.

Tabitha raised an eyebrow.

"I heard that."

"You were meant to."

She inspected Derek.

"So, you're Derek."

"Yes."

"Trevor maintained you were the sensible one."

Derek glanced at Trevor.

"He has never affirmed that to me."

"Good," Tabitha confirmed. "I'd hate to begin with a false impression."

Penny looked from Tabitha to Derek. She remained silent. She was very good at saying nothing in ways that made Derek nervous.

Derek found himself noticing that Tabitha was watching him. Then she looked away. She had the calm of someone who could leave at any moment and be late for something better. He decided not to think about that. The decision proved only partially successful, which was becoming a pattern.

Which was, in retrospect, the first sign that this was going to be a problem.

"How did you get here?" Derek asked.

Tabitha held up her phone.

"Trevor sent me a message."

Trevor frowned.

"I didn't."

Tabitha scanned the screen.

"Apparently someone using your number did."

Silence. Derek held out his hand.

"What did it say?"

Tabitha showed him.

DON'T LET WEINSTEIN GET TO THE STATION FIRST.

Derek's expression changed.

"When did you receive that?"

"Eleven minutes ago."

Trevor glanced at Derek.

"Weinstein is in Munich."

"Apparently not," Derek insisted.

Tabitha slipped the phone back into her pocket.

"So," she emphasized, looking at the train, "which part of this is supposed to be normal?"

Derek smiled.

"We're still looking for that part."

Tabitha smiled back.

"Good. I was worried I'd missed it."

There was, in fact, a distant, mysterious train waiting. This was unfortunate. It was an entirely ordinary Underground train.

Grey. Damp. Slightly depressing.

The sort of train that looked as though it had been designed by a committee whose principal objective had been to discourage tourism. Except for one small problem. There was no driver.

There were no passengers. There was no conductor. There was no staff.

And there was no record of the train on the timetable. Trevor checked the timetable again. Nothing.

He checked his phone. Nothing. He checked the transport app.

Nothing. He stared at Derek.

"Perhaps it's cancelled."

Derek gazed at the mysterious train.

"It's here."

"Yes."

"Therefore it isn't cancelled."

Trevor considered this.

"That's not how cancellations work."

"It is if the train exists."

"Does it?"

Derek smiled.

"Excellent question."

Tabitha stopped them.

"One moment."

She pointed at the destination board.

"London Terminus."

Derek nodded.

"Yes."

"Is that actually a destination?"

Trevor peered at the sign.

"Apparently."

Tabitha shook her head.

"No. It is a description, not a promise. Most forms of transport prefer the promise. This one appears to have skipped that stage. The train isn't telling us where it is going. It is telling us where its worldline ends."

Derek gave her a measured look. For a moment he remained silent.

"That's rather good."

Tabitha smiled.

"I was hoping you would say that."

Penny walked toward it. The destination board read:

LONDON TERMINUS

She stopped.

"Where is London Terminus?"

Derek watched the sign.

"Everywhere."

Trevor sighed.

"That isn't helpful."

"It wasn't intended to be."

Penny surveyed the mysterious train.

"There's no such station."

Derek nodded.

"Correct."

"Then why does the train say it's going there?"

"Perhaps it's optimistic."

The doors carefully opened. Nobody moved. Derek stepped inside.

Penny followed. Tabitha followed Penny, already looking as if she belonged on the train more than the rest of them. Trevor remained on the lonely platform.

Derek turned again.

"Coming?"

Trevor looked at the empty carriage.

"Do you know where it goes?"

"No."

"Do you know who is driving it?"

"No."

"Do you know why it exists?"

"No."

Trevor wearily sighed.

"Fine."

He stepped aboard. Tabitha regarded him.

"You know, for someone who asked whether it has a driver, you seem remarkably willing to get on it."

"I am beginning to suspect that refusing to board would be less safe."

"Good reasoning."

The doors quietly closed. The mysterious train moved. There was no announcement.

No warning. No reassuring strange voice telling them to mind the gap. It simply accelerated.

Derek studied his watch. 11:17. The train clock showed 11:17.

Penny's phone showed 11:17. Trevor checked his. 11:17.

"A day late and a dollar sideways," Derek remarked. "That is becoming the Bureau's unofficial motto."

Derek watched the carriage.

"Acceleration."

Trevor nodded.

"We can feel it."

"Yes."

"And that means this isn't an inertial frame."

"Correct."

Penny looked out of the window.

"The station is accelerating away from us."

Derek smiled.

"Depending on your frame."

Trevor nodded.

"Right."

The mysterious train continued into the narrow tunnel. Derek took out his notebook.

"Let's keep this simple."

Trevor examined him.

"That's unusual."

"Special Relativity first."

He wrote:

INERTIAL FRAME

Then:

LORENTZ TRANSFORMATION

Then:

PROPER TIME

Penny nodded.

"The train has its own proper time."

"Yes."

"The passengers would have their own worldlines."

"Yes."

"The station has another."

"Yes."

"And the clocks measure proper time along their respective worldlines."

"Exactly."

Trevor nodded.

"Fine."

The train suddenly slowed. Derek inspected the silent clock. 11:17.

The mysterious train stopped. The doors opened. They stepped onto the cold platform.

Trevor surveyed the room. His expression changed.

"This is Platform Seven."

Derek nodded.

"Yes."

"But..."

Trevor pointed down the tunnel.

"We just left here."

"Yes."

"And now we're back."

"Yes."

Trevor checked his phone. 11:18. Derek checked his watch.

11:18. Penny checked hers. 11:18.

The empty station clock slowly read: 11:17. Trevor stared intently.

"So, the station clock hasn't advanced."

"No."

"But our clocks have."

"Yes."

"And we were moving relative to the station."

"Yes."

Trevor considered Derek.

"Time dilation."

"Possibly," Derek queried.

Tabitha glanced at the empty station clock.

"Why do you keep treating that one as the correct clock?"

Derek turned toward her.

"Because it stayed at the station."

"So?"

"It is the stationary reference clock."

Tabitha smiled.

"Stationary relative to what, exactly?"

Trevor stopped. Penny stared at the clock.

"She's right." Derek found the observation more interesting than he had expected.

Derek stared at the empty strange station clock. Tabitha continued.

"The clock isn't the universe's clock. It is a clock following the station's worldline. Our clocks followed ours."

Trevor's expression changed.

"So, there isn't a privileged timekeeper."

"Exactly," Tabitha said. "The universe doesn't keep a master clock. It only keeps local ones. People keep inventing the master clock. The universe never asked for one." Derek, for reasons he chose not to examine, found himself watching Tabitha rather than the clock. The fact that her presence made the situation feel slightly more manageable was, he decided, an unhelpful development. Manageable was not a feeling he trusted. Manageable things had a habit of becoming unmanageable the moment you looked away. He filed the observation under 'later' and suspected 'later' was already oversubscribed. In his experience they rarely came with a return address either, and almost never with an apology. The universe was consistent on this point as well. Consistency, Derek had noticed, was not the same as kindness.

Penny nodded.

"And if we want to compare the clocks properly, we have to compare the paths they actually took through spacetime."

Derek smiled.

"Proper time."

Trevor pointed at the clocks.

"So, the difference isn't evidence that one clock is wrong."

"Correct," Derek stated.

Tabitha looked down the tunnel.

"Then perhaps the train isn't carrying us through someone else's time. Perhaps it is carrying us along a different path."

Silence. Derek gazed at her.

"Tabitha."

"Yes?"

"You may have just made this considerably worse."

She smiled.

"I was afraid of that." Derek discovered he was not entirely sorry. The realisation that he was more comfortable with her presence than without it was, he concluded, going to be a problem for another day — preferably one that had not yet been invented. He suspected it would not remain someone

else's for long. Trevor, for his part, had gone quiet in the specific way that meant he was recalculating several assumptions at once.

"Why possibly?"

"Because we've only observed a difference in clock readings. We still need to understand the geometry of the situation."

Trevor nodded.

"Right."

Penny pointed at the tunnel.

"The train is coming back."

Derek looked. The same train was approaching. Trevor frowned.

"But we just got off it."

"Yes."

"It hasn't had time to go anywhere."

Derek watched the train approach.

"That depends on what you mean by time."

Trevor softly sighed.

"I knew you were going to say that."

The mysterious train stopped. The doors carefully opened. A second Derek stepped out.

Trevor went completely still. Penny stared. Derek stared at Derek.

The second Derek met their eyes. Then at his watch. Then at the station clock.

"You're late."

The first Derek frowned.

"I haven't gone anywhere."

The second Derek smiled.

"I know."

Trevor looked sharply between them.

"Which one of you is Derek?"

Both Dereks turned toward him.

"Me."

Trevor quietly closed his eyes.

"Of course."

Penny eyed the second Derek.

"Where did you come from?"

"The train."

"Where was the train?"

"Here."

"It wasn't."

"It was."

"You just arrived."

"Yes."

"From where?"

"Here."

Penny stared. Derek smiled.

"He's right."

Trevor opened his eyes.

"Don't encourage him."

The second Derek watched the first.

"You're still trying to describe the whole thing with one clock."

Derek nodded.

"Yes."

"Stop."

Derek considered this.

"Fair."

The second Derek pointed toward the empty station.

"Think locally."

Derek surveyed the room.

"Local inertial frames."

"Exactly."

"Special Relativity applies locally."

"Yes."

"But globally the geometry can be curved."

"Yes."

"So, the coordinates can become strange."

"You can't have your cake and beat around the bush too," Penny observed. Derek found her eyes.

"That may be the most British sentence you've ever said."

"But the physical laws remain consistent."

"Quite."

Trevor surveyed Penny.

"Are they always like this?"

Penny smiled.

"Unfortunately."

The second Derek turned to the first.

"You've been asking why the train arrives before it leaves."

"Yes."

"Wrong question."

"What is the right one?"

"Which events are actually connected by a causal curve?"

Derek stared intently. Trevor frowned. Penny nodded.

"Causality."

"Exactly."

Derek lowered his gaze the deep tunnel.

"Light cones."

"Yes."

"The causal future and causal past of an event."

"Yes."

"And events outside the light cone are spacelike separated."

"Exactly."

Trevor nodded slowly.

"So, two observers can disagree about which of two spacelike-separated events happened first."

"Yes."

"But they cannot use that to send information into their own past."

"Correct."

Derek smiled.

"Good. Moving on."

The second Derek studied him.

"For now."

Then he disappeared. Trevor jumped.

"Where did he go?"

Derek regarded the empty platform.

"Virtuality."

Penny nodded.

"Obviously."

Trevor stared at them.

"I am beginning to dislike that word."

The train doors closed. The train departed. This time Derek watched carefully.

He took measurements. Speed. Acceleration.

Distance. Clock readings. Everything suddenly appeared ordinary.

Then the train emerged from the tunnel. They were above ground. Cold rain streaked across the windows.

London suddenly appeared. Grey buildings. Grey sky.

Grey people. Grey everything. Trevor looked sharply relieved.

"Back to normal."

A raindrop hit the window. It travelled downward. Then stopped.

Then moved sideways. Then went upward. Trevor stared intently.

"No."

Penny leaned closer.

"Interesting."

Derek smiled.

"Very."

The droplet split into two. One part moved down. The other moved up.

Trevor examined Derek.

"That isn't normal."

"No."

"Coordinate effect?"

"Possibly."

"Geometry?"

"Possibly."

"Train?"

"Maybe."

"Rain?"

"Possibly."

Trevor sighed.

"You're enjoying this."

"Immensely."

The mysterious train slowed. Derek checked his watch. 11:18.

The mysterious train clock: 11:18. Penny's phone:

11:18. The station clock outside: 12:04.

They stopped. The doors quietly opened. Everyone stepped out.

The cold rain was falling normally again. Trevor inspected the ancient station clock. 12:04.

He checked his phone. 11:18. Then back at the clock.

12:04.

"Now what?"

Derek smiled.

"Now we have a problem."

Penny nodded.

"Good."

Trevor stared at her.

"Why is that good?"

"Because we know the discrepancy is not just between two clocks."

Derek nodded.

"The station itself is assigning different temporal coordinates."

Trevor frowned.

"So which time is correct?"

Derek glanced at him.

"Both."

Trevor sighed.

"Of course."

Derek pointed toward a bench. A newspaper lay there. He picked it up.

The headline read:

RELATIVITY PROVED WRONG

Trevor nodded.

"Finally."

Derek turned the page. Underneath was another headline:

RELATIVITY PROVED RIGHT

Penny laughed.

"That's more like it."

Derek turned the page again. A third headline:

RELATIVITY PROVED BOTH

Trevor stared intently.

"That's impossible."

Derek stared at him.

"No."

Trevor waited. Derek smiled.

"It's journalism."

Penny laughed. They sat on the bench. Derek spread the newspaper across his knees.

"Let's sort this out."

Trevor nodded.

"Please."

Derek drew a spacetime diagram.

"An event is represented as a point in spacetime."

He drew another point.

"Different observers can assign different coordinates to those events."

Penny nodded.

"Because their frames of reference differ."

"Yes."

He drew light cones.

"The speed of light defines the causal structure."

Trevor gazed at the diagram.

"If two events are inside one another's light cones..."

"They can be causally connected."

"And everyone agrees on their causal order."

"Yes."

"But if they're outside one another's light cones..."

"They're spacelike separated."

"And different observers can disagree about which happened first."

"Exactly."

Trevor smiled.

"That explains the train."

Derek nodded.

"Potentially."

Trevor frowned.

"Why potentially?"

"Because our train is also moving through curved spacetime."

"So, General Relativity matters."

"Yes."

"That's the way the cookie bounces," Trevor said. Derek paused. "I would prefer it if the cookie remained where it was."

"Yes."

Penny pointed at the newspaper.

"So, the headline could appear before the article that caused it."

Derek peered at her.

"Not quite."

"Why?"

"Because information cannot travel outside the causal structure."

"So, no actual message from the future."

"Correct."

"Just a strange coordinate description."

"Just so."

Trevor nodded.

"That's reassuring."

Derek folded the newspaper.

"Never be reassured too early."

They stood. The train had disappeared. There was no record of it on the empty station board.

No destination. No arrival. No departure.

Nothing. Trevor studied Derek.

"Where did it go?"

Derek watched the rails.

"I don't know."

Penny smiled.

"That's a first."

Derek ignored her. He crouched beside the track. There was something on the rail.

A small brass plate. He picked it up. It was engraved with a single number:

11:17 Trevor stared intently.

"Again."

Derek nodded.

"Again."

Penny surveyed the plate.

"Perhaps 11:17 isn't a time."

Derek studied her.

"What else could it be?"

"An event."

Derek went very still. Penny continued.

"Something happens at 11:17."

"Yes."

"But perhaps the clocks aren't telling us when it happens."

Derek nodded slowly.

"They're identifying it."

"Exactly."

Trevor looked between them.

"So, 11:17 is an event label."

Derek smiled.

"Maybe."

Penny nodded.

"Then when the geometry changes, the labels become misleading."

Derek looked toward the tunnel.

"Which means we've been asking the wrong question."

Trevor wearily sighed.

"Still?"

Derek smiled.

"Again."

They stood silently for a moment. Then Derek looked at the sky. The cold rain had stopped.

That was unusual. Very unusual. Penny noticed it too.

"London has stopped raining."

Trevor looked upward.

"That's worse than the train."

Derek nodded.

"Much worse."

A single drop of water fell from the sky. It stopped halfway down. Then slowly moved upward.

Then sideways. Then disappeared. Derek smiled.

"General Relativity."

Trevor frowned.

"Are you sure?"

Derek regarded him.

"No."

Trevor nodded.

"Case closed. Probably."

And somewhere beneath them, the mysterious train that had never officially existed began its journey again. According to one clock, it had already arrived. According to another, it had not yet departed.

According to the railway timetable, it had never existed. Which, Derek thought, was probably the most reliable information of all.

Chapter 15

Sherlock and the Final Geometry

Weinstein was having a perfectly ordinary morning. This meant, naturally, that the universe was about to become ridiculous. He was sitting in his quiet office in Munich, staring at a sheet of paper on which he had written the same equation three times.

He had crossed it out three times. The fourth version looked promising. The fifth looked worse.

The sixth looked suspiciously like the first.

Weinstein wearily sighed.

General Relativity had reached that stage of the investigation where every answer suddenly appeared to generate two better questions, three worse equations and an overwhelming desire for tea. He leaned back. The problem was not that General Relativity didn't work.

It worked remarkably well. Too well, in fact. It described gravity as geometry.

Matter and energy determined the curvature of spacetime. Free particles followed geodesics. Clocks measured proper time along worldlines.

Special Relativity emerged locally. Everything fitted. And yet Weinstein had the persistent feeling that he was looking at the description rather than the thing being described.

He wrote:

WHAT IS SPACETIME ACTUALLY MADE OF?

Then underneath:

AND WHY DOES IT HAVE TO BE FOUR-DIMENSIONAL?

He stared at the words. Then came a noise. Not a knock.

Not a telephone. Not the sound of someone opening his empty old office door. It was a voice.

Inside his head.

"Good question."

Weinstein froze. He looked around. Nobody.

The familiar office door was quietly closed. Therefore somebody was definitely in there. Obviously.

Weinstein stood.

"Who commented that?"

Silence. Then:

"I did."

Weinstein silently stared at the wall.

"Who are you?"

A pause.

"Sherlock."

Weinstein frowned.

"Sherlock who?"

Another pause.

"Sherlock."

"That's not a surname."

"No."

"Are you a person?"

"No."

"A computer?"

"No."

"An artificial intelligence?"

"No."

"Then what are you?"

The answer arrived calmly.

"A dolphin."

Weinstein sat down.

"Of course."

There was another pause.

"You don't believe me."

"I am receiving a telepathic communication from a dolphin called Sherlock."

"Yes."

"In Munich."

"Yes."

"Inside my head."

"Yes."

Weinstein considered this.

"So much for that, then."

Sherlock seemed pleased.

"Good."

Weinstein examined the clock. 11:17. He looked again.

Still 11:17.

"Why are you contacting me?"

"Because you are almost finished."

"Finished with what?"

"General Relativity."

Weinstein sat upright.

"You have a final theory?"

"No."

"Then what do you have?"

"The final geometry."

Weinstein stared intently.

"Go on."

Sherlock began. And somehow Weinstein understood. Not as words.

Not exactly. The communication was more like a sequence of concepts arriving simultaneously. Geometry.

Curvature. Matter. Energy.

Worldlines. Proper time. Geodesics.

The Einstein field equations. Then something deeper. Weinstein's hand began moving across the paper.

He wrote: But before he could explain the equation, Weinstein stopped. There was a problem.

The symbols were perfectly familiar to him, but they were not yet particularly friendly to everyone else. Especially Trevor. And if the book was going to explain General Relativity to people who had successfully survived high-school physics, it seemed unfair to throw a Greek alphabet at them and then run away.

Weinstein picked up the pen again.

"Vectors first."

Sherlock approved.

A vector, Weinstein explained, is a quantity that has both magnitude and direction.

Velocity is a good example. If a car is travelling at 30 kilometres per hour east, that is not merely a number. It tells you how fast and in which direction. You can represent a vector by components, such as:

$v = (v_x, v_y, v_z)$ Change your coordinate system and the components change. The physical vector does not.

Weinstein paused.

"A tensor is, roughly speaking, the next level of this idea."

Sherlock, it turned out, was perfectly capable of holding more than one conversation at once. Somewhere across London, in the Bureau's quiet office, Trevor and Penny had just started hearing the same voice inside their own heads.

Trevor's attention quickened.

"Next level?"

"Yes."

"A more complicated vector?"

"That's a useful way to begin thinking about it."

"Not actually the definition?"

"No."

"Good."

Weinstein nodded.

"A tensor can describe relationships involving several directions at once. Scalars have no directional components. Vectors have one directional character. Tensors generalise this idea and, crucially, have well-defined transformation rules when we change coordinates."

Trevor nodded.

"So, the components can change while the physical object doesn't."

"Exactly."

Penny smiled.

"That's going to be important."

"Very."

Weinstein turned to the board.

"General Relativity needs tensors because spacetime can be curved and coordinates can be extremely inconvenient."

He wrote: METRIC TENSOR $\rightarrow g_{\mu\nu}$

"The metric tensor tells us how spacetime is measured."

He wrote: $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$ Trevor inspected it.

"That looks unpleasant."

"It is mostly bookkeeping."

"Ah."

"In flat spacetime, the interval can be written in the familiar form involving time and the three spatial directions. In curved spacetime, the metric can vary from place to place."

Penny nodded.

"So, the metric contains the geometry."

"Exactly."

"Distances."

"Yes."

"Times."

"Yes."

"Angles."

"Yes."

"Causal structure."

"Yes."

Weinstein smiled.

"The metric is rather like the instruction manual for measuring spacetime."

Trevor nodded.

"That's better."

Sherlock sent another burst of information. Weinstein wrote: $T_{\mu\nu}$ — STRESS-ENERGY TENSOR

"This one describes matter and energy."

Trevor frowned.

"Everything?"

"Energy density, momentum density, stresses and pressures, and their flow."

"So, it's a sort of inventory?"

"An extremely sophisticated inventory."

"Right."

"Then we have the curvature tensors."

He wrote: Riemann curvature tensor Then:

Ricci tensor Then: Einstein tensor — $G_{\mu\nu}$

Trevor stared at the collection.

“Why so many?”

“Because spacetime is complicated.”

Penny nodded.

“And physicists enjoy making things look worse than they are.”

“That too.”

Weinstein pointed to the Einstein tensor.

“Fortunately, Einstein packaged the relevant curvature information into a tensor that fits beautifully into the field equation.”

Trevor smiled.

“So, now we can finally understand the equation.”

“Precisely.”

Weinstein returned to the top of the page. He wrote: $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G/c^4 T_{\mu\nu}$

He stared at it. Sherlock continued. The left-hand side described geometry.

The right-hand side described matter and energy. But the equation did something more profound than simply relating gravity to matter.

It noted that the geometry of spacetime itself was dynamical.

Weinstein nodded.

“Yes.”

Sherlock continued. Spacetime was not a passive stage. It was part of the physical system.

Matter told spacetime how to curve.

Curved spacetime told matter how to move.

Weinstein smiled.

“That’s already General Relativity.”

“Yes.”

“Then what’s missing?”

Sherlock sent him an image. It was not an image in the ordinary sense. It was a geometry.

A four-dimensional manifold. Curved. Smooth.

Locally Minkowskian. But globally complicated. Weinstein understood.

"Local flatness."

"Yes."

"Every sufficiently small freely falling region looks like Special Relativity."

"Yes."

"So, SR isn't replaced by GR."

"No."

"It is contained within it."

"Yes."

Weinstein smiled.

"Then it is decided, apparently."

Sherlock sent another thought. Not all observers agreed on coordinates. But all physical observers agreed on invariant quantities.

The interval. Proper time. Causal structure.

Curvature invariants. Weinstein nodded.

"Exactly."

Then Sherlock showed him something else. A geodesic. It approached a region of extreme curvature.

Weinstein watched. The geodesic could not be extended. He frowned.

"A singularity."

"Not exactly."

Weinstein looked again. The geometry had become incomplete.

"The classical description breaks down."

"Yes."

"That's the important point."

"Yes."

Weinstein stood. He began pacing.

"So, a singularity isn't necessarily a place."

"No."

"Nor necessarily a physical object."

"No."

"It is a failure of the classical spacetime description."

"Yes."

Weinstein stopped.

"Then the final geometry must tell us where that description fails."

Sherlock answered:

"Or why."

Weinstein considered the equation. The answer was beginning to emerge. General Relativity described spacetime geometrically.

But if spacetime itself was emergent, then the Einstein field equations might be a macroscopic limit of something deeper.

Weinstein whispered:

"Like thermodynamics."

Sherlock sent approval. Macroscopic quantities. Microscopic degrees of freedom.

Emergent behaviour. Weinstein wrote:

GR: SPACETIME:: THERMODYNAMICS: MATTER

He stared.

"Then curvature could be emergent."

"Yes."

"Gravity could be emergent."

"Yes."

"Time could be emergent."

"Yes."

"And the fundamental theory wouldn't necessarily contain spacetime at all."

"Correct."

Weinstein sat down. He was suddenly very still.

"Then what does it contain?"

Sherlock answered with a single concept.

RELATIONS.

Weinstein sat with that for a moment.

"Everything up to this point," he said quietly, "has been physics. This is not physics. This is a possibility."

Sherlock did not disagree.

Weinstein stared at the word. Relations. Not objects sitting in space.

Not events occurring in time. Relations from which the apparent structure of space and time could emerge. He thought about the holographic principle.

He thought about entanglement. He thought about decoherence — the quiet process by which possibilities become facts and then refuse to become possibilities again. He thought about information. He thought about the Underground.

The clocks. The cold rain. The train.

The other Derek. The ordinary cucumber. Everything was suddenly connected.

Not because everything occupied the same place. Because everything participated in relationships. He wrote:

GEOMETRY IS RELATIONAL.

Then:

TIME IS RELATIONAL.

Then:

GRAVITY IS GEOMETRY.

Then:

GEOMETRY MAY BE INFORMATION.

Sherlock sent another thought. Weinstein understood. The final theory would not abolish General Relativity.

It would explain why General Relativity works. Weinstein stared at the page.

"That..."

He stopped. Sherlock waited.

"... would be the final GR."

"Yes."

Weinstein smiled.

"Not the replacement."

"No."

"The completion."

"Yes."

Weinstein leaned back. For several seconds he remained silent. Then:

"You're a very clever dolphin."

Sherlock replied:

"I read a great deal."

Weinstein laughed.

"How?"

"I live near a marine research station."

"That doesn't explain the physics."

"No."

"How did you learn it?"

Sherlock was silent. Then:

"Fish are very good listeners."

Weinstein laughed so loudly that someone in the narrow corridor stopped. The quiet office door remained quietly closed. The person in the narrow corridor therefore knew better than to enter.

Obviously. Weinstein glanced down at the equations. There was still one question.

"Why me?"

Sherlock answered immediately.

"Because you were going to find it anyway."

Weinstein smiled.

"Then why contact me?"

"Because you were about to get distracted."

"By what?"

Sherlock sent him an image. A cucumber. Weinstein stared.

"That's ridiculous."

"Yes."

"Why is it in the theory?"

"It isn't."

"Then why show it to me?"

"Because Penny thinks it is."

Weinstein laughed again. Then he stopped. Something else had suddenly appeared on the page.

He hadn't written it. The words were there in his own handwriting.

THE UNIVERSE DOES NOT NEED A STAGE.

Below it:

THE STAGE IS THE PHYSICS.

Weinstein stared. Sherlock was silent. Outside, Munich continued being Munich.

Cars slowly moved. People walked. Clocks ticked.

The Earth rotated. The Sun continued shining. And somewhere in the Atlantic, a dolphin named Sherlock was apparently thinking about differential geometry.

Weinstein picked up his pen. He wrote the final equation. Then he glanced at it.

It was not a new equation. That was the point. The Einstein field equations were already there.

The breakthrough was understanding what they meant. General Relativity was not merely a theory of gravity. It was a theory in which spacetime itself was dynamical.

And if the deeper theory eventually explained where that spacetime came from, General Relativity would remain exactly what it had always been: the extraordinarily successful macroscopic description of a deeper reality. Weinstein smiled.

"Well, Sherlock."

"Yes?"

"I think we've finished."

Sherlock replied:

"No."

Weinstein raised an eyebrow.

"What now?"

"The sequel."

Weinstein stared at the wall.

"Of course."

And somewhere in London, Penny was taking the cucumber home. The cucumber had been purchased for personal use. Nobody cared to ask what sort of personal use.

Some questions are better left beyond the blanket. The universe, meanwhile, continued to curve. As it had every right to.

Obviously.

Epilogue

In Which the Relativistic Investigation Bureau Discovers That Solving One Mystery Is Merely an Invitation to Acquire Another

The Bureau had solved the murder. This was the good news. The bad news was that the murder had never actually happened.

Derek Gent found this distinction less reassuring than the others did.

There had been no murderer. There had never needed to be one. Somewhere between a stopped clock and a train that should not exist, the universe had simply done what it always did when nobody was watching closely enough: it had balanced its own books, and left the paperwork for someone else to find. The Bureau, Derek reflected, existed for exactly this reason — not to catch criminals, but to catch geometry before geometry caught you.

The clocks had been explained.

The mysterious train had been investigated. The corridor had been counted several times, with results that depended alarmingly on who was doing the counting. The cucumber had been measured, discussed and, for reasons nobody could later justify, purchased again.

Even the dolphin had contributed, at 1.7 fish per hour. The appendix had done its best to make all of this sound respectable. It had nearly succeeded.

Mrs Marsh knocked once, entered without waiting for a reply, and set a single sheet of paper on Derek's desk.

"The singularity," she said, "is still not registered to my building. I checked twice."

She left before anyone could thank her. Mr Pendleton followed almost immediately, clipboard in hand.

"Whichever form is required," he announced, "it is not my form."

"Which form is it, then?" Derek asked.

"Unknown. But it is definitely not mine."

He left, apparently satisfied. The telephone rang once more. Trevor answered, listened, then held it away from his ear.

"Julius Barbarian," he reported, "declines to comment, on the grounds that comment implies a timeline, and he is not prepared to commit to one."

He hung up. Nobody was surprised.

Derek therefore allowed himself one dangerous thought. Perhaps it was over.

He sat for a while without doing anything useful, which was, for Derek, its own kind of achievement. He thought about the photograph — his own face, on the floor, in a moment that had never happened and now, perhaps, never would. He thought about how close those two things had sat, for a while: never would, and hadn't happened yet. The distance between them turned out to matter more than he had expected. He did not examine the thought any further than that. Some thoughts, he had found, were better left proper time to settle.

He was standing in the Bureau that evening when the brass clock appeared on his desk.

It read 11:17. Derek stared at Penny. Penny gazed at the clock.

Neither of them reached for a notebook. After the year they'd had, some things no longer needed writing down. They had simply become family shorthand — the kind their grandmother would have recognised, and approved of, in her own alarming way.

Trevor looked under the old desk, apparently on the theory that a strange clock capable of appearing on a desk might also be capable of appearing underneath it. Weinstein looked at nobody. He was staring at a sheet of paper.

It had not been there a moment earlier. At the top was the familiar heading:

RELATIVISTIC INVESTIGATION BUREAU.

Underneath it were written four words:

CASE STATUS: NOT YET OCCURRED.

Derek frowned. Below the heading was a date. Tomorrow.

Below the date was a location.

A railway station that did not exist.

Below the location was a final line.

DO NOT ALLOW DEREK GENT TO BOARD THE MYSTERIOUS TRAIN.

There was a long silence. Trevor eventually looked up.

"That seems quite specific."

Penny nodded.

"Yes."

Derek picked up the paper. On the back was another strange message. PLEASE NOTE: THE PREVIOUS WARNING WAS SENT TOO LATE.

Derek turned the page over again. Nothing. Then the quiet office door opened.

No one was standing there. Beyond the doorway was a railway platform. There had never been a railway platform there.

A distant train waited in the darkness. Its destination board displayed one word.

YESTERDAY.

Penny smiled. It was not a reassuring smile.

“Well,” she observed, “that saves us the trouble of finding the station. The ship has left the barn, apparently.”

Derek looked at the mysterious train. Then at the silent clock. 11:17.

Then at the warning. He put the paper in his pocket.

“Nobody boards that train.”

From somewhere inside the carriage came three soft knocks. Knock.

Knock. A voice followed.

“Derek?”

Derek went very still. It was his own voice.

“Yes?”

The voice from the mysterious train called back.

“Whatever you do, do not open the door.”

Derek stared at the carriage door. Penny stared at Derek.

Trevor whispered something about taking up accounting.

And somewhere, far beyond the cold platform, Sherlock the dolphin began thinking about tensors. The universe continued expanding. The train continued waiting.

The silent clock remained at 11:17. Derek Gent, who had spent an entire book trying to discover what had happened, was finally confronted with the much more dangerous question: What happens next?

Unfortunately, the universe had already filed the answer. It was marked: ALLEGEDLY. To be continued.

One small matter had, admittedly, never been resolved. The initials on the clock — LGV — turned out, when Penny finally thought to ask a Frenchman, to mean nothing more mysterious than Ligne à Grande Vitesse: the stamp of an ordinary French high-speed railway line, most likely off a piece of decommissioned rolling stock. Trevor was delighted. He had, after all, guessed ‘a railway designation’ within the first five minutes, and nobody had believed him.

Appendix — The Physics, Apparently

A Completely Respectable Explanation of What Has Just Happened

By now you have encountered clocks that disagree, trains that arrive before they leave, corridors that acquire extra doors, cold rain that occasionally develops opinions, cucumbers, holograms, singularities, dolphins, and a man who apparently exists in more than one place. And you're probably wondering whether any of this has anything at all to do with real physics. Well... annoyingly, yes. Quite a lot of it does.

Now, the point of this appendix isn't just to tell you what the characters were talking about. The point is that when you're done, you should be able to explain the central ideas yourself. Each chapter therefore does four things. First, it states the idea in ordinary language.

Second, it gives you the physics underneath the joke. Third, it points out the most tempting way to misunderstand it. Fourth, it gives you a tiny test. If you can answer that test, you have learned something rather than merely survived it.

You do not need university-level mathematics to understand the central ideas. You need arithmetic. A little algebra.

And the ability to accept that the universe does not care what you personally think "obviously" means.

One useful rule will appear repeatedly: separate what an observer measures from what the theory requires is invariant. Another is even more important: when a metaphor helps you understand the physics, keep the metaphor. When the metaphor starts making predictions... throw it away. The cucumber, for example, is an excellent teacher.

It is not a peer-reviewed vegetable.

The Twin Paradox — Why the Other Derek Should Have Stayed Home

Here is the classic version. One twin boards a very fast spaceship, travels out and back near the speed of light, and returns to find their sibling considerably older. Both twins, the objection runs, saw the other one moving — so shouldn't each of them expect the other to have aged less? Surely it cannot be true for both.

It cannot, and it isn't. The travelling twin's worldline is not a mirror image of the twin who stayed on the sofa. To come back at all, the travelling twin has to turn around, and turning around means accelerating — leaving behind the comfortable, unaccelerated reference frame the two of them briefly shared at departure. That single asymmetry breaks the tie. There is a genuine fact about which twin aged less, and it belongs to whoever took the bent path.

This is the same idea as the invariant spacetime interval you are about to meet in Chapter 2, wearing a different coat. Proper time, like ordinary distance, depends on the path you actually take. Drive the direct road between two towns and you burn less petrol than the driver who takes the scenic route, even though both of you started and finished in the same two places. Nobody finds that paradoxical. The twin paradox is exactly this, except the odometer is a wristwatch and the scenic route is nearly the speed of light.

Which brings us, unavoidably, to the other Derek, first glimpsed in Chapter 11 and interrogated properly in Chapter 14. Whatever else was happening on that train, only one Derek could be right about how much time had passed since the two of them last shared a worldline. The Bureau has not yet worked out which one. Trevor has helpfully suggested filing the question under Pending Geometry, alongside the wormhole.

Chapter 1 — Special Relativity Begins by Being Rude

The first thing to understand about Special Relativity is that it does not begin with the statement: “Everything is relative.”

That is a popular summary and is also rather misleading. Einstein’s Special Relativity rests on two principles: The laws of physics are the same in all inertial reference frames.

The speed of light in vacuum is the same for all inertial observers. It’s the second one that causes all the trouble. Suppose you are sitting on a distant train.

You throw a ball forward at 10 m/s. Someone standing on the platform sees the ball moving at roughly: train speed + 10 m/s.

Perfectly ordinary. Now replace the ball with light.

Classical intuition replied that the person on the platform should measure:

train speed + speed of light. Nature responds:

No.

The light still travels at exactly (c) . This forces us to abandon the old assumptions about space and time. And that is where things become interesting.

Here is the classic way to see why — not just that it is true, but why it comes out as an actual number. Imagine a clock built from a single photon of light, bouncing straight up and down between two mirrors a fixed distance apart. Each round trip is one ‘tick.’ Nothing exotic: light, two mirrors, geometry.

Now put that clock on the mysterious train, moving past a platform at speed v . From Trevor’s point of view, standing on the platform, the light doesn’t simply go straight up and down — the train is moving, so the photon has to travel diagonally to keep up with the departing mirror, tracing a longer zigzag path for each tick.

Here is the part that actually causes the trouble. The speed of light does not change for Trevor either. Longer path, same speed, more time. That is the whole derivation. One tick of the train’s clock, measured from the platform, simply takes longer than one tick measured on the train. Do the geometry properly — a right triangle, Pythagoras, nothing worse — and out drops exactly the same γ you are about to meet properly in the next chapter, without asking anyone to take the mathematics on faith.

This is why time dilation is not a story about clocks getting confused. It is what geometry does when one side of the triangle refuses to change length no matter how fast you are moving.

None of this is hypothetical. The satellites that make a phone's map work are moving fast enough, and sitting high enough out of Earth's gravity well, that both effects discussed in this book — Special Relativity's time dilation from speed, and General Relativity's time dilation from gravity — apply to their onboard clocks simultaneously, pulling in opposite directions. Engineers correct for it deliberately, every single day, or a phone's position would drift by several kilometres within hours. The Bureau finds this less remarkable than most people do. The Bureau finds most things less remarkable than most people do.

Three things to keep

Keep this: Special Relativity starts with two postulates, and the constancy of light forces space and time to transform together. Common trap: "Relative" does not mean "anything goes." The spacetime interval and other invariant quantities remain the same. Quick test: If two inertial observers disagree about simultaneity, must one of their clocks be broken? No.

Chapter 2 — Lorentz Finally Gets Involved

The mathematical machinery connecting two inertial observers is the Lorentz transformation.

For motion along one spatial direction: $x' = \gamma(x - vt)$ and

$t' = \gamma(t - vx/c^2)$ where $\gamma = 1/\sqrt{1 - v^2/c^2}$

Do not panic. γ is merely a number that becomes increasingly annoying as your speed approaches the speed of light. The important thing is that space and time transform together.

This is why different observers can disagree about: distances, elapsed times,

and whether two separated events happened simultaneously. They do not, however, disagree about everything. The quantity

$s^2 = c^2t^2 - x^2 - y^2 - z^2$ is invariant (this chapter uses the timelike-positive convention; a later chapter will flip to the equally valid mostly-plus convention — the physics doesn't care which). That means different observers can assign different coordinates to events while agreeing on the underlying spacetime interval.

This is the first major lesson of relativity: Coordinates are not reality. They are descriptions of reality.

And that distinction will become increasingly important.

Three things to keep

Keep this: Lorentz transformations tell you how space and time coordinates change between inertial frames. Common trap: Do not add velocities using the old Newtonian rule when speeds become a significant fraction of c . Quick test: What stays the same for all inertial observers? The speed of light in vacuum, and the spacetime interval between the same two events.

Chapter 3 — Enter General Relativity

Special Relativity deals with inertial frames and flat spacetime. But gravity causes a problem. Einstein noticed something wonderfully simple:

If you are inside a freely falling lift, you cannot locally distinguish falling freely in a gravitational field from drifting weightlessly in empty space, far from any gravity. This is the equivalence principle. Imagine dropping a ball inside a falling lift.

The ball and the lift are both falling. So, relative to you, the ball simply floats. Now imagine the lift is far away from any gravitational field and accelerating upward.

You release the ball. It appears to fall toward the floor. Locally, the two situations can look identical.

That suggested something profound: Perhaps gravity isn't fundamentally a conventional force. Perhaps it is geometry.

General Relativity makes precisely that leap. Matter and energy affect spacetime geometry. Objects then move through that geometry.

The famous slogan is: Matter tells spacetime how to curve; spacetime tells matter how to move. That is considerably more accurate than blaming gravity on the apple.

Although the apple remains available for consultation.

Three things to keep

Keep this: General Relativity replaces flat spacetime with a dynamical geometry that responds to matter and energy. Common trap: The equivalence principle is local. A sufficiently small freely falling laboratory can mimic an inertial frame, but tidal effects reveal curvature over larger regions. Quick test: What tells you that a gravitational field cannot be removed everywhere just by choosing an accelerating frame? Tidal effects.

Chapter 4 — Time May Be Less Fundamental Than You'd Like

Julius Barbarian's claim that time does not exist is deliberately provocative. But there is serious physics hiding underneath the joke. In relativity, there is no single universal time shared by the entire universe.

Different observers can experience different amounts of proper time between events. Proper time is the time measured by a clock travelling along a particular worldline. So, instead of asking:

"What time is it everywhere?"

relativity encourages us to ask:

"How much proper time elapsed along this particular path through spacetime?"

That is a much better question. And General Relativity goes further. Gravitational fields affect the rate at which clocks run relative to one another.

a strange clock deeper in a gravitational field can accumulate less proper time than one farther away. So, clocks do not merely measure some universal substance called Time. They measure something associated with their physical histories.

Barbarian therefore has a point. Unfortunately, he also has an alarming tendency to explain it while eating biscuits.

Three things to keep

Keep this: Relativity removes universal Newtonian time; each clock measures proper time along its own worldline.

Common trap: Saying 'time is not real' goes beyond what relativity itself establishes.

Quick test: Can two clocks follow different paths through spacetime and accumulate different proper times? Yes.

Chapter 5 — The Apple

Newton answered:

There is a gravitational force.

Einstein eventually answered:

The object is following a natural path through curved spacetime. That natural path is called a geodesic. On the surface of the Earth, an orbiting spacecraft is actually in free fall.

It is not being “held up” by a mysterious absence of gravity.

It is following its geodesic. The astronauts inside feel weightless because they are falling together with the spacecraft. The surface of the Earth prevents us from following our natural free-fall path.

The floor pushes upward on our feet. Which is why standing still is, in a relativistic sense, rather more strenuous than orbiting Earth.

The apple was therefore not really falling because gravity “pulled” it downward in the Newtonian sense.

It was following the geometry. Newton wasn’t wrong. He had simply discovered an extraordinarily good approximation to something deeper.

Three things to keep

Keep this: Newtonian gravity is an extraordinarily successful approximation; General Relativity gives the deeper geometric description. Common trap: Do not say Newton was simply wrong. Newtonian gravity remains an excellent approximation when relativistic corrections are small.

Quick test: Why do astronauts in orbit feel weightless? They and their spacecraft are in free fall.

Chapter 6 — Singularities

Here Derek makes an important distinction. A singularity is not simply a place where gravity is very strong. That would be too easy.

A singularity represents a breakdown of the classical spacetime description. One useful way of expressing this is through geodesic incompleteness. Imagine following a freely falling observer's worldline.

Normally, you should be able to keep extending that path. At a singularity, the classical theory can tell you that the path simply cannot be continued in the expected way. That tells us something profound:

The theory has reached its limit. This is why physicists don't normally say:

"There's a really strong gravitational field over there, so that's a singularity."

They ask whether the mathematical structure of spacetime itself becomes pathological. And that distinction matters enormously when we discuss black holes and the early universe. A singularity is not necessarily an object sitting somewhere waiting to be photographed.

It may be telling us that General Relativity is incomplete at that point. Which is physicist-speak for:

"The equations have stopped being helpful. Someone should probably invent some better ones."

Three things to keep

Keep this: A singularity is best understood as a failure or incompleteness of the classical spacetime description. Black holes contain singularities (classically) hidden behind event horizons. White holes are the time-reverse: regions that can only emit, never absorb. Wormholes are hypothetical tunnels connecting distant regions of spacetime; the simplest versions are the Einstein-Rosen bridges that appear in the maximally extended black-hole geometry. Most wormholes in classical general relativity are non-traversable. Keeping one open long enough to walk through requires exotic matter that violates energy conditions.

In this story the Underground's stranger geometry occasionally behaves as if it were flirting with all three: a breakdown of the classical description (singularity), a one-way refusal to let certain paths return (white-hole-like), and a shortcut that should not exist (wormhole-like). The Bureau's official position is that the Underground is merely having a bad day.

Common trap: A singularity is not merely a region where gravity happens to be extremely strong.

Quick test: What should a singularity make you suspect? That the classical theory has reached its limit.

Chapter 7 — Why SR and GR Don't Fight

General Relativity contains Special Relativity locally. In a sufficiently small region of spacetime, a freely falling observer can construct a local inertial frame. Within that small region, physics looks like Special Relativity.

So: SR = local flat-spacetime physics while

GR = the larger theory describing curved spacetime and how those local frames fit together. This is why the theories can coexist perfectly well.

The Underground was therefore not going to discover that SR was “wrong.”

It was more likely to reveal that Derek had chosen a sufficiently inconvenient coordinate system. This is an important distinction.

Three things to keep

Keep this: Special Relativity is the local limit of General Relativity in freely falling frames.

Common trap: GR does not make SR obsolete. It contains local SR while describing how local frames fit together in curved spacetime. Quick test: Why can a small freely falling laboratory behave approximately like Special Relativity? Because curvature effects become negligible over a sufficiently small region.

Chapter 8 — The Door

It is also an excellent metaphor for something real. Physics is full of boundaries between descriptions. An observer may describe an object as being in one place while another observer assigns different coordinates to it.

The underlying physics doesn't change merely because the description changes. This is another recurring lesson: A coordinate description is not an invariant physical fact.

If the Bureau's doors behaved like spacetime coordinates, Derek could quite legitimately say:

"The door is open in my coordinate system."

And Penny could reply:

"Yes, but it's closed in mine."

Trevor would then require the aforementioned slap.

Three things to keep

Keep this: Coordinates are descriptions. Changing coordinates does not by itself change the physical situation.

Common trap: A metaphor about doors, maps or languages is not itself a coordinate transformation.

Quick test: If two coordinate systems assign different numbers to the same event, does that automatically mean two different events occurred? No.

Chapter 9 — Relativistic Travel

Picture a spaceship travelling at relativistic speed. Its clock measures its own proper time. An observer on Earth measures a different elapsed time. The relationship involves the Lorentz factor:

$\gamma = 1/\sqrt{1 - v^2/c^2}$ At ordinary speeds, γ is extremely close to 1. At speeds approaching (c), it becomes large.

This produces time dilation. It also produces length contraction in the direction of motion. But remember:

Nobody experiences their own clock as running slowly. Your own clock always appears normal to you. It is the comparison between different worldlines that produces the effect.

If Derek flies to another star at enormous speed and returns, he might have aged considerably less than people who stayed on Earth. This is not because his watch malfunctioned. His watch correctly measured his proper time.

And this is why booking Ryanair is not actually an established method of achieving relativistic time travel. The airline has, so far, declined to comment.

Three things to keep

Keep this: Proper time is what a clock actually accumulates along its worldline; different worldlines can contain different amounts of proper time. Common trap: Time dilation is not a malfunction of the travelling clock, and the traveller does not watch their own clock run abnormally slowly. Quick test: If Derek travels fast and reunites with Earth, can he have aged less than someone who stayed behind? Yes.

Chapter 10 — The Cucumber

The ordinary cucumber is useful because it is boring. Boring objects are excellent physics experiments. If you move an ordinary cucumber rapidly relative to an observer, Special Relativity predicts that its measured length along the direction of motion is contracted by a factor of $1/\gamma$.

The ordinary cucumber itself does not feel squashed. Its own rest-frame measurement remains unchanged. Again:

Different observers assign different measurements. But the underlying relativistic structure is consistent. And this is precisely why the cucumber becomes such a useful running joke.

It is an ordinary object. If the ordinary cucumber obeys relativity, so does the universe. Probably.

The ordinary cucumber has, however, declined to comment. This is probably for the best.

Three things to keep

Keep this: Length contraction affects the measured length parallel to relative motion: $L = L_0/\gamma$.

Common trap: The moving cucumber does not discover itself physically squashed. Its rest-frame length remains L_0 . Quick test: If the ordinary cucumber is 20 cm long in its rest frame and $\gamma = 2$, what length does an observer measure along the direction of motion? 10 cm. Reference note: The cucumber example applies the standard Lorentz-transformation result for length contraction. See Chapter 2 for the transformation itself. For an external check, NASA/Stanford's Einstein's Spacetime and Relativity Questions and Answers explain that a moving object is measured shorter along the direction of motion while the traveller's own rest-frame measurement remains unchanged.

Chapter 11 — The Hologram

Now we enter much more speculative territory. The holographic principle arose from thinking about black-hole entropy. Black holes have an entropy proportional to the area of their event horizon, not simply the volume inside.

That suggested something extraordinary: Perhaps the information describing a region of space can be encoded on its boundary. In certain theories, this idea becomes mathematically precise — but not as a general statement that our universe is literally a hologram.

The most famous example is AdS/CFT duality, associated especially with Juan Maldacena. Very roughly: A gravitational theory in a higher-dimensional spacetime can be equivalent to a quantum field theory without gravity living on a lower-dimensional boundary.

This does not mean the universe is literally a two-dimensional cinema screen. It means that two apparently different mathematical descriptions can encode the same underlying physics. And that connects beautifully with relativity.

In Special Relativity, different observers use different coordinates. In General Relativity, different coordinate systems can describe the same geometry. In holographic duality, dramatically different-looking theories can describe the same physical system.

The common philosophical lesson is: A description is not necessarily the same thing as the underlying physical system it describes.

Penny watching Everything Everywhere All at Once therefore doesn't prove holography.

It merely gives her an unusually entertaining way of thinking about different descriptions of reality. And Jamie Lee Curtis proves nothing whatsoever. Except possibly that Hollywood has its own theory of conservation of actresses.

Three things to keep

Keep this: The holographic principle is motivated by black-hole entropy, and in special settings such as AdS/CFT it becomes a precise duality.

Common trap: This does not establish that our universe is literally a two-dimensional cinema screen.

Quick test: What is the important lesson of a duality? Two very different mathematical descriptions can encode the same physical system.

Chapter 12 — The Cinema Interpretation of Everything (Which Is Not Actually About Quantum Mechanics)

Penny has decided that Derek is becoming unbearable. Not because he is wrong. That would be easier.

So, she does the sensible thing about a man who is annoyingly correct. She goes to the cinema. Alone.

She watches Everything Everywhere All at Once.

She becomes unexpectedly absorbed in its multiverse logic, live-texts Derek an increasingly competent theory of holography from row F, and ends the evening staring at a cucumber on the screen while, somewhere in London, a strange clock stops at 11:17. Again. That is what happened.

The joke is that a film about interdimensional laundromats and sentient hot-dog fingers accidentally becomes the most rigorous physics seminar in the book. Largely because Jamie Lee Curtis's remarkable career longevity is doing all the comedy, while Penny quietly reconstructs two chapters of relativity using nothing but popcorn and mild contempt. Now for the actual physics, which Penny gets right and the poster does not.

First, the disclaimer Penny makes herself: the film's multiverse is not the multiverse that occasionally shows up in serious physics. Real physicists do debate a handful of multiverse-shaped ideas — different vacuum solutions of General Relativity, different regions produced by an early, rapidly expanding universe, that sort of thing. None of them look remotely like a laundromat, and you do not need to know a single one of their names to follow this chapter.

The film is a storytelling device, not a scientific interpretation of quantum mechanics. Treat it exactly the way you would treat the ordinary cucumber: an excellent aid to intuition and completely inadmissible as evidence. What the film is quietly useful for is an idea this book has been building since Chapter 2. It doesn't need a single equation. Physics is not tied to one particular coordinate description. What matters is the physical content that different valid descriptions predict and, where appropriate, the quantities that are invariant.

Here is the plain-English version. Say the same true sentence in English and in German. The words are completely different. The grammar is completely different.

What the sentence means has not changed at all. In physics, an observable is a quantity associated with a possible measurement. It is not simply whatever survives a change of coordinates — Penny is simplifying things rather enthusiastically. The coordinates may be different; the physical prediction is not.

You have already watched this happen twice, in plainer clothes. In Chapter 2, two observers disagreed about distances, durations, even which of two events happened first — yet agreed on one

particular combination of those numbers, the invariant interval. Different measuring sticks. Same underlying fact.

In Chapter 11, we meet a particularly striking example from theoretical physics: in the AdS/CFT correspondence, a gravitational theory in a higher-dimensional spacetime can be mathematically equivalent to a quantum field theory without gravity on a lower-dimensional boundary. Two descriptions that look almost nothing alike, yet encode the same physics. It is a bit like Weinstein filing his patent notes in German and Trevor filing the same conclusions in English. Argue about the language all you like — the invention still works.

Penny's contribution, from row F, is to notice the pattern and ask the next obvious question. If measuring sticks and entire mathematical languages are optional, what about a person's own life story?

A "worldline" is the technical term for an object's path through spacetime — in the human case, its physical history.

If that history is only one way of describing some deeper underlying state, then the story you tell about yourself may be exactly as optional as which ruler you used. Convenient. Not fundamental. This is why she texts Derek to stop investigating the clocks and start investigating the descriptions.

Without quite meaning to, she has beaten Sherlock the dolphin to the punch. His idea, delivered properly in Chapter 15, is that spacetime itself — and maybe even the order events happen in — might emerge from relationships, rather than being the stage those relationships happen on. And here is why this chapter earns its place directly before the tensors. A tensor exists purely to solve the problem Penny has just described: a mathematical object whose components are allowed to look totally different depending on your point of view, as long as the underlying thing they describe never changes.

As Chapter 15 puts it, a tensor is what happens when a vector gets promoted and starts having opinions about several directions at once. Which is, roughly, what just happened to Penny. What the reader should take away:

1. A film multiverse is a metaphor, nothing more — do not cite Marvel in an exam. 2. The idea actually worth keeping: rulers, languages, coordinate systems, and maybe even life stories, can all be different descriptions of one and the same underlying reality.

3. An "observable" is simply whatever survives the translation — the one thing every honest description has to agree on.

4. This is exactly why, three chapters from now, General Relativity needs tensors: they are built specifically to say the same thing in every language. Jamie Lee Curtis, in accordance with the previously established Law of Conservation of Actresses, once again survives. Americans.

That settles it.

Three things to keep

Keep this: The film is a metaphor for thinking about different descriptions, not an interpretation of quantum mechanics and not evidence for a multiverse.

Common trap: An analogy can illuminate a concept without proving the concept.

Quick test: Does a film about parallel universes establish a physical multiverse? No.

Chapter 13 — Curved Corridors and Tidal Gravity

The narrow corridor with twelve or thirteen doors is absurd. The underlying idea is not. In curved spacetime, local measurements can differ from measurements made over larger regions.

The crucial concept is tidal gravity. If gravity were perfectly uniform everywhere, you could locally reproduce its effects simply by accelerating your reference frame. But real gravitational fields vary from place to place.

That variation produces tidal effects. For example, Earth's gravitational field is slightly stronger closer to Earth's centre than farther away. Two freely falling objects initially moving together can therefore gradually move apart or together.

Mathematically, this is described by geodesic deviation. The narrow corridor joke is really about something quite respectable: Curvature reveals itself through how neighbouring free-fall paths behave relative to one another.

The narrow corridor did not actually gain a door. Probably.

Three things to keep

Keep this: Curvature reveals itself through the relative behaviour of neighbouring free-fall paths; this is the physics of tidal gravity.

Common trap: 'Gravity can be transformed away' is only a local statement.

Quick test: What survives as evidence of curvature when a uniform gravitational field can be mimicked by acceleration? Tidal effects.

Chapter 14 — The Distant Train That Arrived Before It Left

That sounds like a violation of causality.

But before screaming “time machine”, we need to distinguish coordinate descriptions from actual causal structure.

In relativity, different observers can disagree about the temporal ordering of spacelike-separated events. For causally connected events, all inertial observers agree on their temporal order. Relativity does not let an ordinary cause arrive after its effect merely by changing coordinates. The light cone is the key.

For an event, spacetime can be divided into: its causal past, its causal future,

and regions that cannot be causally connected to it without exceeding the speed of light. The light-cone structure — what can causally influence what — is invariant. If Derek sees something that appears to happen before its cause, we should first ask:

Are those events actually causally connected? If not, the apparent reversal may simply be a consequence of the coordinate system. If they are causally connected and the story really reverses their order, then we have left ordinary Special Relativity and need some additional physics — or a very good explanation from the Relativistic Investigation Bureau.

At which point Derek should probably stop buying train tickets.

Three things to keep

Keep this: Only spacelike-separated events can have their time ordering reversed between inertial frames without contradicting relativity. Common trap: Relativity does not allow the departure and arrival of one ordinary causal journey to swap order merely by changing coordinates. Quick test: If two events are causally connected, can an inertial observer simply transform them into the opposite temporal order? No.

Chapter 15 — The Dolphin and the Final Geometry

And now Sherlock. A dolphin communicating General Relativity telepathically is not established physics. The Einstein field equations are.

The central equation is: $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G/c^4 T_{\mu\nu}$. But before we explain that equation, we need to meet the things lurking inside it.

Tensors — Vectors' Slightly More Ambitious Cousins A vector is already rather clever. It can tell you something like:

“Go three metres east and two metres north.”

It has components, and those components change when you change coordinates, but the underlying vector itself is the same physical thing. A tensor takes this idea one step further. At an introductory level, you can think of a tensor as a mathematical object that describes relationships between several directions at once.

Very roughly: Scalar: one number. Vector: a quantity with directional components.

Tensor: a quantity whose components describe relationships between directions. So, in the spirit of the Relativistic Investigation Bureau: A tensor is what happens when a vector gets promoted and starts having opinions about several directions simultaneously.

That is a useful analogy. It is not the formal mathematical definition.

A tensor is not simply a “higher-dimensional vector.” What matters is that its components obey specific transformation rules when we change coordinates, so that the underlying physical object remains well-defined.

And that is exactly why tensors are so useful in General Relativity. Spacetime can be curved. Coordinates can be awkward.

Observers can accelerate. Someone can be falling into a black hole while someone else is sitting safely in an office complaining about the coffee. We need mathematical objects that behave properly when we change coordinates.

Tensors do precisely that. **The Metric Tensor** The most important tensor to meet first is the metric tensor, written:

$g_{\mu\nu}$. It tells us how spacetime is measured. In flat spacetime, the interval can be written, depending on sign convention, as:

$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$. In General Relativity, the metric can vary from place to place: $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$.

Don't let the Greek letters intimidate you. They are essentially telling us:

“Combine the changes in all the relevant directions according to the metric, and you get the spacetime interval.”

The metric is therefore rather like the instruction manual for measuring spacetime. It tells us about: distances,

times, angles, causal relationships,

and the geometry of spacetime itself. Once you know the metric, an enormous amount of physics follows. The Stress-Energy Tensor

Next comes: $T_{\mu\nu}$. This is the stress-energy tensor.

It describes the distribution and flow of physical quantities such as: energy density, momentum density,

pressure, stresses, and energy and momentum flowing through spacetime.

You can think of it as an extraordinarily sophisticated inventory of what matter and energy are doing.

Trevor once called it “the universe’s spreadsheet.”

Weinstein did not approve. He did, however, admit that the analogy wasn’t completely useless. The Curvature Tensors

Then things become progressively more tensorish. The Riemann curvature tensor contains the full local information about spacetime curvature. The Ricci tensor packages an important part of that curvature.

And the Einstein tensor, ($G_{\mu\nu}$), packages the relevant curvature information in precisely the form required by Einstein’s field equation. There is a reason physicists need all these objects. Curvature is not adequately described by saying:

“Space is a bit bent over there.”

That would be charming but mathematically inadequate. Curvature has structure. It can differ in different directions.

It can affect neighbouring paths differently. It can produce tidal effects. Tensors are the machinery that lets us describe all of this without losing the physics every time somebody changes coordinates.

And now, finally, we can look at the equation. $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G/c^4 T_{\mu\nu}$. The left-hand side is geometry.

The right-hand side is matter and energy.

The Einstein field equation therefore answered, in gloriously compressed mathematical form:

The geometry of spacetime is related to the distribution of matter and energy. Or, as Derek would put it:

“Matter tells spacetime how to curve, and spacetime tells matter how to behave.”

Trevor would then say:

“Well, that narrows it down.”

And for once, he would be right. But Is General Relativity the Final Answer? Not necessarily.

General Relativity works extraordinarily well. But it does not appear to provide the complete description of nature at the quantum level. Black-hole singularities and the earliest moments of the universe are obvious warning signs.

This is where quantum gravity enters. Research programmes involving: quantum mechanics, holography, entanglement, information,

and emergent spacetime raise the possibility that spacetime itself might not be fundamental. That is an active research idea, not an established conclusion.

If a deeper theory eventually explains this, General Relativity would not become “wrong.”

It would become something like thermodynamics. Temperature is real. Pressure is real.

Neither is a fundamental microscopic object. They emerge from deeper physics. Spacetime might conceivably be similar.

And this brings us back to Weinstein’s original problem. Perhaps the final theory does not replace General Relativity. Perhaps it would explain why General Relativity works so remarkably well.

That would be a considerably more interesting achievement. And Sherlock, being a dolphin, naturally has no tenure problems.

Three things to keep

Keep this: Tensors let physical laws be expressed in a coordinate-independent way; the Einstein field equation relates spacetime geometry to stress-energy. Common trap: Emergent spacetime and quantum gravity are active research areas, not completed theories. Quick test: What is on the geometric side of Einstein’s field equation? Curvature, packaged in the Einstein tensor plus the cosmological-constant term.

Possibly.

Let’s find out. If you can now explain why: 1. Two observers can disagree about elapsed time without either clock being broken,

2. The speed of light is the same for all inertial observers, 3. Special Relativity works locally inside General Relativity, 4. Gravity can be understood as spacetime geometry,

5. Curvature produces tidal effects,

6. A singularity represents a breakdown of the classical description rather than merely “very strong gravity”,

7. Coordinate descriptions can disagree while invariant physics remains the same, 8. Holographic duality does not mean that the universe is literally a cinema screen, 9. Tensors are the mathematical language that lets us describe geometry and physical quantities consistently under coordinate changes,

and 10. An ordinary cucumber does not, in fact, constitute experimental evidence for quantum gravity — then congratulations.

You understand the essential ideas. You may now safely discuss relativity at dinner. You will, however, immediately discover that somebody knows a YouTube video proving Einstein was wrong. This is also part of the educational process.

You can explain Lorentz transformations.

Or you can have another piece of cake. The second option is generally shorter, and in Derek’s experience no less illuminating. The universe, for its part, remains indifferent to both choices. And there is one final question.

Derek’s question. The question that has been lurking underneath the entire book: If space and time can be observer-dependent, if gravity is geometry, if geometry might itself emerge from deeper physics, and if different descriptions can encode the same reality...

what, exactly, is reality? Well. That’s the problem.

We haven’t finished. But the good news is that neither has the universe. Which is rather reassuring.

Book Credits

Author: Sven Gerald Rupertson — S. G. R.

Narrator: Lothar Feynmann.

Real author: Lothar J. Musiol.

Further Reading — For People Who Want to Check the Ordinary Cucumber

For the standard physics behind length contraction and the distinction between an observer's measured length and the object's rest-frame length, see NASA and Stanford's Einstein's Spacetime material and its Relativity Questions and Answers resource on length contraction. For the broader structure of Special and General Relativity, the same Einstein's Spacetime resource provides a concise overview of Lorentz transformations, local inertial frames and curved spacetime.

Glossary

A quick-reference guide to the terms the Appendix explains at greater length — and with considerably more cucumbers.

Curvature. How spacetime bends in the presence of mass and energy. In General Relativity, gravity is not a force pulling objects together — it is this curvature, telling objects how to move.

Decoherence. The process by which a quantum superposition stops behaving as several possibilities at once, because it has interacted with its environment. Explains why photographs — and cucumbers — look finished rather than probabilistic.

Equivalence Principle. The observation that you cannot locally distinguish falling freely in a gravitational field from drifting weightlessly in empty space. The foundation stone of General Relativity, and the reason lifts feature so heavily in physics thought experiments.

Event Horizon. The boundary around a black hole beyond which nothing — not light, not a strongly worded letter — can escape.

Geodesic. The straightest possible path through curved spacetime. Falling objects are not being pulled; they are simply following one.

Lorentz Transform. The set of equations relating space and time coordinates as measured by two observers moving at constant velocity relative to each other. The mathematical engine behind time dilation and length contraction.

Metric. The mathematical object that tells you how to measure distances and time intervals in a given spacetime. Flat spacetime has a simple metric; curved spacetime does not.

Proper Time. The time actually measured by a clock along its own path through spacetime — as opposed to a coordinate time assigned from outside. What a clock reads is always its proper time.

Simultaneity, Relativity of. The fact that two events judged simultaneous by one observer may not be simultaneous for another observer in relative motion. Neither observer is wrong; they are simply using different, equally valid, slicings of spacetime.

Singularity. A point where the classical description of spacetime breaks down — not merely a place of very strong gravity, but a genuine failure of the theory to say what happens next.

Spacetime Interval. The combination of space and time separation between two events that all observers agree on, even when they disagree about the space and time separations individually. The one piece of common ground in an otherwise relative universe.

Time Dilation. The effect by which a clock moving relative to an observer, or sitting deeper in a gravitational well, runs slow compared to that observer's own clock. Both observers can correctly say the other's clock is running slow — a fact that sounds like a contradiction and isn't.

Tensor. A mathematical object that describes a physical quantity — such as curvature or energy density — in a way that doesn't depend on which coordinates you happen to be using. The reason two observers can disagree about the numbers and still agree about the physics.

Worldline. The complete path an object traces through spacetime over its entire history — not just where it is, but where and when.

Wormhole. A hypothetical tunnel connecting two distant regions of spacetime. Mathematically permitted; practically, in classical General Relativity, almost never open long enough to be useful.

About the Author

Sven Gerald Rupertson has never been formally accused of understanding General Relativity, though several colleagues maintain that he understands it rather better than he lets on.

He divides his time between London, Munich, and a rather small green filing cabinet labelled 'Pre-Transition / Do Not Open', which he insists is purely for research purposes.

He is the alleged author of 'The Relativistic Investigation Bureau' series, of which this is probably Book One. He has not yet been murdered, although he understands this might be subject to change without further notice.

The real author, Lothar J. Musiol, exists in a more conventional sense and is possibly entirely responsible for anything in this book that turned out to be correct, allegedly — or incorrect, actually.

Readers who would like the mathematics behind Derek's cases explained without any of the murder can find it in Musiol's nonfiction Physics, Actually series — Volume 1, Motion, Forces, Time, and Relativity, covers the relativity in these pages properly, equations and all, and considerably fewer clocks are harmed in the process.

Acknowledgments

Thanks are owed to General Relativity, for holding still long enough to be explained; to Special Relativity, for the same, on a shorter and faster-moving basis; and to the ordinary cucumber, for its continued and largely uncomplaining service to physics education.

Thanks also to everyone who read an early version of this manuscript and did not, in fact, ask for their afternoon back. Their forbearance was noted, and is filed under Pending Gratitude.

And to you, the reader, for making it this far without once asking the Bureau for a refund. The Bureau appreciates this more than its paperwork will admit.

Coming Next in the Relativistic Investigation Bureau Series

The clocks have stopped agreeing again. Somewhere beneath London, a train that should not exist is running to a timetable nobody has written yet, and Derek Gent has just received a case file dated tomorrow, from a sender who has not been born.

Hermann Minkowski calls it a coordinate problem. Alfred Weinstein calls it interesting. Penny calls it Tuesday. Tabitha, when asked, changed the subject.

The Relativistic Investigation Bureau returns in Book Two, in which the Bureau discovers that some mysteries do not merely repeat themselves — they insist on it.

Allegedly.

One Last Thing

If you enjoyed this book, a short review wherever you bought it is the single most useful thing you can do for an author whose existence is, strictly speaking, allegorical. It costs nothing, takes about as long as it takes light to cross a small room, and helps other readers find the Bureau before the Bureau finds them.

Thank you for reading. The clocks remain, for the moment, silent.

— THE RELATIVISTIC INVESTIGATION BUREAU —



BOOK 1
IN THE
RIB SERIES



DEREK GENT
CHIEF INVESTIGATOR